

EE114 - Power Engineering - 1

Magnetic circuits - 50-60 Hz - Power Frequency
we will discuss

High voltage long-distance power transmission -
DC has certain advantages.

DC - Power Electronics, e.g., mobile chargers,
solar cells/panels etc

Disadvantage -

Power Electronic machinery is light, hence has
less inertia $\rightarrow$ hence generation system will
become less stable

Voltage

Power transmission is carried out at high voltage.

Why?

At same power, high voltage $\Rightarrow$ low current ($P = VI$)

Advantages -

- Less power loss in transmission lines ($P = I^2 R$)
- Less voltage drop across transmission lines ($V = IR$)
- Less current $\Rightarrow$ we can have less cross section of
Transmission wires ($R \propto 1/A$)
 $\rightarrow$ Less cost of wire

Disadvantage of high voltage -

- Needs high clearance (safety distance) from ground
 $\Rightarrow$ High tower and thick insulation $\Rightarrow$ High cost

Frequency

Some countries have 50 Hz and some have 60 Hz.
 (India) (USA)

Advantage and of higher frequency -

- Power output $\propto$ no. of turns $\times$ frequency.
 $\therefore$ For same power, more frequency $\Rightarrow$ less turns
 $\Rightarrow$ less cost

Disadvantage

- Skin effect (due to internal magnetic flux)
 Magnetism - Internal flux exists
 Electricity - No internal flux
- $R_{ac} > R_{dc}$ due to inductive effect
 inductive effect $\propto$ frequency, eddy current loss $\propto f^2$

Some statistics regarding power generation

In 2024, total generation was
 453 GW, out of which

202 GW (46%) is through renewable methods



Solar (91 GW)

Wind (47 GW)

Large Hydro (47 GW)

Small Hydro (5 GW)

Biomass (11 kW)

Nuclear (8.18 GW)

Solar and wind sources also have environmental disadvantages along with advantages

In India, only 16% of total energy consumption is of electrical nature.

Electrical power - also has environmental damage to some extent
 (Lithium)

Future of energy? $\rightarrow$ Hydrogen economy

AC Waveform

$$V = V_p \sin(\omega t + \phi)$$

Why do we prefer sinusoids -

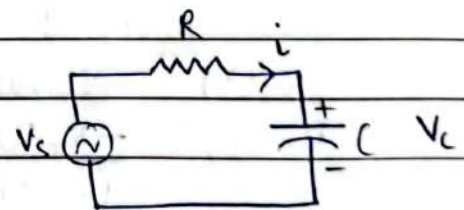
- Addition, subtraction, differentiation and integration also result into sinusoids. Hence, due to capacitors & inductors, the ~~net~~ circuit output also becomes sinusoidal.
- Due to Fourier series, sinusoids are convenient
- In power generation, sinusoids are favourable, since they don't have high harmonics.
 High harmonics (high frequency) $\rightarrow$ lead to high eddy current loss ($\propto f^2$)
- Sinusoids are commonly encountered in nature (e.g. pendulum)

Harmonics

Component ~~Integers~~ Sinusoids with frequency that is an integer multiple of the fundamental frequency.

$$V_s = 6 \sin 2t$$

$$V_c(t) = ? \quad (\text{Forced Response})$$



$$V_s = Ri + V_c$$

$$V_s = RC \frac{dV_c}{dt} + V_c$$

v_c and i_L are called state variables

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YOUVA

$$\text{Let } v_c(t) = A \cos 2t + B \sin 2t$$

$$6 \sin 2t = RC(-2A \sin 2t) + RC(2B \cos 2t) + A \cos 2t + B \sin 2t$$

$$A = -2RCB$$

$$B - 2ARC = 6$$

$$B + 2RC(2RCB) = 6$$

$$B(1 + 4R^2C^2) = 6$$

$$B = \frac{6}{1 + 4R^2C^2}$$

$$A = \frac{-12RC}{1 + 4R^2C^2}$$

$$\text{For } R = 1 \Omega, C = \frac{1}{2} F$$

$$A = -3, B = 3$$

$$v_c(t) = -3 \cos 2t + 3 \sin 2t = 3\sqrt{2} \sin(2t - \pi/4)$$

$$P_s(t) = v_s(t)i(t) = 6 \sin 2t \times 3\sqrt{2} \sin(2t - \pi/4) \\ = -9\sqrt{2} \cos(4t + \pi/4) + 9$$

$$\langle P_s(t) \rangle = 9 \text{ W}$$

Hence,

$$\langle P_R(t) \rangle = 9 \text{ W}$$

$$P_c(t) = v_c(t)i_c(t) = -9 \cos 4t$$

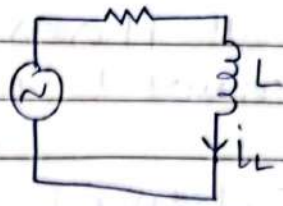
Generalizing the above computation

$$v_s(t) = V_m \sin \omega t$$

$$v_c(t) = \frac{-V_m \omega RC}{1 + \omega^2 R^2 C^2} \cos \omega t + \frac{V_m}{1 + \omega^2 R^2 C^2} \sin \omega t$$

$$V_s(t) = V_m \cos \omega t$$

$$i_L(t) = \frac{V_m R}{R^2 + \omega^2 L^2} \cos(\omega t) + \frac{V_m \omega L \sin(\omega t)}{R^2 + \omega^2 L^2}$$



Let

$$i_L(t) = A \cos(\omega t - \theta)$$

$$A \cos \theta = \frac{V_m R}{R^2 + \omega^2 L^2}, \quad A \sin \theta = \frac{V_m \omega L}{R^2 + \omega^2 L^2}$$

$$\tan \theta = \frac{\omega L}{R}$$

$$\cos \theta = \frac{R}{\sqrt{R^2 + \omega^2 L^2}}$$

$$A = \frac{V_m}{\sqrt{R^2 + \omega^2 L^2}}$$

$$i_L(t) = \frac{V_m}{\sqrt{R^2 + \omega^2 L^2}} \cos\left(\omega t - \tan^{-1} \frac{\omega L}{R}\right)$$

Phasors

Due to linearity of R , L and C , "real" sources (i.e. real part of $e^{j\omega t}$) will produce "real" output. "Imaginary" sources will produce "imaginary" output.

Hence replacing $\cos \omega t$ with $e^{j\omega t}$ will not affect "real" output.

In the equation, since all terms will contain $e^{j\omega t}$, they will cancel out.

$$i_L(t) = \text{Re} \{ A m_s \times e^{j\omega t} \}$$

$A m_s$ will be of form $A e^{-j\phi}$

$$\therefore i_L(t) = \text{Re} \{ A e^{-j\phi} e^{j\omega t} \} = \text{Re} \{ A e^{j(\omega t - \phi)} \}$$

$= A \cos(\omega t - \phi)$ which is the form derived

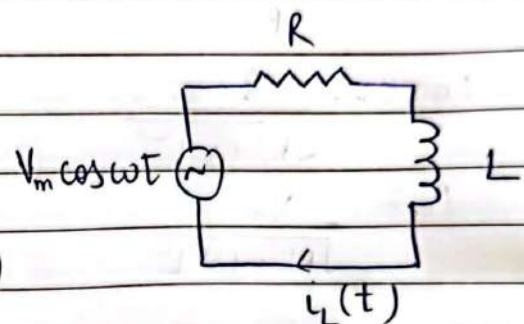
- With phasors, differential equations in real domain convert to algebraic equations in complex domain

Impedance (Z) is not a phasor since it does not rotate on the complex plane.

Power is not a phasor since it does has twice the frequency.

Forced Response to AC

$$Ri + L \frac{di}{dt} = V_m \cos \omega t$$



Assuming $i = I_m \cos(\omega t + \phi)$

$$\text{Let } \tilde{I} = I_m e^{j\omega t} e^{j\phi}$$

$$\frac{d\tilde{I}}{dt} = I_m j\omega e^{j\omega t} e^{j\phi}$$

$$\therefore R I_m e^{j\omega t} e^{j\phi} + L j\omega I_m e^{j\omega t} e^{j\phi} = V_m e^{j\omega t}$$

$$\therefore \frac{I_m (R + j\omega L) e^{j\phi}}{I_m e^{j\phi}} = \frac{V_m}{R + j\omega L}$$

Natural Response

Response with zero forcing function (i.e., source-free response)

$$iR + L \frac{di}{dt} = 0$$

$$\therefore i = I_0 e^{-Rt/L}$$

Complete Response to DC

$$i(t) = \frac{V}{R} (1 - e^{-Rt/L})$$

The length of a phasor is the RMS value of the quantity.

Complete Response to AC

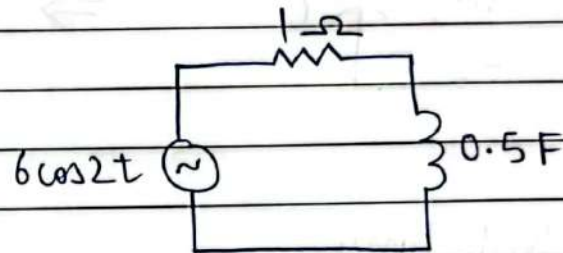
$$i(t) = \underbrace{K e^{-Rt/L}}_{\text{Natural Response}} + \underbrace{\frac{V_m}{\sqrt{R^2 + \omega^2 L^2}} \cos(\omega t - \tan^{-1} \omega L/R)}_{\text{Forced Response}}$$

K depends on initial current $i(0)$

When $i(0) = 0$, $K = \frac{-V_m}{\sqrt{R^2 + \omega^2 L^2}} \cos[-\tan^{-1} \omega L/R]$

Example

Find $i(t)$, $v(t)$



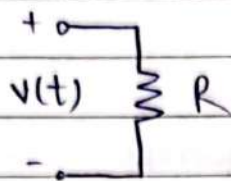
Phasors are represented as $\tilde{V}$, $\tilde{I}$ etc.

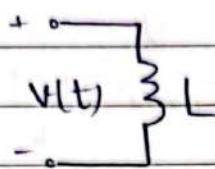
How to compare phasors?

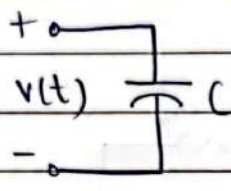
- Same Frequency
- Same sign
- Both either sin or both cos.

$$I_m \cos(\omega t + \phi) \Leftrightarrow \frac{I_m}{\sqrt{2}} e^{j\phi} = \frac{I_m}{\sqrt{2}} \angle \phi$$

Phasor Analysis of Some Simple Circuits


 $\Rightarrow \begin{aligned} v(t) &= R i(t) \\ \tilde{V} &= R \tilde{I} \end{aligned}$


 $\Rightarrow \begin{aligned} v(t) &= L \frac{di(t)}{dt} \\ \tilde{V} &= j\omega L \tilde{I} \end{aligned}$


 $\Rightarrow \tilde{V} = \frac{\tilde{I}}{j\omega C}$

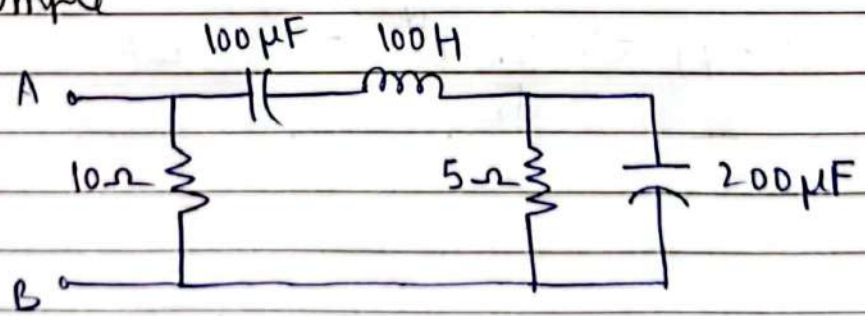
Impedance

- Not applicable in time domain

$L \rightarrow Z_L = \frac{\tilde{V}}{\tilde{I}} = j\omega L = \omega L e^{j\pi/2}$

$C \rightarrow Z_C = \frac{\tilde{V}}{\tilde{I}} = \frac{1}{j\omega C} = \frac{1}{\omega C} e^{-j\pi/2}$

Example



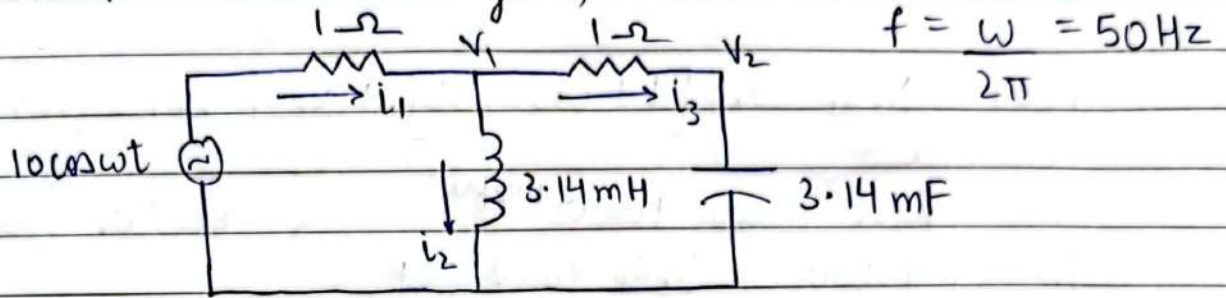
$f = 50 \text{ Hz}$, Find Z_{AB}

10mA — Fatal current for body (human body)

Good Book - Electromagnetics Explained - Schmitt

$$\tilde{Z}_{\text{series}} = \tilde{Z}_1 + \tilde{Z}_2$$
$$\frac{1}{\tilde{Z}_{\text{parallel}}} = \frac{1}{\tilde{Z}_1} + \frac{1}{\tilde{Z}_2}$$

Example (Nodal Analysis)



Ans.

$$\tilde{I}_1 = 4.472 \angle -26.56^\circ \text{ A}$$
$$\tilde{I}_2 = 6.32 \angle -71.56^\circ \text{ A}$$
$$\tilde{I}_3 = 4.472 \angle 63.43^\circ \text{ A}$$
$$\tilde{V}_1 = 6.32 \angle 18.43^\circ \text{ V}$$
$$\tilde{V}_2 = 4.472 \angle -26.56^\circ \text{ V}$$

When switching off an electric appliance, if at the instant of switching off, current was high, the inductor would have high $\frac{1}{2}LI^2$ energy.

When switched off, the break in the circuit at the switch acts as a capacitor. ~~Very~~ Instantaneously, the energy $\frac{1}{2}LI^2$ is converted to $\frac{1}{2}CV^2$ across capacitor. This causes high V across capacitor, leading to breakdown and sparking.

Power

$$p(t) = v(t) i(t)$$

$$P = \frac{1}{T} \int_0^T v(t) i(t) dt = \frac{V_p I_p \cos \phi}{2}$$

$$p(t) = V_p \cos(\omega t) I_p \cos(\omega t + \phi)$$

$$= \frac{1}{2} V_p I_p \{ \cos(2\omega t + \phi) + \cos \phi \}$$

$$p(t) = \underbrace{\frac{V_p I_p \cos \phi}{2}}_{\text{Real / Active Power (useful)}} + \underbrace{\frac{V_p I_p \sin \phi}{2} \sin 2\omega t}_{Q = \text{Reactive Power}}$$

$$\text{Average Power} = \frac{V_p I_p \cos \phi}{2} = P$$

$$\text{Reactive Power} = Q = \frac{V_p I_p \sin \phi}{2}$$

In practical circuits, even in purely resistive circuits, L and C components are present.

Because of current magnetic flux gets induced in conductors $\Rightarrow$ leading to inductance.

Two wires of the same circuit lying side-by-side $\Rightarrow$ leading to stray capacitance (only becomes significant at high frequencies).

P is present because of R; Q is present because of L and C.

Power for resistive circuit

$$\phi = 0$$

$$v = V_p \cos \omega t, \quad i = I_p \cos \omega t = \frac{V_p \cos \omega t}{R}$$

$$P = \frac{V_p^2}{2R} = \frac{V_{rms}^2}{R}$$

RMS Values

$$I_{rms} = \sqrt{\frac{1}{T} \int_0^T i^2(t) dt}$$

I_{rms} is that effective ^{DC} current which will produce same amount of heating.

For AC,

$$I_{rms} = \frac{I_p}{\sqrt{2}}, \quad V_{rms} = \frac{V_p}{\sqrt{2}}$$

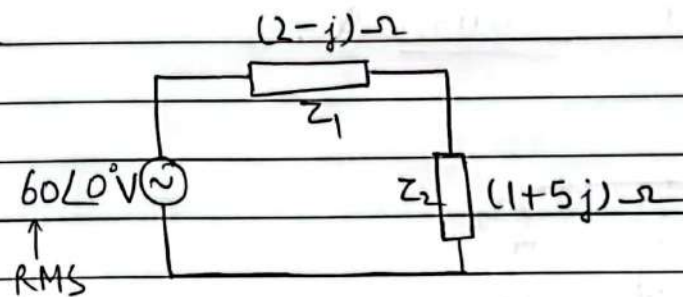
This equation is for analytic calculation.

For numeric calculation,

Since $i^2(t)$ is same in $[0, T/2]$ and $[T/2, T]$, we can sample the $i^2(t)$ ^{from $[0, T/2]$} waveform into a number of discrete samples and take an average.

$$I_{rms} \approx \sqrt{\frac{i^2(t_1) + i^2(t_2) + \dots + i^2(t_n)}{n}}$$

Example



$$\tilde{I} = \frac{60}{3+4j} = 12\angle -0.9273^\circ \text{ A}$$

$$I_{rms} = 12 \text{ A}$$

$$\text{Average power taken by } Z_1 = I_{rms}^2 R = (12)^2 \times 2 = 288 \text{ W}$$

$$\text{" " " } Z_2 = I_{rms}^2 R = (12)^2 \times 1 = 144 \text{ W}$$

$$\text{Total power} = P = 432 \text{ W}$$

~~Reactive Power~~

$$\text{Apparent Power} = V_{\text{rms}} \times I_{\text{rms}} = 60 \times 12 = 720 \text{ VA}$$

$$\text{P.F.} = \frac{P}{|S|} = \frac{432}{720}$$

$$P = VI \cos \phi$$

$$Q = VI \sin \phi$$

$$P = V_{\text{rms}} I_{\text{rms}} \cos \phi = \text{Re}\{VI^*\}$$

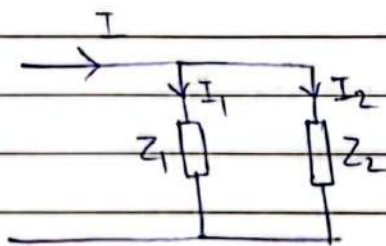
$$Q = V_{\text{rms}} I_{\text{rms}} \sin \phi = \text{Im}\{VI^*\}$$

$$|S| = V_{\text{rms}} I_{\text{rms}} = |VI^*|$$

$$\bar{S} = P + jQ = \tilde{V}\tilde{I}^*$$

conventionally,
For inductive circuit, $Q > 0 \Rightarrow Q$ is absorbed
For capacitive circuit, $Q < 0 \Rightarrow Q$ is supplied

Both L and C require Q .



$$\bar{S} = \tilde{V}\tilde{I}^* = \tilde{V}(\tilde{I}_1^* + \tilde{I}_2^*) = \tilde{V}\tilde{I}_1^* + \tilde{V}\tilde{I}_2^*$$

$$\bar{S} = \bar{S}_1 + \bar{S}_2$$

For inductor,

$$\tilde{V} = VL0'$$

$$\tilde{I} = I \angle -\pi/2$$

$$\tilde{V}\tilde{I}^* = jVI \Rightarrow Q > 0 \text{ (Q absorbed)}$$

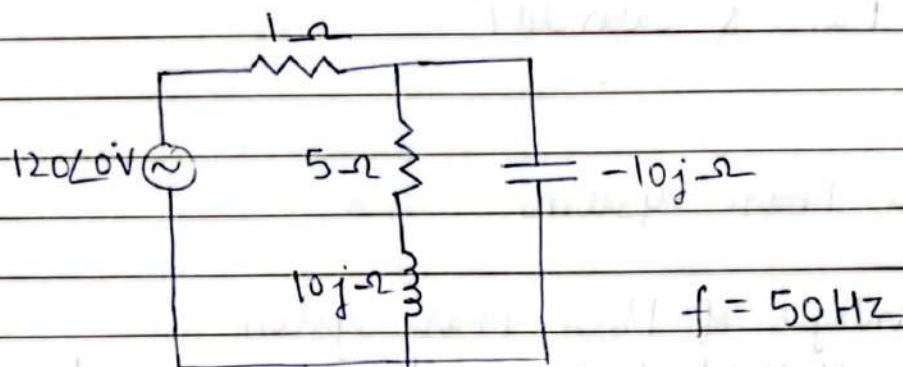
For capacitor,

$$\tilde{V} = V \angle 0, \tilde{I} = I \angle \pi/2$$

$$\tilde{V} \tilde{I}^* = -jVI \Rightarrow \phi < 0 \quad (\text{supplied})$$

The notions of ϕ absorbed and supplied does not make sense for purely L or purely C circuits. It makes sense for LC circuits.

Example

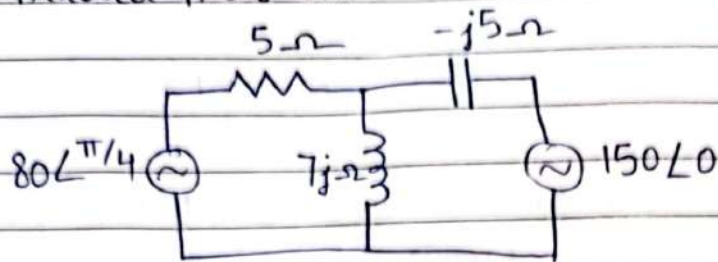


Find complex power absorbed by

- i) 1Ω resistor $\Rightarrow (27 + j0) \text{ VA}$
- ii) $(5 + j10)\Omega$ $\Rightarrow (532 + j1065) \text{ VA}$
- iii) $-j10\Omega$ $\Rightarrow (0 - j1331) \text{ VA}$
- iv) 120 V source $\Rightarrow (-559 + j266) \text{ VA}$

In last point P is negative because source is actually supplying power, but we want power absorbed.

Practice Problem



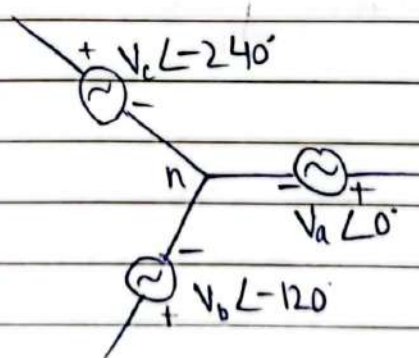
- Calculate complex power given out by each source.
- Calculate average power absorbed by 5Ω resistor.
- Calculate Q absorbed by $7j\Omega$.
- Calculate Q delivered by $-5j\Omega$.
- Are P and Q balanced?

Three-Phase Systems

Advantages of Three-Phase System

- More efficient of power transmission than single phase, i.e.; lower costs.
- Total instantaneous power delivered is constant in time.
- Single-Phase ^{motors} don't have a self-starting torque. 3- ϕ motors do.

Star-Connected Three-Phase System



This is called a-b-c sequence

Conventionally, we call
 $a-b-c \rightarrow$ positive sequence
 $a-c-b \rightarrow$ negative sequence
 zero sequence $\rightarrow$ no phase difference b/w a, b and c

Phase Voltages

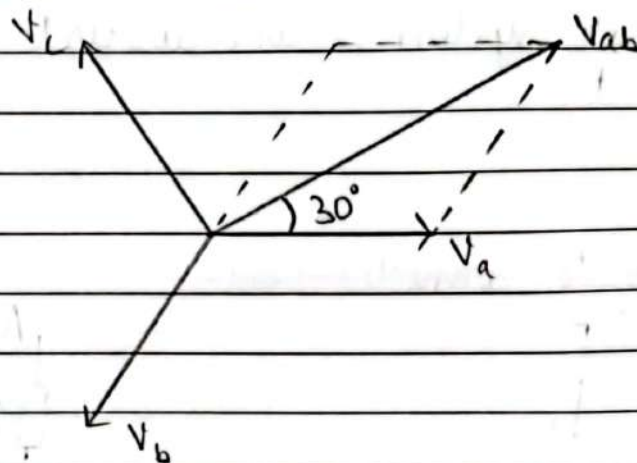
$$V_a = V_{an}$$

$$V_b = V_{bn}$$

$$V_c = V_{cn}$$

$$V_a + V_b + V_c = 0 \text{ at every instant}$$

Line Voltages



$V_{ab} = \sqrt{3} V_a$ and V_{ab} is 30° ahead of V_a .
 $V_{ab} + V_{bc} + V_{ca} = 0$ at every instant.

For balanced loads

$$\tilde{I}_a = \frac{\tilde{V}_{an}}{\bar{Z}}, \quad \tilde{I}_b = \frac{\tilde{V}_{bn}}{\bar{Z}}, \quad \tilde{I}_c = \frac{\tilde{V}_{cn}}{\bar{Z}}$$

$$S_a = \tilde{V}_{an} \tilde{I}_a^* = \tilde{V}_{an} \left(\frac{\tilde{V}_{an}}{\bar{Z}} \right)^* = \frac{\tilde{V}_{an}^2}{\bar{Z}^*} = \frac{V_{an}^2}{Z} \angle \phi$$

where $\bar{Z} = Z \angle \phi$

$$S_b = \frac{V_{bn}^2}{Z} \angle \phi, \quad S_c = \frac{V_{cn}^2}{Z} \angle \phi$$

$$\therefore S_a + S_b + S_c = \frac{3 V_{\text{m}}^2}{Z} \angle \phi = P + jQ$$

where V_{m} = line-to-neutral voltage

$$\therefore P = \frac{3 V_{\text{m}}^2}{Z} \cos \phi = 3 V_{\text{m}} I_{\text{L}} \cos \phi$$

$$Q = \frac{3 V_{\text{m}}^2}{Z} \sin \phi = 3 V_{\text{m}} I_{\text{L}} \sin \phi$$

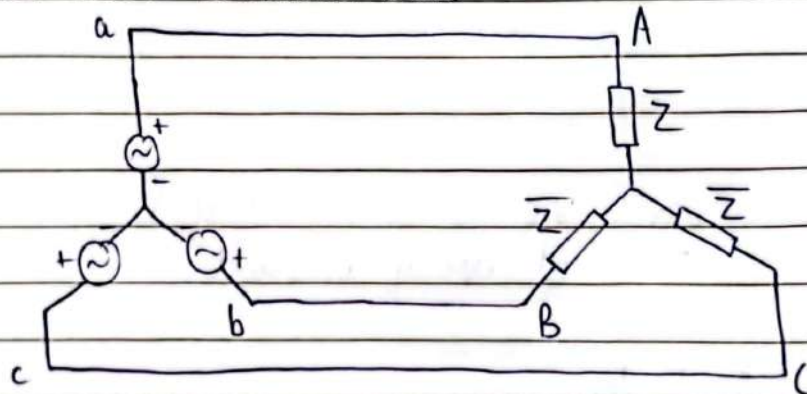
where $I_{\text{L}} = I_a = I_b = I_c$

$$\therefore P = 3 V_{\text{pn}} I_{\text{pn}} \cos \phi = \sqrt{3} V_{\text{LL}} I_{\text{L}} \cos \phi$$

where V_{LL} = line-to-line voltage.

* For star system, line current is same as phase current.

~~For Delta-connected Load~~

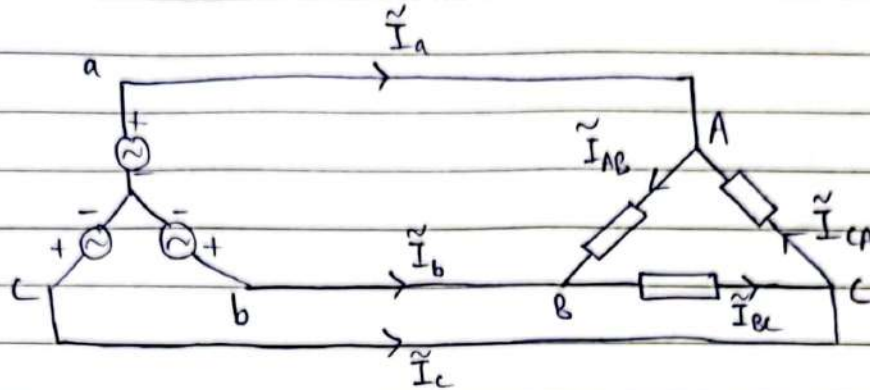


Star-connected load

Star-Delta Transformations

$$Z_{\Delta} = 3Z_{\Delta}$$

For delta-connected load



$$\tilde{I}_a = \tilde{I}_{AB} - \tilde{I}_{CA} = \frac{\tilde{V}_{ab}}{Z} - \frac{\tilde{V}_{ca}}{Z} = \frac{+ \sqrt{3} V_m}{Z} \left[\right]$$

$$\text{Let } \tilde{V}_a = V_a \angle 0^\circ$$

$$\therefore \tilde{I}_a = \frac{\sqrt{3} V_m}{Z} \left[1 \angle \pi/6 - 1 \angle 5\pi/6 \right]$$

Symmetrical components

Decompose unbalanced system into superposition of +ve, -ve and zero 3- ϕ sequence. (Due to linearity and superposition)

Derivation

$$\text{Define operator } a = 1 \angle 120^\circ = e^{j(2\pi/3)} = \frac{-1 + \sqrt{3}j}{2}$$

$$a^2 = 1 \angle -120^\circ = e^{-j(2\pi/3)} = \frac{-1 - \sqrt{3}j}{2}$$

$$a^3 = 1$$

a and a^2 are complex conjugate.

Suppose V_{a1}, V_{b1} and V_{c1} are positive sequence.
 V_{a2}, V_{b2} and V_{c2} are negative sequence.
 V_{a0}, V_{b0}, V_{c0} are zero sequence.

$$\therefore V_{b1} = a^2 V_{a1}$$

$$V_{c1} = a V_{a1}$$

$$V_{b2} = a V_{a2}$$

$$V_{c2} = a^2 V_{a2}$$

$$V_{a0} = V_{b0} = V_{c0}$$

Now, $V_a = V_{a1} + V_{a2} + V_{a0}$

$$V_b = V_{b1} + V_{b2} + V_{b0} = a^2 V_{a1} + a V_{a2} + V_{a0}$$

$$V_c = V_{c1} + V_{c2} + V_{c0} = a V_{a1} + a^2 V_{a2} + V_{a0}$$

$$\therefore \begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} V_{a0} \\ V_{a1} \\ V_{a2} \end{bmatrix}$$

Let A be the coefficient matrix.

$$A^{-1} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix}$$

$$\therefore \begin{bmatrix} V_{a0} \\ V_{a1} \\ V_{a2} \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix}$$

$$\therefore V_{a0} = \frac{1}{3} (V_a + V_b + V_c)$$

$$V_{a1} = \frac{1}{3} (V_a + a V_b + a^2 V_c)$$

$$V_{a2} = \frac{1}{3} (V_a + a^2 V_b + a V_c)$$

Zero sequence component can ~~have~~ be present in phase voltage like V_{an} , but not in line-to-line voltage V_{ab} , because it gets cancelled from $V_a - V_b = V_{ab}$.

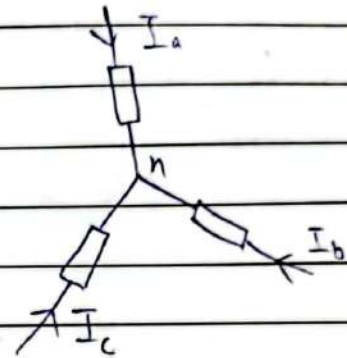
Similarly for current,

$$\begin{bmatrix} I_{a0} \\ I_{a1} \\ I_{a2} \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix}$$

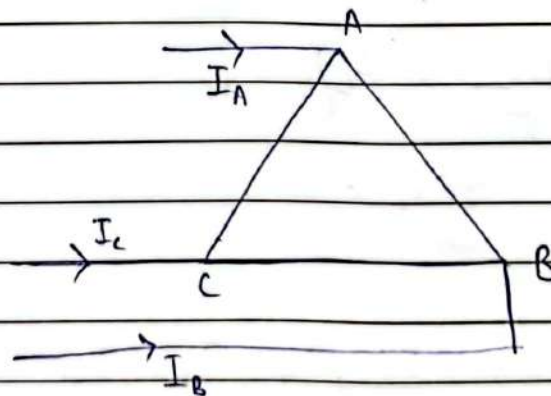
If neutral of load is not grounded,

$$I_a + I_b + I_c = 0$$

$$\therefore I_{a0} = 0$$



For Δ -connected load, zero sequence component of current flows only in Δ -loop. It doesn't flow in the lines.



$$I_A + I_B + I_C = 0$$

always (KCL)

Power with Symmetrical Components

$$S = P + jQ = \tilde{V}_a \tilde{I}_a^* + \tilde{V}_b \tilde{I}_b^* + \tilde{V}_c \tilde{I}_c^*$$

$$= [\tilde{V}_a \ \tilde{V}_b \ \tilde{V}_c] \begin{bmatrix} \tilde{I}_a^* \\ \tilde{I}_b^* \\ \tilde{I}_c^* \end{bmatrix} = (AV)^T (AI)^*$$

$$(AV)^T = V^T A^T$$

$$(AI)^* = A^* I^*$$

$$\text{where } V = [\tilde{V}_{a0} \ \tilde{V}_{a1} \ \tilde{V}_{a2}]^T$$

$$I = [\tilde{I}_{a0} \ \tilde{I}_{a1} \ \tilde{I}_{a2}]^T$$

$$\therefore S = V^T A^T A^* I^*$$

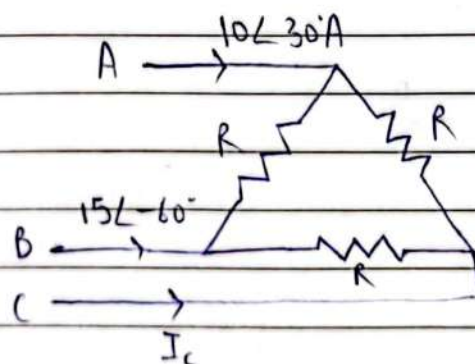
$$A^T = A$$

$$AA^* = 3I$$

$$\therefore S = 3 V^T I^* = 3 [\tilde{V}_{a0} + \tilde{V}_{a1} + \tilde{V}_{a2}] \begin{bmatrix} \tilde{I}_{a0}^* \\ \tilde{I}_{a1}^* \\ \tilde{I}_{a2}^* \end{bmatrix}$$

$$\therefore S = 3 (\tilde{V}_{a0} \tilde{I}_{a0}^* + \tilde{V}_{a1} \tilde{I}_{a1}^* + \tilde{V}_{a2} \tilde{I}_{a2}^*)$$

Example



Find $I_c, I_{a1}, I_{a2}, I_{a0}, I_{b1}, I_{b2}, I_{c1}, I_{c2}$

$$I_c = 18 \angle 154^\circ$$

$$I_{A1} = \frac{1}{3} [I_A + a I_B + a^2 I_C] = 14 \angle 42^\circ \text{ A}$$

$$I_{A2} = \frac{1}{3} [I_A + a^2 I_B + a I_C] = 4.65 \angle 248^\circ \text{ A}$$

$$I_{A0} = 0 \quad \left(= \frac{1}{3} [I_A + I_B + I_C] \right)$$

$$I_{B1} = a^2 I_{A1} = 14 \angle 282^\circ \text{ A}$$

$$I_{C1} = a I_{A1} = 14 \angle 162^\circ \text{ A}$$

$$I_{B2} = a I_{A2} = 4.65 \angle 8^\circ \text{ A}$$

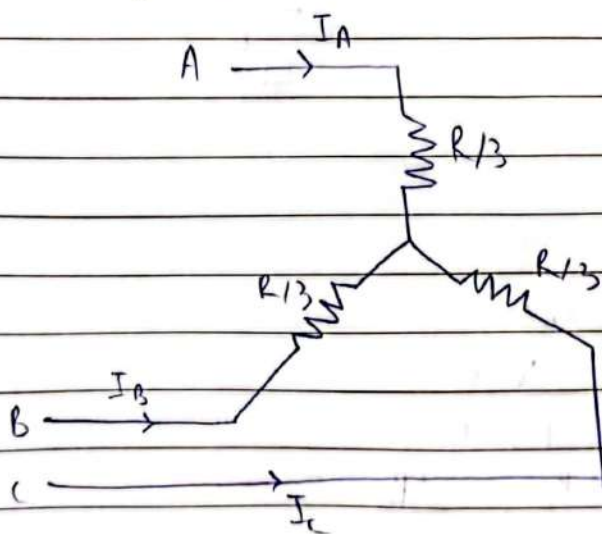
$$I_{C2} = a^2 I_{A2} = 4.65 \angle 128^\circ \text{ A}$$

$$I_{AB} = \frac{1}{3} (I_A - I_B)$$

$$I_{BC} = \frac{1}{3} (I_B - I_C)$$

$$I_{CA} = \frac{1}{3} (I_C - I_A)$$

For To arrive at this relation, consider
 equivalent Δ load. A load.



$$\begin{aligned} V_{AB} &= \frac{R I_A}{3} - \frac{R I_B}{3} \\ &= \frac{R}{3} (I_A - I_B) \end{aligned}$$

PTD

$$\text{Now, } I_{AB} = \frac{V_{AB}}{R} = \frac{1}{3} (I_A - I_B) \text{ and so on.}$$

Using these equations,

$$I_{AB} = 6 \angle 86^\circ \text{ A}$$

$$I_{BC} = 10.5 \angle -41.5^\circ \text{ A}$$

$$I_{CA} = 8.3 \angle 173^\circ \text{ A} \quad (\text{Unbalanced!!!})$$

$$I_{AB1} = 8 \angle 72^\circ \text{ A}$$

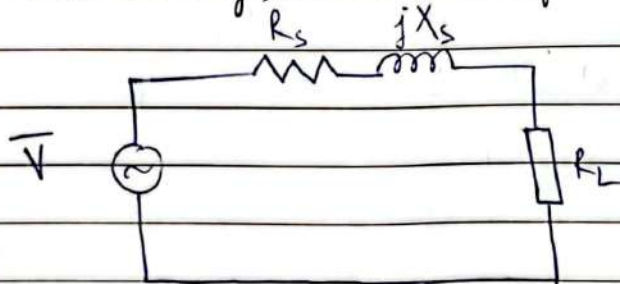
$$I_{AB2} = 2.7 \angle 218^\circ \text{ A}$$

$$I_{AB0} = 0 \text{ A}$$

Poynting vector is direction of flow of energy power
 $\vec{P} = \vec{E} \times \vec{H}$

Poynting vector has 3 terms - $I^2 R$ loss,
 Electrostatic ($\propto E^2$) &
 Magnetostatic ($\propto B^2$)

Maximum (Average) Power Transfer Theorem



$$\vec{I} = \frac{\vec{V}}{(R_s + R_L) + jX_s}$$

$$P_L = I^2 R_L = \frac{V^2 R_L}{(R_s + R_L)^2 + X_s^2}$$

For maximum P_L , $\frac{dP_L}{dR_L} = 0$

$$\Rightarrow R_L^2 = R_s^2 + X_s^2 \Rightarrow R_L = \sqrt{R_s^2 + X_s^2} = |Z_s|$$

If load is of the form $R_L + jX_L$

$$\bar{I} = \frac{\bar{V}}{(R_s + R_L) + j(X_s + X_L)}$$

$$P_L = \frac{V^2 R_L}{(R_s + R_L)^2 + (X_s + X_L)^2}$$

For maximum power transfer,

$$\frac{\partial P_L}{\partial X_L} = 0 \Rightarrow X_L = -X_s$$

$$\frac{\partial P_L}{\partial R_L} = 0 \Rightarrow R_L = R_s$$

$$\frac{\partial P_L}{\partial R_L} = 0 \Rightarrow R_L = R_s$$

$$\frac{\partial P_L}{\partial R_L} = 0 \Rightarrow R_L = R_s$$

$$\therefore Z_L = Z_s^*$$

Magnetic Fields and Circuits (and Materials)

Magnetostatics

Lorentz Force Equation

$$\vec{F} = q(\vec{v} \times \vec{B}) + q\vec{E}$$

B vs H

B is magnetic flux density. Its unit is Wb/m^2 .

It is flux per unit cross sectional area.

H can be thought of as the effect of source at a point. Its unit is A/m .

B is microscopic. H is macroscopic.

Biot-Savart Law

$$dH = \frac{1}{4\pi} \frac{I dl \sin \alpha}{r^2}$$

$$d\vec{H} = \frac{1}{4\pi} \frac{I d\vec{l} \times \hat{r}}{r^2}$$

Current can be of three types -

Line current (I), Surface current density (K),
Volume current density (J)

For the

($I d\vec{l}$) with unit Am (I is scalar)

($\vec{J} dV$) with unit Am ($\vec{J}$ is vector)

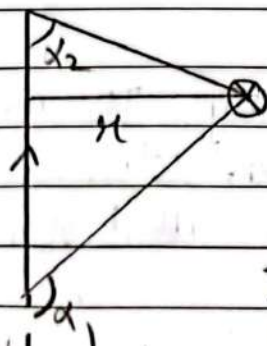
($\vec{K} dS$) with unit Am ($\vec{K}$ is vector)

Here unit of -

$$I = A$$

$$\vec{J} = A/m^2$$

$$\vec{K} = A/m \rightarrow \text{relevant in high-frequency electromagnetics}$$



$$\vec{H} = \frac{I}{4\pi r} (\cos \alpha_2 - \cos \alpha_1) \hat{a}_\phi$$

In the limit as the conductor becomes infinite.

$$H = \frac{I}{2\pi r}$$

Ampere's Circuital Law

$$\oint \vec{H} \cdot d\vec{l} = I \quad \text{--- (1)}$$

The corresponding Maxwell's equation is $\nabla \times \vec{H} = \vec{J}$ --- (2)

These equations are equivalent.

Equation (2) is in point form, i.e., applicable at a single point in space.

Maxwell's 4 equations were originally in point form.

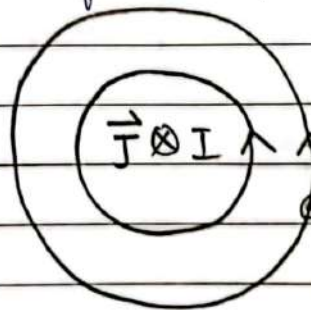
Taking surface integral of (2),

$$\iint_S (\nabla \times \vec{H}) \cdot d\vec{s} = \iint_S \vec{J} \cdot d\vec{s}$$

By Stokes Theorem,

$$\oint \vec{H} \cdot d\vec{l} = I$$

We get ~~to~~ back equation (1), which is the integral form of the equation.



At this point, $\vec{H} \neq 0$,
 But $\nabla \times \vec{H} = 0$ because
 $\vec{J} = 0$ at this point

$\vec{H}$ in magnetics is analogous to $\vec{E}$ as in electrostatics.

$$\vec{D} = \epsilon \cdot \vec{E}$$

$$\vec{B} = \mu_0 \vec{H}$$

$$\nabla \cdot \vec{B} = 0$$

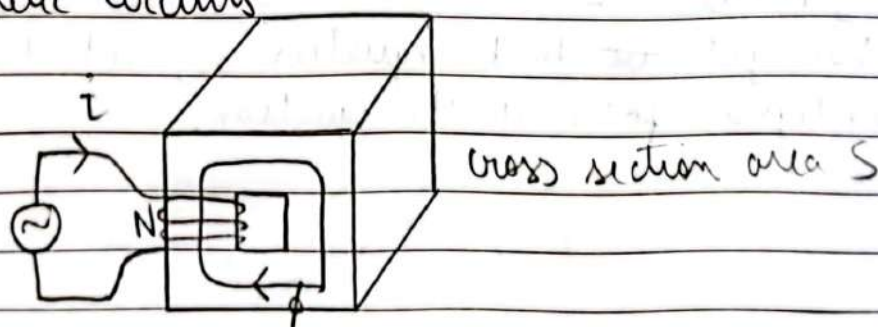
$$\nabla \cdot \vec{D} = \rho_v \rightarrow \text{Free volume charge density}$$

$$\oint \vec{B} \cdot d\vec{S} = 0 \quad (\text{Closed Surface Integral})$$

$$\oint \vec{B} \cdot d\vec{S} = \phi \quad (\text{Flux})$$

$\nabla \cdot \vec{B} = 0$ and $\oint \vec{B} \cdot d\vec{S} = 0$ equations are equivalent. (Using Gauss Divergence Theorem)
They tell us that magnetic monopoles do not exist.

Magnetic Circuits



$$\oint \vec{H} \cdot d\vec{l} = Ni$$

Assuming $\vec{H} \parallel d\vec{l}$ everywhere and H constant

$$Hl = Ni$$

$$\phi = BS$$

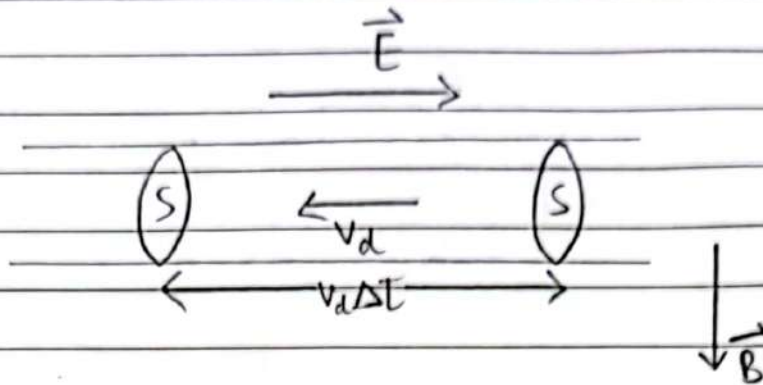
$$B = \mu H \quad (\mu = \mu_0 \mu_r)$$

Copper & Aluminium have very low μ_r (almost the same 1, same as air)

Unit of μ_0 is H/m

Unit of ϵ_0 is F/m

μ_r is relative permeability.



$$\text{Volume} = S v_d \Delta t$$

n = number density of free electrons

$$\therefore \Delta Q = (n S v_d \Delta t) e$$

$$I = \frac{\Delta Q}{\Delta t} = n e S v_d$$

$$J = n e v_d$$

$$v_d = \frac{e E \tau}{m}$$

$$\therefore J = \left(\frac{n e^2 \tau}{m} \right) E$$

$$\therefore J = \sigma E$$

This is the point form of Ohm's Law.

$V = IR$ is the integral form of $J = \sigma E$.

Each free electron experiences magnetic force

$$\vec{F} = -e \vec{v}_d \times \vec{B}$$

Force on length $d\vec{l}$ of conductor,

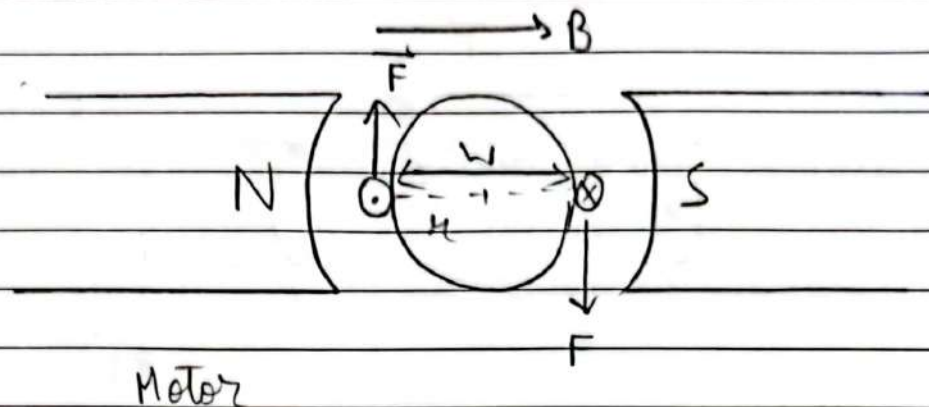
$$d\vec{F} = (n S d\vec{l}) (-e \vec{v}_d \times \vec{B})$$

$$d\vec{F} = n S e v_d d\vec{l} \times \vec{B}$$

$$d\vec{F} = I d\vec{l} \times \vec{B}$$

$$\therefore d\vec{F} = I d\vec{l} \times \vec{B}$$

The term $I d\vec{l}$ is called current element.



$$F = B I l \sin \alpha = B I l \quad (\alpha = 90^\circ)$$

$$T = |\vec{r} \times \vec{F}| = 2 F \frac{w}{2} = B I l w$$

$$\therefore T = B I l w \sin \alpha$$

$$\therefore T = B I S \sin \alpha$$

S is the conductor loop area.

Magnetic Dipole Moment

$$\vec{m} = I S \hat{a}_n$$

Magnetization

$\vec{M} =$ Magnetic Dipole Moment per unit volume.

$$\vec{M} = \frac{\vec{m}}{V}$$

Unit of $\vec{m}$ is $A m^2$

Unit of $\vec{M}$ is $\frac{A}{m}$ (Same as $\vec{H}$)

H is external magnetic field. M is internal magnetic field that aids H .
 (inside the conductor)

Magnetic Materials

2 Types - Soft and Hard Magnetic Materials
↳ (Permanent Magnets)

Soft Magnetic Materials -

Diamagnetic $\Rightarrow \mu_r \approx 1.0$ $\mu_r \lesssim 1.0$

Paramagnetic $\Rightarrow \mu_r \gtrsim 1.0$

Ferromagnetic $\Rightarrow \mu_r \gg 1$
(Roughly in range 1000 to 50000)

Three types of magnetic sources in materials

- Electron orbital motion
- Electron spin (stronger than orbital)
- Nuclear spin (insignificant)

Diamagnetic Materials (E.g. Cu)

All electrons antiparallely paired (due to Pauli's Exclusion Principle). Hence their spin cancels.

However, Lenz law causes eddy currents, due to which internal magnetic field ~~attract~~ opposes applied field.

Paramagnetic Materials (E.g. Al)

Unpaired e^- (s).

Magnetic energy tries to align the individual dipoles to ~~field~~ applied field.

Thermal energy tries to randomize dipole orientation.

Thermal effect dominates at room temperature.

$$\vec{B} = \mu_0 (\vec{H} + \vec{M})$$

$$\text{Also, } \vec{M} \propto \vec{H} \Rightarrow \vec{M} = \chi_m \vec{H}$$

$$\therefore \vec{B} = \mu_0 (1 + \chi_m) \vec{H}$$

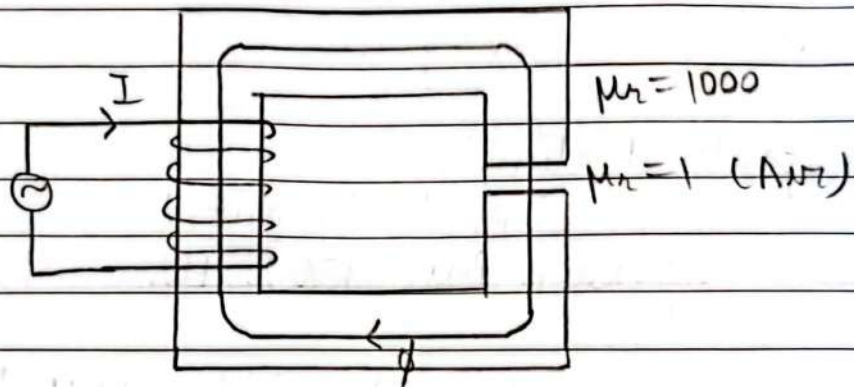
$$\vec{B} = \mu_0 \mu_r \vec{H}$$

$$\vec{B} = \mu \vec{H}$$

Ferromagnetic Materials (E.g. ~~Fe~~ Fe)

Thermal Energy only starts to dominate at very high temperature.

Otherwise magnetic energy dominates.



$$\phi = BS$$

$$= \mu HS$$

$$= \mu \frac{NI S}{l}$$

$$\phi = \mu \frac{\text{MMF} S}{l}$$

$$\text{Magnetomotive Force (MMF)} = NI$$

$$= \phi \cdot \left(\frac{l}{\mu S} \right)$$

$$= \phi R$$

$$\text{MMF} = \phi R$$

Here R is magnetic reluctance.

$$R = \frac{l}{\mu S}$$

Analogy b/w Electric and Magnetic domains

Electric	Magnetic
σ (unit C/m)	μ (unit H/m)
$\vec{E}$	$\vec{H}$
$I = \int \vec{J} \cdot d\vec{s}$	$\phi = \int \vec{B} \cdot d\vec{s}$
$J = I/s = \sigma E$	$B = \phi/s = \mu H$
$R = 1/(\sigma s)$	$R = 1/(\mu s)$
(KCL) $\sum I = 0$ at a node	$\sum \phi = 0$ at a node
$\sum V - \sum RI = 0$	$\sum MMF - \sum R\phi = 0$

Energy stored in air gaps in magnetic circuit
vs electric circuit (capacitance)

$$\text{Energy in electric circuit} = \frac{1}{2} CV^2$$

$$= \frac{1}{2} \int \epsilon E^2 dV$$

$$\text{Energy in magnetic circuit} = \frac{1}{2} \int \mu H^2 dV$$

$$= \frac{1}{2} \int \frac{B^2}{\mu} dV$$

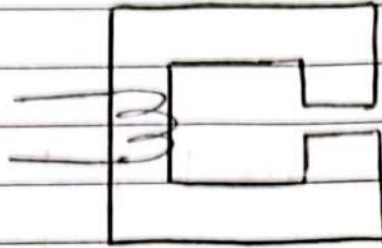
10 Marks — MCQ (10 Questions)
— 30 Mins —> Syllabus upto Tuesday

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Example with Practical values



Maximum Permissible $B = 2.0 \text{ T}$

Maxwell's Equations

$$\nabla \cdot \vec{D} = \rho_v$$

$$\nabla \cdot \vec{B} = 0$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$$

We cannot always apply analytical techniques to solve systems because of their complexity.
 $\therefore$ Need to resort to numerical techniques.

For $\frac{\partial \vec{D}}{\partial t} = 0$

$\nabla \times \vec{H} = \vec{J}$

In integral form, $\oint \vec{H} \cdot d\vec{l} = I$

Nabla operator, $\nabla = \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z}$

Its unit is m^{-1}

$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$

$\vec{D} = \epsilon_0 \vec{E}$, unit of D is C/m^2
 $\vec{E} = -\nabla V$

Units of

$D = C m^{-2}$

$H = A m^{-1}$

$B = Wb m^{-2}$

$J = A m^{-2}$

$\vec{D} = \epsilon_0 \vec{E}$

$\vec{B} = \mu \vec{H}$

$\vec{J} = \sigma \vec{E}$

Continuity Equations

$\nabla \cdot \vec{J} = -\frac{\partial \rho_v}{\partial t}$

Lorentz - Force Equation

$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$

Poisson's Equation in Electrostatics

$$\nabla \cdot \vec{D} = \rho_v$$

$$\nabla \cdot (\epsilon \vec{E}) = \rho_v$$

$$\epsilon \nabla \cdot (-\nabla V) = \rho_v$$

$$\therefore \nabla^2 V = -\frac{\rho_v}{\epsilon}$$

This is called Poisson's equation.

∇^2 is the Laplacian operator.

In case $\rho_v = 0$,

$$\nabla^2 V = 0$$

This is called Laplace's Equation.

Magnetic Vector Potential ($\vec{A}$)

Maxwell called it Electromagnetic Momentum Vector

$$\vec{B} = \nabla \times \vec{A}$$

For a charge moving, its momentum is given by

$$p = mu + eA$$

Consider

$$\nabla \cdot \vec{B} = 0$$

Apply volume integral,

$$\iiint \nabla \cdot \vec{B} dV = 0$$

By divergence theorem,

$$\oiint \vec{B} \cdot d\vec{S} = 0$$

$$\phi = \iint \vec{B} \cdot d\vec{S} \quad (\text{open surface})$$

$$\therefore \phi = \iint (\nabla \times \vec{A}) \cdot d\vec{S}$$

By Stokes' Theorem,

$$\phi = \oint \vec{A} \cdot d\vec{l} \quad (\text{closed line integral})$$

According to Biot-Savart Law,

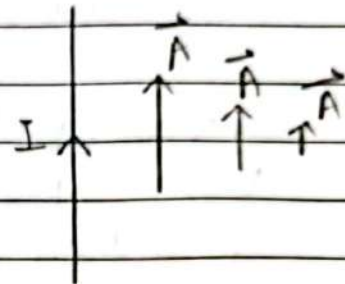
$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{I d\vec{l} \times \vec{r}}{r^2}$$

We get using Biot-Savart Law,

$$\vec{A} = \frac{\mu_0}{4\pi} \int \frac{I d\vec{l}}{R}$$

Analogous to

$$V = \frac{1}{4\pi\epsilon_0} \int \frac{q dV}{R}$$

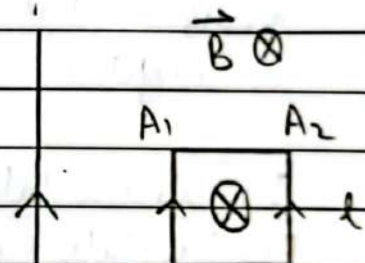


Let us analyze using $\phi = \oint \vec{A} \cdot d\vec{l}$

Flux through the rectangle,

$$\phi = \oint \vec{A} \cdot d\vec{l}$$

$$\phi = (A_1 - A_2)l$$



Here, we assumed that the current is infinitely extending.
 (assumed 2-D)

If we account for bending-off the wire after finite length, the problem turns 3-D $\Rightarrow$ Very Difficult.

Hence we try to make 2-D assumption wherever possible.

Poisson's Equation for Magnetostatics

It can be shown that

$$\nabla^2 A = -\mu J$$

Finite Element Method

- Divide entire domain into small elements.
(Continuous $\rightarrow$ Discrete)
- Convert Equation

$$\nabla^2 A = -\mu J$$
 to matrix form

$$[K]\{A\} = \{b\}$$
- To calculate field distribution, we have to assume that the least-energy distribution is adopted.

In electrostatic case,

$$\text{Electrostatic energy, } E = \int \frac{1}{2} \epsilon E^2 dV - \int \rho_v V dV$$

$$\therefore E = \int \frac{1}{2} \epsilon |\nabla V|^2 dV - \int \rho_v V dV$$

The second term represents energy expended in bringing free charges into the configuration

In 2-D,

If we take linear approximation,

$$A = a + bx + cy$$

Since it has 3 unknowns (a, b and c), each element must have 3 nodes.



3 nodes
(triangular)

Entire geometry is broken down into large number of finite elements (triangular in this case).

The values a, b and c are different for different elements.

Calculate energy of each elements in terms of its potentials (A) at 3 nodes.
Sum them to get total energy.

To minimise total energy, differentiate w.r.t all 16 nodal potentials, and equate to 0.
 $\Rightarrow$ 16 equations.

Homework-1

1. (a) Rectangular Form = $0 + j$
Polar Form = $1 \angle 90^\circ$

(b) Rectangular Form = ~~$-7.58 - 4.71j$~~ $-2.95 - 5.43j$
Polar Form = ~~$8.92 \angle -2.58^\circ$~~ $6.183 \angle -118.52^\circ$

(c) Rectangular Form = $6.83 - 5.21j$
Polar Form = $8.59 \angle -37.36^\circ$

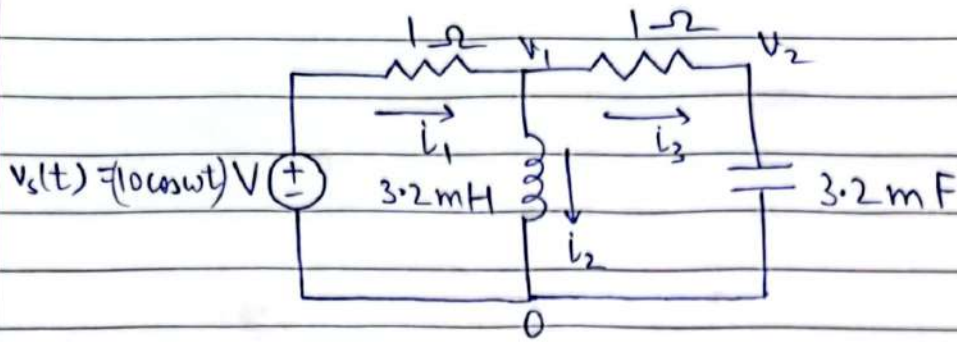
(d) Rectangular Form = $1.38 - 0.15j$
Polar Form = $1.386 \angle -6.31^\circ$

2. (a) $-10 \cos(\omega t + \pi/3) - 10 \cos(\omega t - \pi/6 + \pi/2)$
= $-20 \cos(\omega t + \pi/3)$
= $20 \cos(\omega t - 2\pi/3)$
Phasor Form = $20 \angle -120^\circ$

(b) $3 \cos(\omega t + \pi/3) - 5 \sin(\omega t)$
= $3 \cos(\omega t + \pi/3) - 5 \cos(\omega t - \pi/2)$
= $3 \cos(\omega t + \pi/3) + 5 \cos(\omega t + \pi/2)$
Phasor Form = $3 \angle 60^\circ + 5 \angle 90^\circ$
= $7.74 \angle 78.83^\circ$

(c) $5 \cos(\omega t + \pi/2) + 10 \cos(\omega t - \pi/6 - \pi/2)$
= $5 \cos(\omega t + \pi/2) + 10 \cos(\omega t - 2\pi/3)$
Phasor Form = $5 \angle 90^\circ + 10 \angle -120^\circ$
= $6.19 \angle -143.79^\circ$

3. (a)



$$\omega = 2\pi \times 100 \text{ rad s}^{-1}$$

$$\tilde{V}_s = 10 \angle 0^\circ$$

$$\bar{X}_C = \frac{-j}{\omega C} = -0.497j \Omega$$

$$\bar{Z}_{RC} = (1 - 0.497j) \Omega$$

$$\bar{X}_L = j\omega L = 2.01j \Omega$$

$$\begin{aligned} \bar{Z}_{nt} &= 1 + \frac{(2.01j)(1 - 0.497j)}{2.01j + 1 - 0.497j} \\ &= (2.23 + 0.15j) \Omega \end{aligned}$$

$$\therefore \tilde{I}_1 = \frac{\tilde{V}_s}{\bar{Z}_{nt}} = \frac{10 \angle 0^\circ}{2.23 + 0.15j} = (4.467 - 0.304j) \text{ A}$$

Ans: $\tilde{I}_1 = 4.477 \angle -3.891^\circ \text{ A}$

By current divider formula

$$\tilde{I}_2 = \frac{(1 - 0.497j)}{2.01j + 1 - 0.497j} \times (4.467 - 0.304j)$$

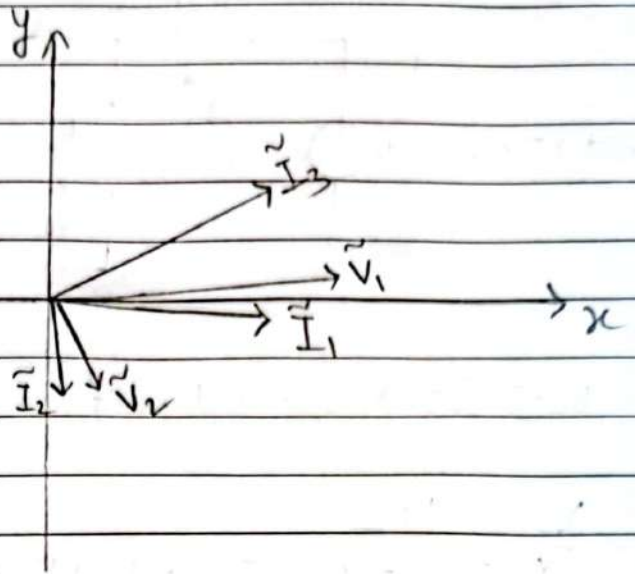
Ans: $\tilde{I}_2 = 2.756 \angle -86.856^\circ \text{ A}$

Ans: $\tilde{I}_3 = \tilde{I}_1 - \tilde{I}_2 = 4.962 \angle 29.573^\circ \text{ A}$

$$\begin{aligned} \tilde{V}_1 &= \tilde{I}_2 \bar{X}_L = (2.756 \angle -86.856^\circ) \times (2.01j) \\ &= 5.531 \angle 3.144^\circ \text{ V} \end{aligned}$$

Ans: $\tilde{V}_2 = \tilde{I}_3 \bar{X}_C = 2.466 \angle -60.427^\circ \text{ V}$

Ans: $\tilde{V}_2 = \tilde{I}_3 \bar{X}_C = 2.466 \angle -60.427^\circ \text{ V}$



(b) $v_1(t) = 5.54 \cos(\omega t + 3.144^\circ) \text{ V}$
 $v_2(t) = 2.466 \cos(\omega t - 60.427^\circ) \text{ V}$
 $i_1(t) = 4.477 \cos(\omega t - 3.891^\circ) \text{ A}$
 $i_2(t) = 2.756 \cos(\omega t - 86.856^\circ) \text{ A}$
 $i_3(t) = 4.962 \cos(\omega t + 29.573^\circ) \text{ A}$

(c) $\omega = 2\pi \times 200 \text{ rad s}^{-1}$
 $\overline{X_C} = \frac{-j}{\omega C} = -0.249j \Omega$

$\overline{X_L} = j\omega L = 4.02j \Omega$

$\overline{Z_{RC}} = (1 - 0.249j) \Omega$

$\overline{Z_{int}} = 1 + \frac{(4.02j)(1 - 0.249j)}{4.02j + 1 - 0.249j}$
 $= \frac{1}{(2.062 + 0.016j)} \Omega$

$\tilde{I}_1 = \frac{10}{2.062 + 0.016j} = (4.849 - 0.038j) \text{ A}$

$\therefore \tilde{I}_1 = 4.85 \angle -0.448^\circ \text{ A}$

$$\tilde{I}_2 = \frac{(1 - 0.249j)}{4.02j + 1 - 0.249j} \times 4.85 \angle -0.448^\circ$$

$$\therefore \tilde{I}_2 = 1.281 \angle -89.58^\circ \text{ A}$$

$$\tilde{I}_3 = 4.84 \angle 14.40^\circ \text{ A}$$

$$\tilde{V}_1 = \tilde{I}_1 \bar{X}_L = 19.497 \angle 89.552^\circ \text{ V}$$

$$\tilde{V}_1 = \tilde{I}_2 \bar{X}_L = 5.149 \angle 0.42^\circ \text{ V}$$

$$\tilde{V}_2 = \tilde{I}_3 \bar{X}_L = 1.243 \angle -75.6^\circ \text{ V}$$

$$4. (a) \bar{X}_{C1} = \frac{-j}{\omega C} = -20j \Omega, \bar{X}_{C2} = -2j \Omega$$

$$\bar{X}_L = +j\omega L = 0.5j \Omega$$

$$\therefore \bar{Z}_1 = \frac{10(-20j)}{10 - 20j} = \frac{-20j}{1 - 2j} = (8 - 4j) \Omega$$

$$\bar{Z}_2 = -2j + 0.5j + 8 - 4j = 8 - 5.5j$$

$$\bar{Z}_{eq} = \frac{10(8 - 5.5j)}{18 - 5.5j} = 5.158 \angle -17.518^\circ \Omega$$

$$(b) \bar{Z}_1 = \frac{10(-10j)}{10 - 10j} = \frac{-10j}{1 - j} = (5 - 5j) \Omega$$

$$\bar{Z}_2 = -j + j + 5 - 5j = (5 - 5j) \Omega$$

$$\bar{Z}_{eq} = \frac{10(5)(1 - j)}{15 - 5j} = \frac{10(1 - j)}{3 - j} = 4.472 \angle -26.56^\circ \Omega$$

$$5 \quad \tilde{I}_{z1} = \frac{\tilde{V}_{s1}}{\bar{Z}} = 2.776 \angle -0.018^\circ \text{ A}$$

$$\tilde{I}_{z2} = 2.776 \angle -60.018^\circ \text{ A}$$

$$\tilde{I}_{z3} = 4.808 \angle -30.018^\circ \text{ A}$$

$$\therefore \tilde{I}_1 = 7.344 \angle -19.12^\circ \text{ A}, \tilde{I}_2 = 7.344 \angle 139.088^\circ \text{ A}$$

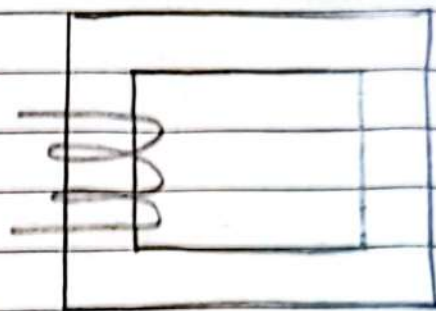
Magnetic Circuits

Inductance (L)

$$\lambda = \text{Flux Linkage} = N\phi$$

$$\lambda \propto i$$

$$N\phi = \lambda = Li$$



$$\phi = \frac{\mu HS}{l} = \frac{\mu NiS}{l} = \frac{F}{R}$$

$$\therefore L = \frac{N}{i} \left(\frac{\mu NiS}{l} \right) = \frac{\mu N^2 S}{l} = \frac{N^2}{R}$$

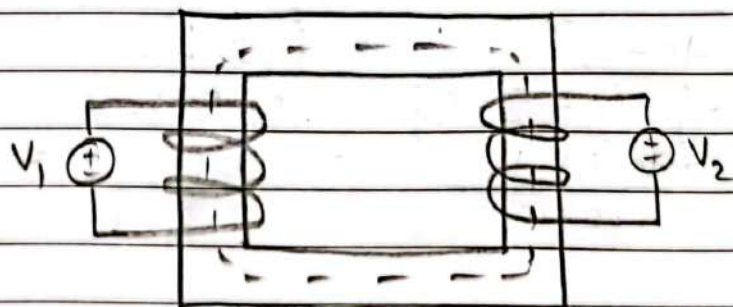
Faraday's Law

$$e = -N \frac{d\phi}{dt} = -L \frac{di}{dt}$$

-ve sign is necessary for field-view point.

From ~~circuit~~ circuit viewpoint,

$$e = L \frac{di}{dt}, \text{ which opposes source voltage.}$$



$$\lambda_1 = N_1 \phi_1$$

$$\lambda_2 = N_2 \phi_2$$

$$\phi_1 = \phi_2$$

$$e_1 = \frac{d\lambda_1}{dt} = L_1 \frac{di_1}{dt}$$

$$\text{Also, } \lambda_{12} =$$

A source is strong if it can maintain a constant terminal voltage irrespective of current through it. (Source Resistance = 0)

Also, $\lambda_1 = L_{11} i_1 + L_{12} i_2$

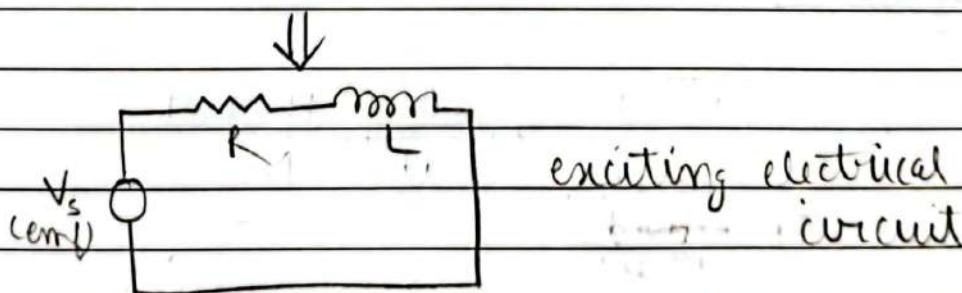
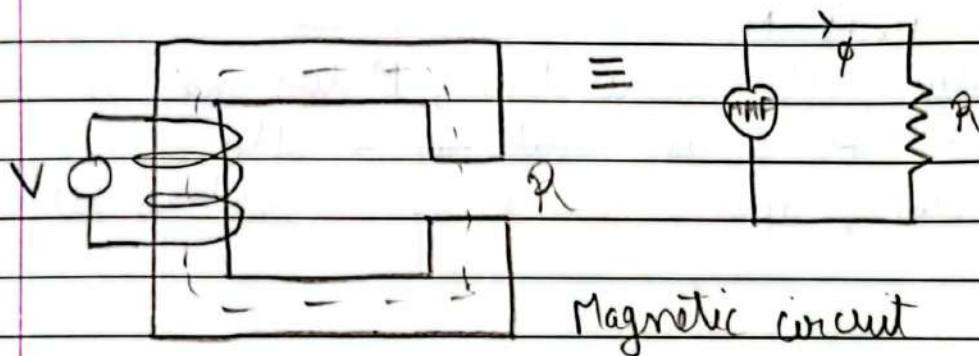
$$\lambda_1 = L_{11} i_1 + L_{12} i_2$$

$$\therefore e_1 = \frac{d\lambda_1}{dt} = L_{11} \frac{di_1}{dt} + L_{12} \frac{di_2}{dt}$$

We assumed L_1 & L_2 is constant.

It may not be the case if μ is not constant.
 $\therefore B$ vs H is non-linear.

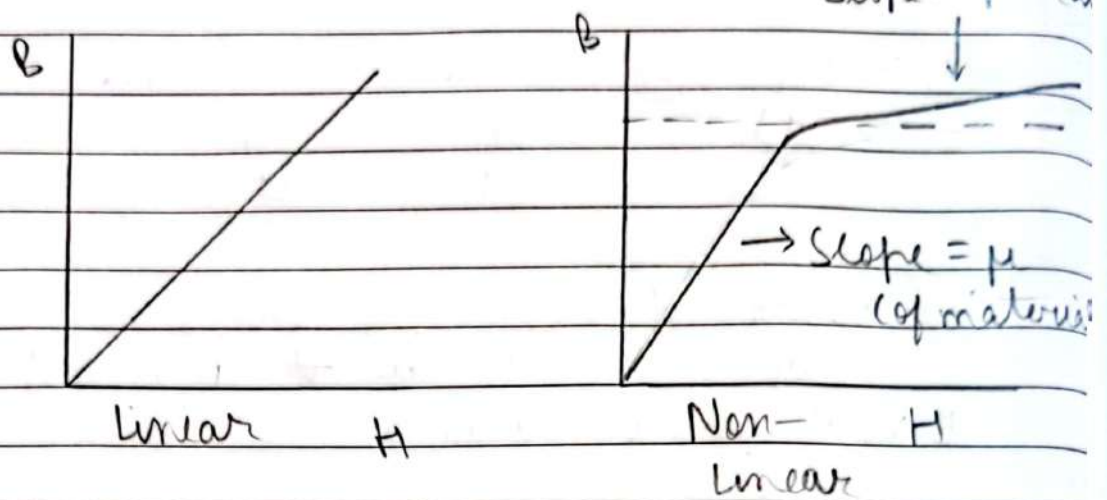
Similarly, $e_2 = L_{22} \frac{di_2}{dt} + L_{21} \frac{di_1}{dt}$



If airgap is increased, $R \uparrow \Rightarrow \phi \downarrow \Rightarrow L \downarrow \Rightarrow X_L \Rightarrow I \uparrow \Rightarrow \phi \uparrow$

Intuitively, we know that ϕ must be constant because it only depends on N, f and V_s .

B-H characteristics



When magnetic material saturates, all internal dipoles are aligned.

$$B = \mu_0(H + M)$$

At saturation After saturation, M does not increase further. Hence the material after saturation, material becomes as good as air.

Hysteresis

$$\begin{aligned} \text{Energy} &= \int v_i dt = \int \left(N \frac{d\phi}{dt} \right) \left(\frac{H l}{N} \right) dt \\ &= \int H l d\phi \end{aligned}$$

$$\phi = BS \Rightarrow d\phi = S dB$$

$$\begin{aligned} \therefore \text{Energy} &= \int H B H l S dB \\ &= \int H V dB \end{aligned}$$

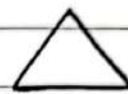
Finite Element Method (Contd.)

$$F = \text{Energy} = \frac{1}{2} \int_s \frac{1}{\mu} |\nabla A_z|^2 ds - \int_s J_z A_z ds$$

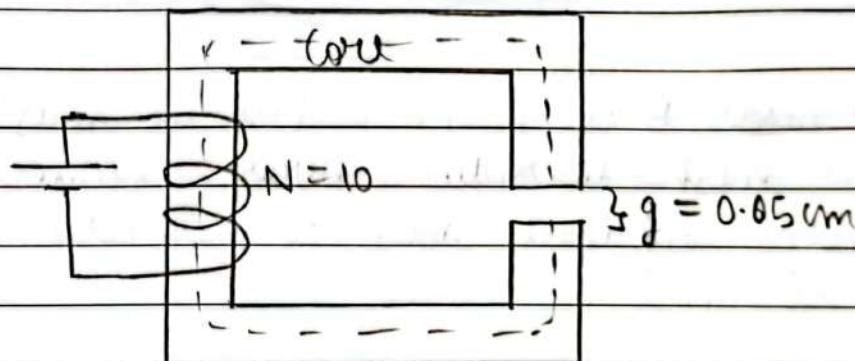
$$F = \text{Energy} = \frac{1}{2} \int_s \frac{1}{\mu} |\nabla A_z|^2 ds - \int_s J_z A_z ds$$

2D FEM: Linear (1st order elements)

$$A_z = a + bx + cy$$



Example Problem using FEM



$$A_c = A_g = 9 \text{ cm}^2$$

$$l_c = 30 \text{ cm}$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$$

$$\mu_r = 70000 \text{ (core)}$$

$$R_c = \frac{l_c}{\mu_r \mu_0 A_c} = \frac{0.3}{70000 \times 4\pi \times 10^{-7} \times 9 \times 10^{-4}} = 3.79 \times 10^3 \frac{\text{A turns}}{\text{Wb}}$$

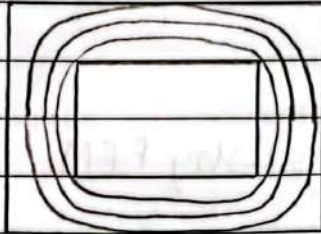
$$(\mu_c = \mu_r \mu_0)$$

$$R_g = \frac{l_g}{\mu_0 A_g} = 4.43 \times 10^5 \frac{\text{A turns}}{\text{Wb}}$$

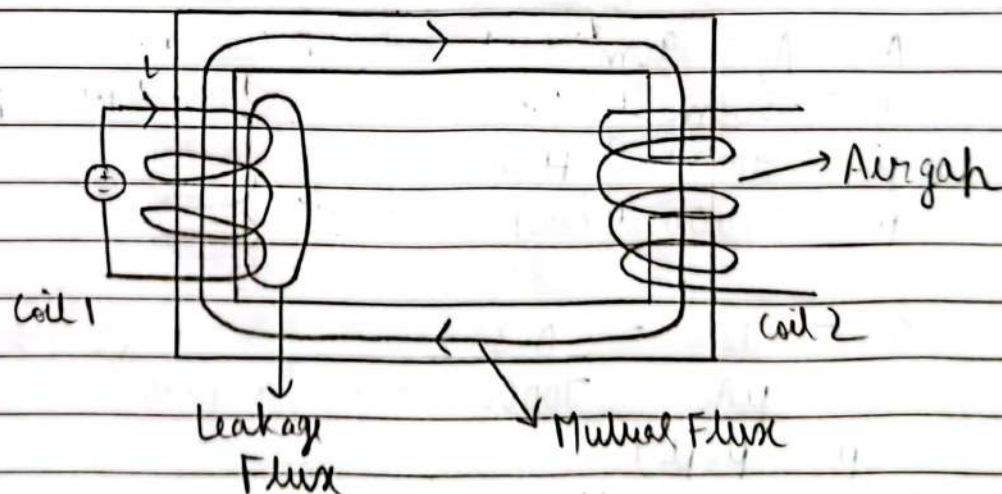
P.T.O.

If $B = 1 \text{ T}$, $\phi = B \times A = 1 \times 9 \times 10^{-4} = 9 \times 10^{-4} \text{ Wb}$
 $\therefore \text{MMF} = \phi(R_c + R_g) = Ni$
 $\therefore i = \frac{\phi(R_c + R_g)}{N} = 40 \text{ A}$

In this computation, we neglected fringing of flux density in the air gap, and also neglected leakage flux due to MMF coils.



Wherever B is stronger, there we must have finer mesh to reduce error, because these regions contribute more to ~~the~~ total energy.
 $(E \propto B^2)$



Leakage flux $\propto$ Reluctance of path of flux
 $\propto$ Airgap

Inductance of coil, $L_1 = \frac{N \phi_{\text{total}}}{i}$

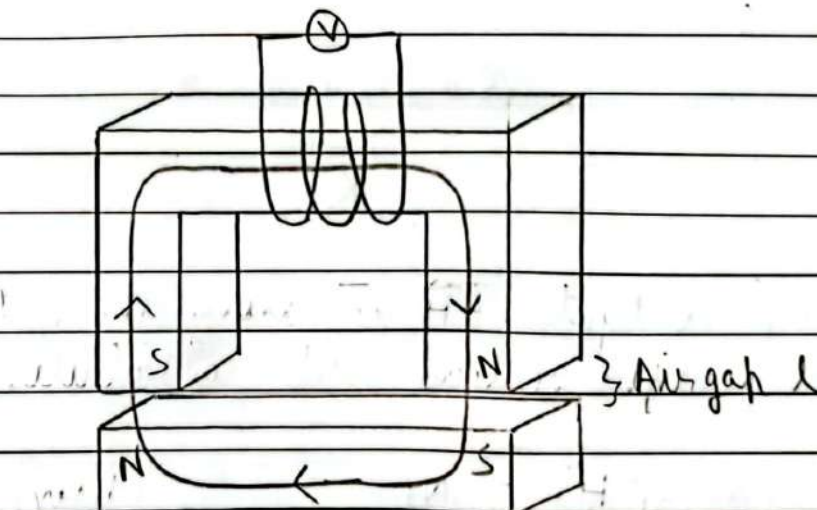
$$\phi_{\text{total}} = \phi_{\text{mutual}} + \phi_{\text{leakage}}$$

$$\therefore L_1 = \frac{N \phi_{\text{mutual}}}{i} + \frac{N \phi_{\text{leakage}}}{i}$$

$$L_1 = \frac{N^2}{R_{\text{mutual}}} + \frac{N^2}{R_{\text{leakage}}}$$

$$L_1 = L_{\text{mutual}} + L_{\text{leakage}}$$

Here R_{mutual} and R_{leakage} are in parallel.
 However, L_{mutual} and L_{leakage} are in series.



Let the lower piece move up by $(-dl)$
 $(dl < 0)$

$$\text{Work done} = -F dl = dW_m$$

$$\text{change in magnetostatic energy} = 2 \left[\frac{B^2}{2\mu_0} S dl \right]$$

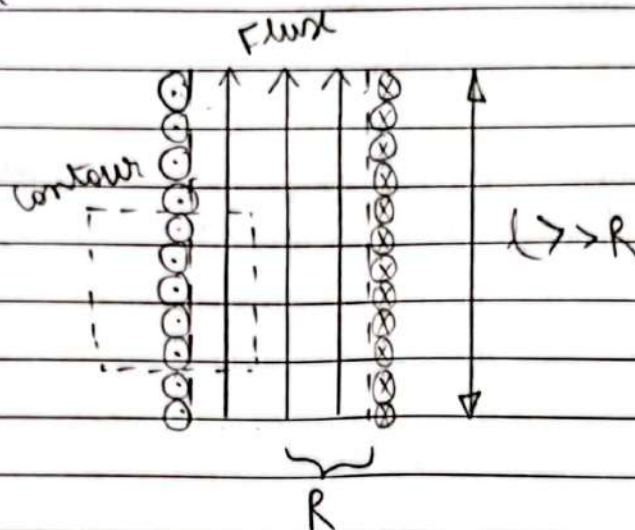
Neglecting change in B due to change in R

$$\therefore F dl = \frac{B^2 S}{\mu_0} dl$$

$$\therefore F = \frac{B^2 S}{\mu_0}$$

F_{avg}	$= \frac{B^2 S}{2\mu_0}$
Pressure	$= \frac{B^2}{2\mu_0}$

Solmoid



Let's evaluate $\oint \vec{H} \cdot d\vec{l}$ along the contour
only right vertical side contributes

$$\oint \vec{H} \cdot d\vec{l} = Hl = Ni \quad (\text{Using Maxwell's H-i relation})$$

$$H = \frac{Ni}{l}$$

$$B = \frac{\mu_0 Ni}{l}$$

Let $n = N/l$

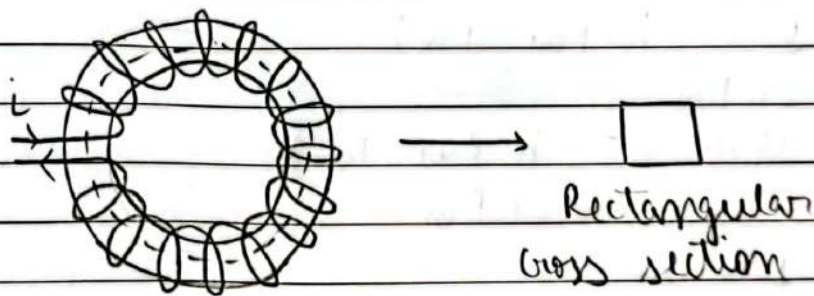
$$\begin{aligned} \therefore H &= ni \\ B &= \mu_0 ni \\ \phi &= \mu_0 ni \pi R^2 \\ L &= \frac{N\phi}{i} = \frac{n l \mu_0 n i \pi R^2}{i} = \frac{\mu_0 N^2 \pi R^2}{l} \end{aligned}$$

For a finite solenoid, let us place origin at center of solenoid, and z axis along axis of solenoid, then

$$B_z = \left(\frac{\mu_0 Ni}{l} \right) \cdot \frac{1}{2} \left[\frac{(l/2 - z)}{\sqrt{(l/2 - z)^2 + R^2}} + \frac{(l/2 + z)}{\sqrt{(l/2 + z)^2 + R^2}} \right]$$

This reduces to $B = \frac{\mu_0 Ni}{l}$ for $l/2 \gg z$

Toroid



r_{in} = Inner radius

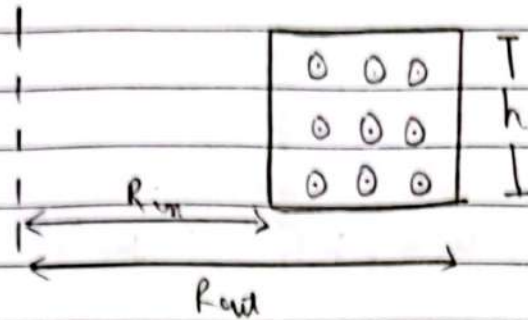
r_{out} = Outer radius

$$\oint \vec{H} \cdot d\vec{l} = H(2\pi r) = Ni$$

$$H = \frac{Ni}{2\pi r}$$

(Flux)

Let us evaluate $\iint \vec{B} \cdot d\vec{s}$ over cross section



$$\iint \vec{B} \cdot d\vec{s} = \int_{R_{in}}^{R_{out}} \left(\frac{\mu_0 N i}{2\pi r} \right) h dr$$

$$\phi = \frac{\mu N i h}{2\pi} \ln \frac{R_{out}}{R_{in}}$$

$$L = \frac{N\phi}{i}$$

$$L = \frac{\mu N^2 h}{2\pi} \ln \frac{R_{out}}{R_{in}}$$

In case $R_{in} \gg R_{out} - R_{in}$

$$\ln \left(1 + \frac{R_{out} - R_{in}}{R_{in}} \right) \approx \frac{R_{out} - R_{in}}{R_{in}}$$

$$\therefore L = \frac{\mu N^2 h (R_{out} - R_{in})}{2\pi R_{in}}$$

Note that $S = h(R_{out} - R_{in})$

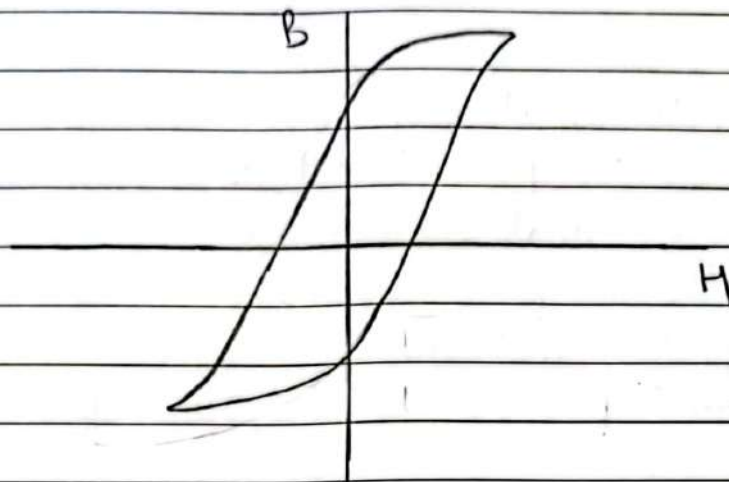
$$l = 2\pi R_{in}$$

$$L = \frac{\mu N^2 S}{l}$$

B-H curve and Hysteresis

~~B~~ $\propto V$ and $H \propto i$ in AC machines
 Hence B-H curve has same shape as V-i curve.

B-H curve is experimentally measured by measuring V-i



$$\begin{aligned}
 \text{Energy} &= \int V i \, dt \\
 &= \int \frac{N d\phi}{dt} \cdot \frac{H l}{N} \, dt \\
 &= \int H l \, d\phi \\
 &= \int H l \, S \, dB
 \end{aligned}$$

$$\begin{aligned}
 \text{Energy at a point} &= \int H V \, dB \quad (V = lS)
 \end{aligned}$$

$$\text{Energy over space} = \iint H \, dV \, dB$$

Energy per unit volume = $\int H dB$

$\therefore$ Hysteresis loss is equal to area of B-H curve.

Magnetic Materials

Hard magnetic - Permanent Magnets

Soft magnetic -

i) Diamagnetic material

E.g. Cu $\Rightarrow \mu_r = 0.99945$

ii) Paramagnetic material

E.g. Al $\Rightarrow \mu_r = 1.0056$

iii) Ferromagnetic material

E.g. Fe $\Rightarrow \mu_r = 1000$ to 50,000 for steel

Orbital motions of e's acts as bound currents

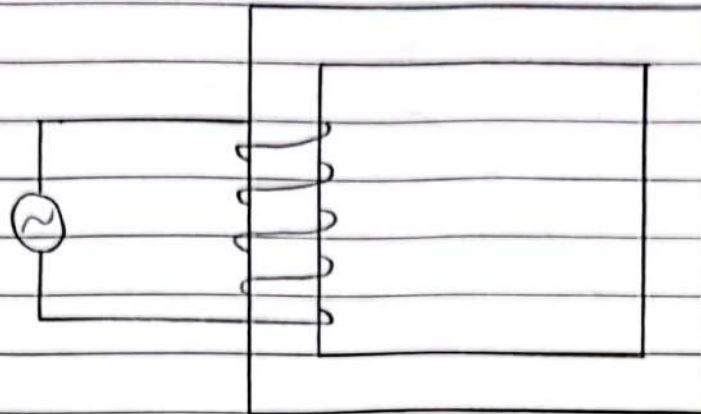
$$\vec{B} = \mu_0 (\vec{H} + \vec{M})$$

$$\vec{B} = \mu_0 (1 + \chi_m) \vec{H}$$

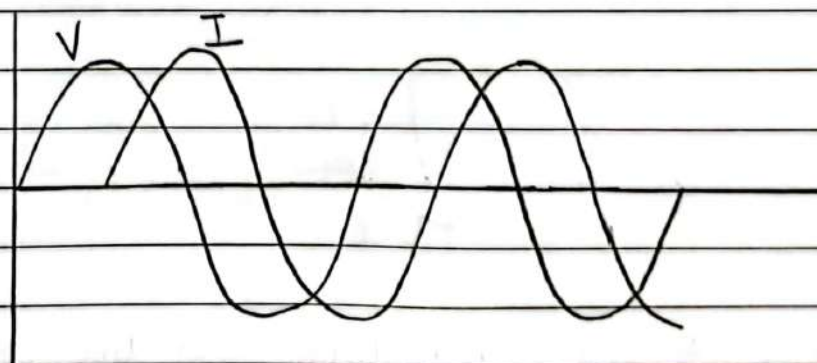
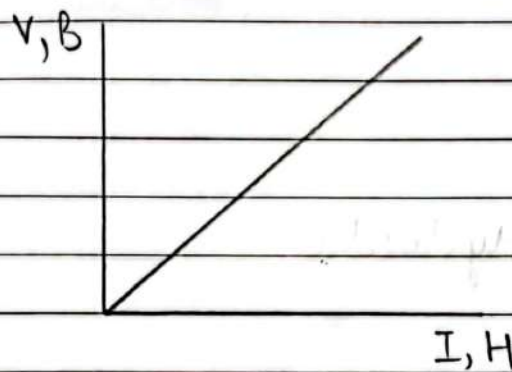
χ_m = Magnetic Susceptibility

V-I characteristics for magnetic circuits

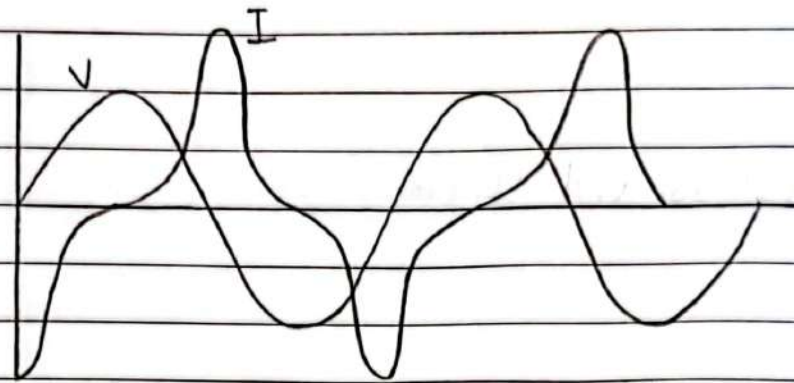
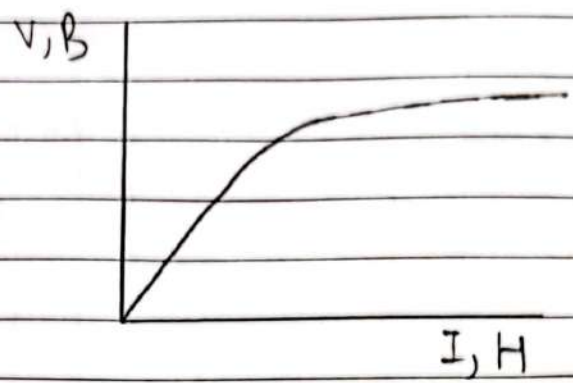
Case 1)



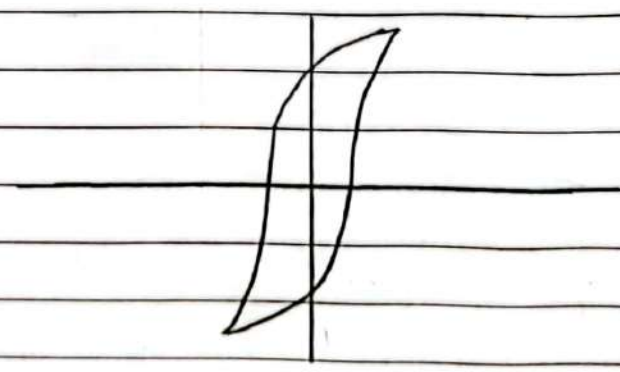
Case 1) Lossless and linear



Case 2) Lossless & Nonlinear



Case 3) Lossy and Hysteretic



$V-I$ and V and I waveform has two components -
 i_m (magnetic component, lossless, 90° out of phase with v)
 i_n (hysteretic component, lossy, in phase with v)

Permanent Magnets (Hard Magnetic Materials)

Permanent Magnet acts as a source of flux in magnetic circuit.

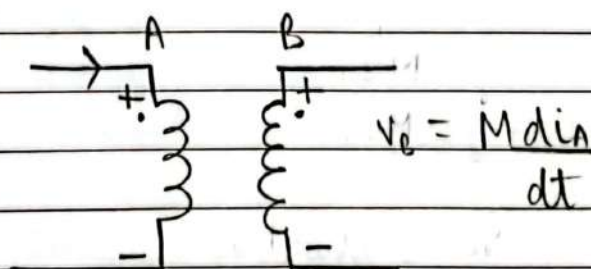
Keeping a permanent magnet in air is equivalent to an open magnetic circuit. However, there will be flux set up, because air is not ~~will~~ a perfect resistor magnetic.

Adding a path made up of magnetic material $\approx$ shorting. However no material has ~ 0 ~~max~~ reluctance.

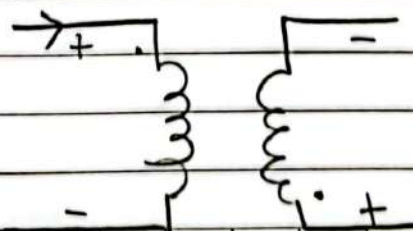
Permanent Magnets operate in II ~~quadrant~~ quadrant of B-H curve.

Dot Convention

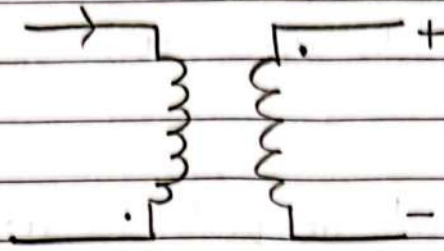
Case 1)



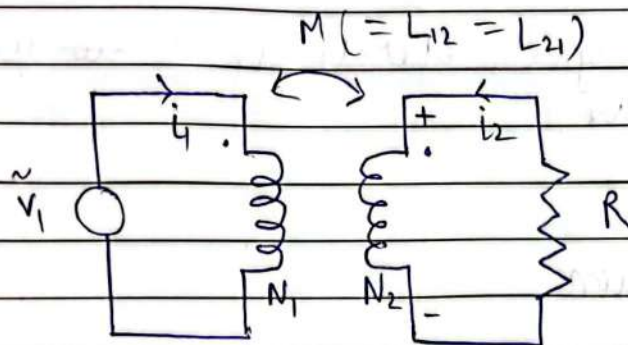
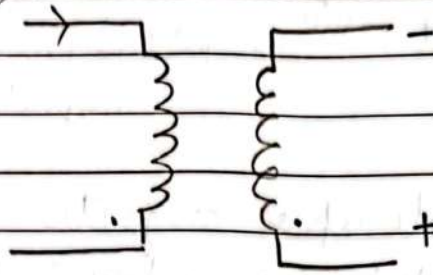
Case 2)



Case 3)



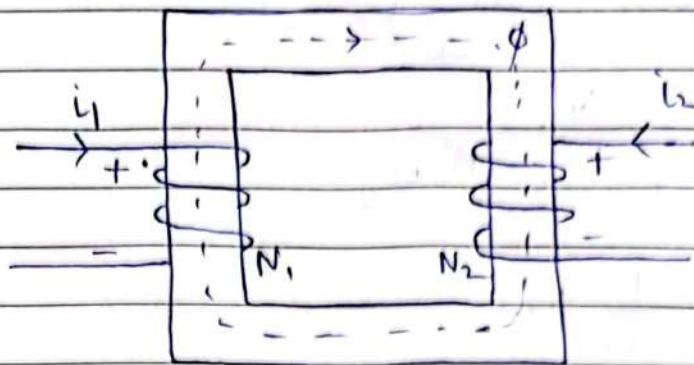
Case 4)



$$\begin{aligned}\tilde{V}_1 &= j\omega L_1 \tilde{I}_1 + j\omega M \tilde{I}_2 \\ \tilde{V}_2 &= j\omega L_2 \tilde{I}_2 + j\omega M \tilde{I}_1\end{aligned}$$

If $R=0$, i.e., $\tilde{V}_2$ is shorted
 $\tilde{V}_2 = 0$

$$\begin{aligned}\therefore j\omega L_2 \tilde{I}_2 + j\omega M \tilde{I}_1 &= 0 \\ \tilde{I}_2 &= -\frac{M}{L_2} \tilde{I}_1\end{aligned}$$



$$L_1 = \frac{\mu N_1^2 A}{l}, \quad L_2 = \frac{\mu N_2^2 A}{l}$$

$$\frac{L_1}{L_2} = \frac{N_1^2}{N_2^2}$$

$$\lambda_1 = L_1 i_1, \quad \lambda_2 = L_2 i_2$$

$$\phi_1 = \frac{L_1 i_1}{N_1}$$

$$\phi_2 = \frac{L_2 i_2}{N_2}$$

$$\text{But } \phi_1 = \phi_2 = \phi$$

$$M_{12} = \frac{\lambda_2}{i_1} = \frac{N_2 L_1}{N_1} = \sqrt{\frac{L_2}{L_1}} \cdot L_1 \quad \text{--- ①}$$

$$\therefore M_{12} = \sqrt{L_1 L_2}$$

Here we made assumption ①, by neglecting leakage flux. If we don't make this assumption,

$$M_{12} = K \sqrt{L_1 L_2}, \quad \text{where } K = \text{coefficient of coupling} \quad (K \leq 1)$$

$$\therefore M \leq \sqrt{L_1 L_2}$$

$$\therefore \tilde{V}_1 = [j\omega L_1 - j\omega \frac{M^2}{L_2}] \tilde{I}_1$$

$$\tilde{I}_1 = \frac{\tilde{V}_1}{j\omega [L_1 - M^2/L_2]}$$

$$\tilde{I}_2 = -\frac{M}{L_2} \tilde{I}_1$$

If ϕ_{11} = leakage flux

ϕ_{12} = mutual flux

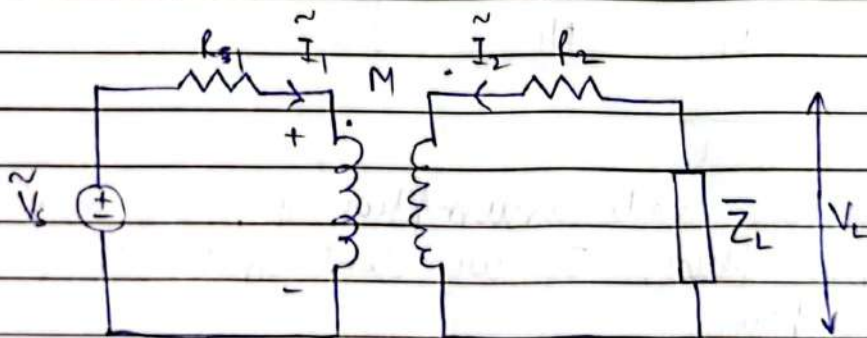
ϕ = total flux

$$\phi_{12} = K\phi$$

$$\phi_{11} = (1-K)\phi$$

$$M = K\sqrt{L_1 L_2}$$

Leakage flux in ② coil causes leakage inductance and limits current even though resistance is zero.



$$\tilde{V}_1 = (R_1 + j\omega L_1) \tilde{I}_1 + j\omega M \tilde{I}_2$$

$$(j\omega L_2 + R_2 + \bar{Z}_L) \tilde{I}_2 + j\omega M \tilde{I}_1 = 0$$

$$\therefore \tilde{I}_2 = \frac{-j\omega M \tilde{I}_1}{j\omega L_2 + R_2 + \bar{Z}_L}$$

$$\frac{\tilde{V}_s}{\tilde{I}_1} = \bar{Z}_{in} = \bar{Z}_{11} + \frac{\omega^2 M^2}{\bar{Z}_{22}}$$

where $\bar{Z}_{11} = R_1 + j\omega L_1$

$\bar{Z}_{22} = R_2 + j\omega L_2 + \bar{Z}_L$

Here, leakage is considered to be present.
 $\frac{\omega^2 M^2}{\bar{Z}_{22}}$ is called reflected impedance

~~Ideal~~

Ideal Transformer

- Losses are neglected $\Rightarrow R_1 = R_2 = 0$
- Leakage flux $= 0 \Rightarrow M = \sqrt{L_1 L_2}$

$$\therefore \bar{Z}_{in} = \bar{Z}_{11} + \frac{\omega^2 L_1 L_2}{\bar{Z}_{22}} \quad (K=1)$$

Let us define $\frac{L_2}{L_1} = \frac{N_2^2}{N_1^2} = a^2$

$$\therefore a = \frac{N_2}{N_1}$$

$$\therefore \bar{Z}_{in} = \bar{Z}_{11} + \frac{\omega^2 a^2 L_1^2}{\bar{Z}_{22}}$$

$$\bar{Z}_{in} = j\omega L_1 + \frac{\omega^2 a^2 L_1^2}{j\omega L_2 + \bar{Z}_L} \quad (R_1 = R_2 = 0)$$

$$\therefore \bar{Z}_{in} = \cancel{j\omega L_1} + j\omega L_1 + \frac{\omega^2 a^2 L_1^2}{j\omega a^2 L_1 + \bar{Z}_L}$$

$$\therefore \bar{Z}_{in} = \frac{\bar{Z}_L}{(\bar{Z}_L / j\omega L_1) + a^2} \approx \frac{\bar{Z}_L}{a^2} \quad \left(\text{Assuming } \bar{Z}_L \ll \omega L_1 \right)$$

Summary of Magnetism

- Lorentz Force

$$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$$

$$\vec{F}_m = \vec{J} \times \vec{B}$$

$$\vec{F} = \mu_0 I \vec{L} \times \vec{B}$$

- Biot - Savart Law

$$dH = \frac{1}{4\pi} \frac{I dl \sin \theta}{r^2}$$

$$H = \frac{I}{2\pi r}$$

$$\oint \vec{H} \cdot d\vec{l} = I$$

$$\vec{\nabla} \times \vec{H} = \vec{J}$$

$$\nabla \cdot \vec{B} = 0$$

$$\oint_S \vec{B} \cdot d\vec{S} = 0, \quad \oint_S \vec{B} \cdot d\vec{S} = \phi$$

$$\vec{B} = \nabla \times \vec{A}$$

$$\phi = \oint \vec{A} \cdot d\vec{l} = (A_1 - A_2) l$$

$$\vec{A} = \int \frac{\mu_0 I d\vec{l}}{4\pi r}$$

Poisson's Equation -

$$\nabla^2 A_z = -\mu_0 J_z$$

- Magnetic Materials

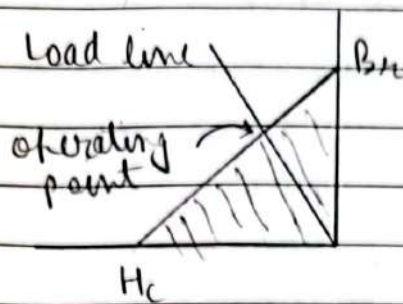
Soft - Ferromagnetic - μ_r high ($\gg 1$)

$$\vec{B} = \mu_0 (\vec{H} + \vec{M})$$

- Hysteresis

- V-I waveforms for linear / non-linear / hysteresis B-H curve

- Permanent Magnets
 (operate in II quadrant of B-H curve)



Greater the shaded area, stronger the magnet

- $MMF = \text{Flux} \times \text{Reluctance}$
 Reluctance and reactance vary in inverse proportion.

$$(X_L \propto L \propto \phi \propto 1/R)$$

$$\therefore X_L \propto \frac{1}{R}$$

$$\lambda = N\phi, \quad L = \frac{N\phi}{i}, \quad L = \frac{N^2}{R}, \quad L = \frac{\mu N^2 S}{l}$$

l is that length ~~not~~ which decides the flux

$$\text{Energy density} = \frac{B^2}{2\mu_0}$$

- Self and Mutual Inductance

- Dot convention

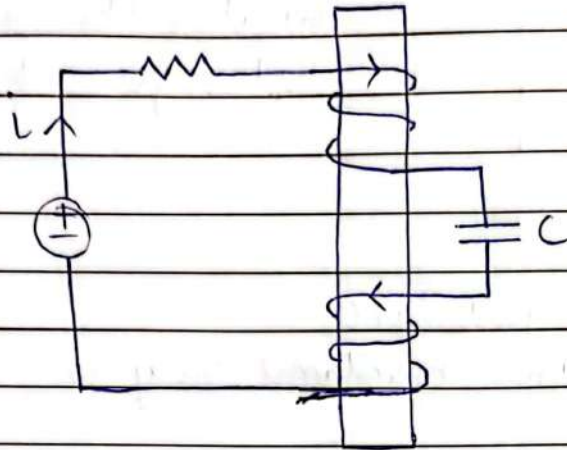
- Z_{in} when leakage is not neglected

- $Z_{in} = \frac{Z_L}{n^2}$ in ideal case ($R_1 = R_2 = 0$, $K=1$, μ_r is high)

In case of two mutually coupled coils,
 Total Energy,

$$W = \frac{1}{2} L_1 i_1^2 + \frac{1}{2} L_2 i_2^2 \pm M i_1 i_2$$

when current in both coils enters the dot,
~~#~~ the +ve sign is taken.



Apply KVL,

$$R i + L_1 \frac{di}{dt} - M \frac{di}{dt} + \frac{1}{C} \int i dt + L_2 \frac{di}{dt} - M \frac{di}{dt} = V$$

$$L_{eff} = L_1 + L_2 - 2M$$

$$\therefore R i + L_{eff} \frac{di}{dt} + \frac{1}{C} \int i dt = V$$

Transformers

At no load,

$$\begin{aligned} e_1 &= N_1 \frac{d\phi}{dt} \\ &= N_1 \frac{d(\phi_m \sin \omega t)}{dt} \\ &= N_1 \phi_m \omega \cos \omega t \end{aligned}$$

$$E_{1(rms)} = \frac{(e_1)_{max}}{\sqrt{2}} = \frac{2\pi N_1 \phi_m f}{\sqrt{2}} \cos \omega t$$

$$\therefore E_{1(rms)} = 4.44 \phi_m f N_1$$

$$E_{2(rms)} = 4.44 \phi_m f N_2$$

$$\therefore \frac{E_2}{E_1} = \frac{N_2}{N_1} = a$$

At no load $\tilde{I}_1 = \tilde{I}_m$ (magnetising current, used to set up ϕ)

With load,

$$\tilde{I}_1 = \tilde{I}_1' + \tilde{I}_m$$

$\tilde{I}_1'$ is reflected current.

$\tilde{I}_m$ is no-load current

If $\tilde{I}_m \approx 0$, for μ_r very high,
 $\tilde{I}_1 = \tilde{I}_1'$

$$\therefore I_1 N_1 = I_2 N_2$$

$$\therefore \frac{V_1}{V_2} = \frac{E_1}{E_2} = \frac{N_1}{N_2} = \frac{I_2}{I_1}$$

$$Z_2 = \frac{V_2}{I_2} = \frac{(N_2/N_1) V_1}{(N_1/N_2) I_1} = \left(\frac{N_2}{N_1} \right)^2 \frac{V_1}{I_1} = a^2 Z_2'$$

$$\therefore Z_2' = \frac{Z_2}{a^2} \quad (Z_2' \text{ is reflected impedance in Primary})$$

Homework 4-A

1.

$$(a) R = \frac{l}{\mu A} = \frac{2(16+14) \times 10^{-2}}{1500 \times 12 \times 10^{-4} \mu_0}$$

$$R = \frac{2 \times 30 \times 10^{-2}}{15 \times 12 \times 4\pi \times 10^{-7}} = \frac{1}{12\pi \times 10^{-7}}$$

$$= 2.65 \times 10^5 \text{ A turns}$$

$$\phi = \frac{Ni}{R} = \frac{200 \times 2}{R} = 4800\pi \times 10^{-7}$$

$$= 1.508 \times 10^{-3} \text{ Wb}$$

$$L = \frac{N\phi}{i} = \frac{200\phi}{2} = 0.1508 \text{ H}$$

$$(b) R = \frac{56 \times 10^{-2}}{1500 \times 12 \times 10^{-4} \mu_0} + \frac{4 \times 10^{-2}}{150 \times 12 \times 10^{-4} \mu_0}$$

$$= \left(\frac{56}{15 \times 12 \times 4\pi} + \frac{40}{15 \times 12 \times 4\pi} \right) \times 10^7$$

$$= \frac{896 \times 10^7}{15 \times 12 \times 4\pi} = \frac{2}{15\pi} \times 10^7$$

$$= 424.41 \times 10^3 \text{ A/Wb}$$

$$\phi = \frac{400}{R} = 9.42 \times 10^{-4} \text{ Wb}$$

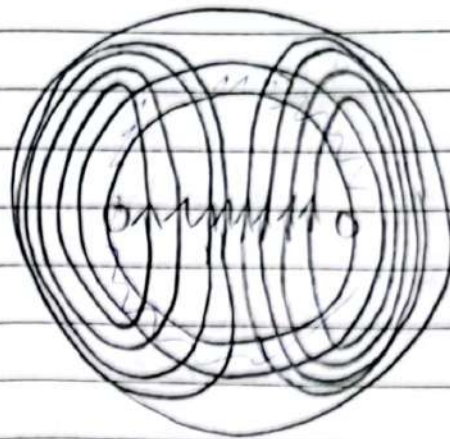
$$L = \frac{200\phi}{2} = 942 \times 10^{-4} \text{ H}$$

2. $R_1 = \frac{g_1}{A_1 \mu_0}, R_2 = \frac{g_2}{A_2 \mu_0}$

$$R = \frac{1}{\mu_0} \left(\frac{A_1}{g_1} + \frac{A_2}{g_2} \right)$$

$$L = \frac{N^2}{R} = N^2 \mu_0 \left(\frac{A_1}{g_1} + \frac{A_2}{g_2} \right)$$

3. (a)



(b) $L = \frac{N\phi}{i}$

$$\phi = \frac{Ni}{R}$$

$$\frac{\phi}{i} = \frac{N}{R}$$

$$L = \frac{N^2}{R}$$

$$l_m = 2g$$

$$\mu = \mu_0$$

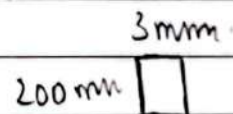
$$A = \pi r^2 l$$

$$R = \frac{2g}{\mu_0 \pi r^2 l}$$

$$L = \frac{N^2 \pi \mu_0 r^2 l}{2g}$$

Homework 4-B

2. $\phi = 0.2 \times 0.002 = 4 \times 10^{-4} \text{ Wb}$
 $= 0.4 \text{ mWb}$



3. $E = 3x^2 - 2xy + 2y^2 - 5x - 2y$

$$\frac{\partial E}{\partial x} = 6x - 2y - 5 = 0$$

$$\frac{\partial E}{\partial y} = -2x + 4y - 2 = 0$$

$$12x - 4y - 10 = 0$$

$$-2x + 4y - 2 = 0$$

$$10x = 12 \Rightarrow x = \frac{6}{5}$$

$$4y = 2 \times \frac{11}{5} = \frac{22}{5} \Rightarrow y = \frac{11}{10}$$

$$E = \frac{144}{100} - \frac{264}{100} + \frac{242}{100} - \frac{600}{100} - \frac{220}{100}$$

$$= -6.98 \text{ units}$$

$$4. \quad \vec{B} = \vec{\nabla} \times \vec{A}$$

$$\vec{A} = (0.025x - 0.017y)\hat{i} + (0.03y - 0.061z)\hat{j} + (0.01z - 0.1x)\hat{k}$$

$$\vec{\nabla} \times \vec{A} = \hat{i}(-0.1 + 0.061) - \hat{j}(-0.1) + \hat{k}(0 + 0.017)$$

$$= 0.061\hat{a}_x + 0.1\hat{a}_y + 0.017\hat{a}_z$$

$$|\vec{B}| = 0.118 \text{ Wb m}^{-2}$$

$$5. \quad \left(\frac{2}{3}, \frac{1}{3}\right)$$

$$a = 2$$

$$2 + b = 4 \Rightarrow b = 2$$

$$2 + 2(1) + c(1) = 6 \Rightarrow c = 2$$

$$A = 2 + 2 = 4 \frac{\text{Wb}}{\text{m}}$$



$$6. \quad E = \int H dB$$

$$B_1 = 1$$

$$B_2 = 1 + (0.0016)(625) = 2$$

$$\therefore E = 0.032 \int_1^2 \left(\frac{B}{0.0016} - \frac{1}{0.0016} \right) dB$$

$$= 20 \int_1^2 (B - 1) dB$$

$$= 20 \left[\frac{B^2}{2} - B \right]_1^2 = 10 \text{ J}$$

7.

$$B = \mu_0 n_2 i_2$$

$$A = \pi r_1^2$$

$$\phi_1 = \mu_0 n_2 i_2 \pi r_1^2$$

$$M = \frac{N_1 \phi_1}{i_2} = \mu_0 n_2 N_1 \pi r_1^2$$

$$= \frac{4\pi \times 10^{-7} \times 796 \times 796 \times \pi \times 9 \times 10^{-4}}{1}$$

$$= \frac{2.25 \text{ mH}}{4} = 0.56 \text{ mH}$$

8. ~~Q. =~~

~~$$L = \frac{\mu_1 A_1}{l_1} + \frac{\mu_2 A_2}{l_2}$$

$$= \left(\frac{50}{1} + \frac{1}{10^{-3}} \right) \times 25 \times 10^{-4} \times 4\pi \times 10^{-7}$$

$$= (1050) \times 10^{-1} \pi$$

$$L = \frac{N^2}{\frac{1050\pi \times 10^{-1}}{25 \times 10^4 \times 1050\pi \times 10^{-9}}} = \frac{25 \times 10^4 \times 10^9}{1050\pi} = 7.579 \times 10^{10}$$

$$L = 25 \times 10^4 \times 1050\pi \times 10^{-9}$$

$$= 82466.8 \times 10^{-5}$$

$$= 824.67 \text{ mH}$$

$$R = \frac{\rho l}{A} = \frac{0.0211 \times 10^{-6} \times 150}{10^{-6}} = 3.165 \Omega$$

$$|X_L| = \omega L = 2\pi f L = 100\pi \times L = 82.467 \pi$$

$$= 259.078 \Omega$$

$$Z = 259.097$$

$$I = 0.385 \text{ A}$$~~

$$R = \frac{1}{\frac{50 \mu\text{A}}{\pi} + \frac{10^{-3}}{\mu\text{A}}} = \frac{(0.021)}{4\pi \times 10^{-7} \times 25 \times 10^{-4}} = \frac{0.021}{\pi \times 10^{-9}}$$

$$= \frac{21 \times 10^6}{\pi} = 6.684 \times 10^6$$

$$L = \frac{25 \times 10^{-7}}{6.684 \times 10^6} = 3.74 \times 10^{-12}$$

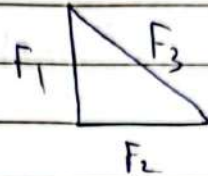
$$|X_L| = 100\pi \times 3.74 \times 10^{-12} = 3.74 \times \pi = 11.749 \Omega$$

$$R = 3.165 \Omega$$

$$Z = 14.914$$

$$I = 8.218 A$$

9. $\vec{F}_1 = I(\vec{r} \times \vec{B})$
 $= 2 \times (-5\hat{j} \times 30\hat{k})$
 $= -300\hat{i}$



$$\vec{F}_2 = 2 \times (5\hat{i} \times 30\hat{k})$$

$$= -300\hat{j}$$

$$\vec{F}_3 = 2(-5\hat{i} + 5\hat{j})(30\hat{k})$$

$$= 300(\hat{j} + \hat{i})$$

$$\vec{F}_{net} = 0$$

10. $B = \frac{\mu N i}{2\pi r} = \frac{4\pi \times 10^{-7} \times 200 \times i \times 10}{2\pi \times 1}$

$$= 4000 \times 10^{-7} i = 0.4 i$$

$$\phi = 0.4 i \times \pi \times 10^{-7} = 0.5 \times 10^{-4}$$

$$i = \frac{5}{0.4\pi} = 3.978 A$$

Homework-5

1. a) $Z_{in} = \frac{Z_L}{a^2}$

$$a = 2 \Rightarrow Z_{in} = \frac{100}{2^2} = 25$$

$$I_1 = \frac{50}{26} = 1.923 A$$

$$I_2 = 0.49 A$$

$$P = 24.01 W$$

$$Z_{in} = Z_1 + \frac{\omega^2 M^2}{Z_2}$$

$$b) Z_{in} = 1 + \frac{100}{400} = 1.25$$

$$I_1 = \frac{50}{1.25} = 40$$

$$I_2 = \frac{40}{20} = 2 \text{ A}$$

$$P = 4 \times 100 = 400 \text{ W}$$

$$c) Z_{in} = 1 + \frac{100}{a^2}$$

$$I_1 = \frac{50}{1 + \frac{100}{a^2}}$$

$$I_2 = \frac{50}{a + \frac{100}{a}}$$

$$\Rightarrow P \propto \frac{1}{(a + \frac{100}{a})^2}$$

$$1 - \frac{100}{a^2} = 0$$

$$a = 10$$

$$N_2 = 1000 \text{ turns}$$

$$P_{max} = \frac{2500}{20 \times 400} \times 100 = 625 \text{ W}$$

$$2. Z_{in} = 1 + j(2\pi \times 50 \times 0.001) + j(2\pi \times 50 \times 0.1) Z_{ref}$$

$$Z_{ref} = \frac{4 + X_c}{a^2} = \frac{4 + X_c}{4}$$

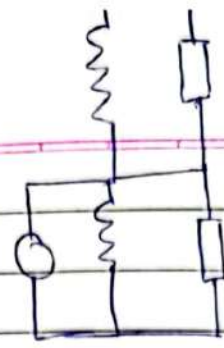
$$L_2 = a^2 = 4$$

$$L_1$$

$$Z_{in} = 2 + j(\pi \times 0.1) + \frac{X_c}{4}$$

$$I_1 = \frac{V}{|Z_{in}|} = 50$$

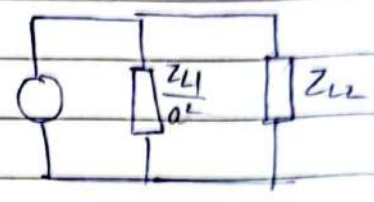
$$I_2 = \frac{I_1}{a}$$



$$X_L = +j10$$

$$\frac{1}{100\pi C} = \frac{2\pi}{5}$$

$$C = \frac{5}{200\pi} = 2.53 \text{ mF}$$



$$I_1 = 25$$

$$I_2 = 12.5$$

$$P = (12.5)^2 \times 4 = 625 \text{ W}$$

3. $Z_{in} = Z_{in} = \frac{(Z_L)_2}{a^2}$

$$\frac{V_2}{V_1} = a = \frac{1}{10}$$

$$\therefore (Z_L)_2 = a$$

$$(Z_L)_1$$

$$(Z_L)_2 + (Z_L)_1 = Z_L$$

$$(Z_L)_1 = \frac{Z_L}{a+1}$$

$$(Z_L)_2 = \left(\frac{a}{a+1} \right) Z_L = \left(\frac{\frac{1}{10}}{\frac{11}{10}} \right) Z_L = \frac{Z_L}{11}$$

$$Z_{in} = \frac{100}{11} Z_L = 9.09 \angle \pi/6 \Omega$$

$$V_s = V_1 = 100 \angle 0^\circ \text{ V}$$

$$V_2 = 10 \angle 0^\circ \text{ V}$$

$$V_L = 110 \angle 0^\circ \text{ V}$$

$$\bar{S}_L = \bar{V}_L \bar{I}_L^* = \frac{\bar{V}_L \cdot \bar{V}_L^*}{\bar{Z}_L^*} = \frac{|\bar{V}_L|^2}{\bar{Z}_L^*}$$

$$\bar{S}_L = \frac{(110)^2}{1 \angle -\pi/6} = 12100 \angle \pi/6 \text{ VA}$$

$$\bar{S}_s = \frac{V_s^2}{\bar{Z}_{in}^*} = \frac{10000 \times 11 \angle \pi/6}{100} = 1100 \angle \pi/6 \text{ VA}$$

$$Z_{in} = \frac{10 Z_L}{11} \left(\frac{10}{11} \right) = \frac{100}{121} Z_L$$

4. $Z_{in} = j\omega L \frac{\pi}{6}$

$$a = \frac{\sqrt{L_2}}{\sqrt{L_1}} = \frac{1}{\sqrt{2}}$$

$$Z_{in} = \frac{Z_L}{j\omega L_1 + a^2}$$

$$\frac{Z_L}{j\omega L_1} = Z_{in} = 1.9355 \angle 33.0586^\circ \Omega$$

$$\frac{V_0}{Z_{in}} = \bar{I}_1 = 5.166 \angle -33.0586^\circ \text{ A}$$

$$\bar{I}_2 = \sqrt{2} \bar{I}_1 = 7.306 \angle -33.0586^\circ \text{ A}$$

$$\bar{S}_L = V_L \bar{I}_L^* = \bar{I}_L \bar{I}_L^* Z = 53.377 \angle 30^\circ \text{ VA}$$

$$\bar{S}_S = V_S \bar{I}_S^* =$$

$$Z_{in} = 2 Z_L = 2 \angle 30^\circ$$

$$\bar{I}_1 = 5 \angle -30^\circ \text{ A}$$

$$\bar{I}_2 = 5\sqrt{2} \angle -30^\circ \text{ A}$$

$$\bar{S}_L = 50 \times \angle 30^\circ \text{ A VA}$$

$$\bar{S}_S =$$

5. $Z_{in} = \frac{j\omega L_1 + \omega^2 M^2}{R + j\omega L_2}$

$$= \frac{j(20) + \frac{100}{10 + 10j}}{10 + 10j}$$

$$= 20j + \frac{10}{1+j} = 20j + 5 - 5j = 5 + 15j$$

$$\bar{I}_1 = \frac{142.3}{5 + 15j} = 8.99 \angle -71.56^\circ$$

$$\therefore = 8.99 \angle -71.56^\circ$$

6. $\phi_1 = \phi_{11} + \phi_{12} = 0.6 \text{ mWb}$

$\frac{\phi_{12}}{\phi_1} = k = \frac{2}{3}$

$L_1 = \frac{N_1 \phi_1}{i_1} = \frac{500 \times 0.6 \times 10^{-3}}{5}$
 $= 0.6 \times 10^{-1} = 0.06$

$L_1 = 60 \text{ mH}$

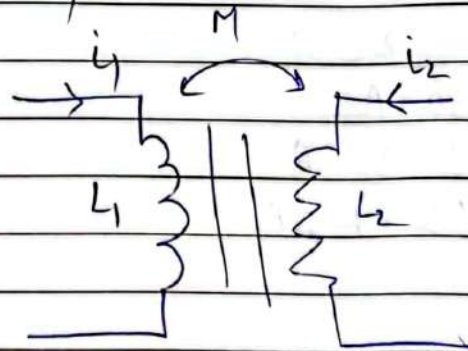
$M = \frac{N_2 \phi_{12}}{i_1} = \frac{1500 \times 0.4 \times 10^{-3}}{5}$
 $= 300 \times 0.4 \times 10^{-3}$

$M = 120 \text{ mH}$

$\frac{M^2}{k^2 L_1} = L_2 = \frac{0.0144 \times 9}{4 \times 0.06}$ ~~$\frac{144 \times 9}{4 \times 36}$~~

$L_2 = \frac{0.0144 \times 9}{4 \times 0.06} = \frac{1.44 \times 9}{24}$
 $= 0.06 \times 9 = 540 \text{ mH}$

7. ~~$\phi_2 = \phi_1 = \phi$~~



$N_2 \phi_2 = L_2 i_2 + M i_1$

$V_2 = L_2 \frac{di_2}{dt} + M \frac{di_1}{dt}$

$\tilde{V}_2 = j\omega L_2 \tilde{I}_2 + j\omega M \tilde{I}_1 = 0$
 $\tilde{I}_2 = -\frac{M}{L_2} \tilde{I}_1$

$$M = \frac{\sqrt{50 \times 50 \times 4}}{2} = 50 = L_1$$

$$\tilde{I}_2 = -\tilde{I}_1$$

$$\tilde{V}_2 = 0.05 \times j \times 400 \times 5 = 100j$$

$$v_2 = 100 \sin(400t + 90^\circ)$$

$$N_1 \phi_1 = L_1 I_1$$

$$\phi_1 =$$

$$N_2 \phi_2 = M I_1 = 0.25 \sin(400t)$$

$$\phi_2 = 0.25 \sin 400t \text{ mWb}$$

$$\phi_2 = \frac{\phi_1}{2}$$

$$\begin{aligned} \phi_1 &= 0.5 \sin 400t \times 10^{-3} \text{ Wb} \\ &= 5 \times 10^{-4} \sin 400t \text{ Wb} \end{aligned}$$

$$8. \quad N_1 \phi_1 = L_1 i - M_{12} i + M_{13} i$$

$$\tilde{V}_1 = j\omega L_1 (L_1 - M_{12} + M_{13})$$

$$\tilde{V}_2 = j\omega L_2 (-L_2 + M_{12} + M_{23})$$

$$\tilde{V}_3 = j\omega L_3 (L_3 + M_{13} - M_{23})$$

$$j\omega X \quad jX_L =$$

$$X_L = 3 - 5 + 6 + 8 = 12 \Omega$$

$$v_1(t) = N_1 \frac{d\phi_1}{dt}$$

$$i(t) = I_m \sin \omega t$$

$$I = I e^{j\omega t}$$

$$\frac{dI}{dt} = j\omega I e^{j\omega t}$$

Midsem - 2024

$$\begin{aligned}
 1. \quad v(t) &= 100 (\cos \omega_1 t - \cos \omega_2 t) \\
 &= \cos(A+B) - \cos(A-B) \\
 &= \cos A \cos B - \sin A \sin B - (\cos A \cos B + \sin A \sin B) \\
 &= -2 \sin A \sin B
 \end{aligned}$$

$$v(t) = -200 \sin\left(\frac{(\omega_1 + \omega_2)t}{2}\right) \sin\left(\frac{(\omega_1 - \omega_2)t}{2}\right)$$

Let us assume impedance X_L

$$(a) \quad i(t) = \frac{200}{X_L} \sin\left(\frac{(\omega_1 + \omega_2)t}{2}\right) \sin\left(\frac{(\omega_1 - \omega_2)t}{2}\right)$$

$$(b) \quad i_{\text{max}} = \frac{200}{2 \times \pi \times 50 \times 0.01} = \frac{200}{\pi}$$

(c)

2.

$$\begin{aligned}
 (a) \quad V &= 100 (\angle 0 - \angle 2\pi/3) = 173.2 \angle -30^\circ \\
 P &= \frac{V^2}{R} = \frac{299}{100} = 2.99 \text{ kW}
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad V_x - V_1 + V_x - V_2 + V_x - V_3 &= 0 \\
 10j \quad 10j \quad 10j \\
 V_x &= \frac{V_1 + V_2 + V_3}{3} = \frac{100 + 100 \angle 120^\circ + V_3}{3} \\
 I_3 &= \frac{V_x - V_3}{10j} = \frac{100 + 100 \angle 120^\circ - 2V_3}{30j} \\
 P &= V_3 I_3 = V_3 (a + bV_3) = aV_3 + bV_3^2 \\
 \frac{\partial P}{\partial V_3} &= a + 2bV_3 = 0 \\
 V_3 &= \frac{-a}{2b} = \frac{-100}{2 \times 10j} = +100 \angle 120^\circ
 \end{aligned}$$

$$V_3 = 50 \angle 60^\circ$$

$$i_3 = \frac{100 + 100 \angle 120^\circ - 100 \angle 60^\circ}{30j}$$

$$i_3^* = \frac{100 + 100 \angle -120^\circ - 2V_3^*}{-30j}$$

$$S = v(a + bv^*)$$

$$= av + b|v|^2$$

$$P = \frac{b|v|^2}{2}$$

$$S = \left(\frac{100 + 100 \angle -120^\circ}{-30j} \right) V_3 + \frac{2 \times 10000}{30j}$$

$$P =$$

$$V_n - 100 + V_n - 100 \angle 120^\circ = 0$$

$$V_n = 50(1 + \angle 120^\circ)$$

$$\left(\frac{V_n - V_3}{j15} \right)^* V_3 = V_3 \left(\frac{V_3^* - 50 \angle -\pi/3}{j15} \right)$$

$$\frac{10000}{15j} - \frac{50 \angle -\pi/3 \times V_3}{j15}$$

$$\frac{50 \angle \pi/6 \times 100 \angle \delta}{15}$$

$$= \frac{5000 \angle \frac{\pi}{6} + \delta}{15}$$

$$\delta = -\pi/6$$

$$3. \quad \tilde{I}_n = \sqrt{2} \angle -90^\circ = -j\sqrt{2}$$

$$\tilde{V}_n = -j\sqrt{2} \times j \times 100 \times 0.05 = j5\sqrt{2} = 5\sqrt{2}$$

$$5\sqrt{2}j - i_2(2+j) - V_2 = 0$$

$$i_2 = \frac{5\sqrt{2}(1-j)}{5+2.5j}$$

$$i_2 = i_1 - i_n = 1.897 \angle -116.56^\circ$$

$$\bar{S}_2 = -V_2 I_2^*$$

$$V_2 = 5\sqrt{2} - (2+j)I_2$$

$$\bar{S}_2 = -V_2 I_2^* = (13.2 - 8.4j) \text{ VA}$$

$$\bar{S}_1 = V_1 I_1^* = 5$$

$$\bar{S}_1 = 0$$

$$V_2 = (\sqrt{2} \angle -90^\circ)(2+j) = 6\sqrt{2} - 2\sqrt{2}j$$

$$\hat{I}_2^* = \sqrt{2} \angle 90^\circ$$

$$\bar{S}_2 = (4+12j) \text{ VA}$$

$$4. \quad \bar{V}_2 = \frac{\bar{X}_c}{R_1 + \bar{X}_c} \bar{V}_1 = \left(\frac{-j/2}{1 - j/2} \right) \bar{V}_1$$

$$\bar{Z}_{c1} = -j/2$$

$$\bar{Z}_{c2} = -j$$

$$\bar{Z}_1 = (R_2 + \bar{Z}_{c2}) \parallel \bar{Z}_{c1} = \frac{(2-j) \cdot (-j/2)}{2-j-j/2} = 0.08 - 0.44j$$

$$\bar{V}_2 = \frac{\bar{Z}_1}{1 + \bar{Z}_1} \bar{V}_1 = \left(\frac{7-11j}{39} \right) \bar{V}_1$$

$$\bar{V}_2 = \left(\frac{-j}{2-j} \right) (\dots) \bar{V}_1$$

$$H = \frac{\bar{V}_2}{\bar{V}_1} = 0.1715 \angle -120.96^\circ$$

$$\bar{V}_2 = 17.15 \angle -120.96^\circ$$

$$6. \quad L_1 \hat{I}_1 + M \hat{I}_2 = 0$$

$$\hat{I}_1 = -\frac{M}{L_1} \hat{I}_2$$

$$\begin{aligned}\bar{Z}_{in} &= \bar{Z} + j\omega L_2 + \frac{\omega^2 M^2}{j\omega L_1} \\ &= 1\angle -30^\circ + j\omega L_2 + \frac{\omega^2 M^2}{j\omega L_1}\end{aligned}$$

$$\bar{I}_2 = 0.34 \angle -88.31^\circ$$

$$\bar{I}_1 = 0.34 \angle 91.69^\circ$$

$$(1) \quad I = \frac{(V_1 - V_2)}{X_L}$$

$$P = \operatorname{Re} \left(\frac{V_2}{X_L^*} (V_1^* - V_2^*) \right)$$

$$= -\operatorname{Re} (V_1^* V_2)$$

$$\bar{V}_1 = 100 \angle 0$$

$$\bar{V}_2 = 100 \angle -\theta$$

$$P = \operatorname{Re} \left(\frac{100 \angle -\theta (100 - 100 \angle \theta)}{j\omega L} \right)$$

$$= \operatorname{Re} \left(\frac{10000 \angle -\theta - 10000}{j\omega L} \right)$$

$$= \operatorname{Re} \left(\frac{10000 \angle -\theta - 90^\circ}{\omega L} \right)$$

$$= \frac{10000}{\omega L} \cos(\theta + 90^\circ)$$

$$100 - j\omega L_1 (\bar{I}_s - \bar{I}_L) + j\omega M \bar{I}_L = 0$$

$$j\omega L_1 (\bar{I}_s - \bar{I}_L) - 100 - j\omega L_2 \bar{I}_L + j\omega M (\bar{I}_s - \bar{I}_L) - \bar{I}_L \bar{Z}_L = 0$$

$$\frac{j\omega M (\bar{I}_s - \bar{I}_L) - j\omega L_2 \bar{I}_L}{100} = \frac{1}{10}$$

$$100 + 10 = \bar{I}_L \bar{Z}_L$$

$$\bar{I}_L = 110 \angle -30^\circ \text{ A}$$

$$\text{Assume } \bar{Z}_L \ll \omega L_1$$

$$j\omega M (\bar{I}_s - \bar{I}_L) = (j\omega L_2 - \bar{Z}_L) \bar{I}_L - 100$$

$$100 - \frac{L_1}{M} [(j\omega L_2 - \bar{Z}_L) \bar{I}_L - 100] + j\omega M \bar{I}_L = 0$$

$$\bar{V}_s \left(1 + \frac{L_1}{M}\right) + \bar{I}_L \left(j\omega M - \frac{L_1}{M} (j\omega L_2 - \bar{Z}_L)\right) = 0$$

$$\frac{\bar{V}_s}{\bar{I}_L} = \left(\frac{M}{M+L_1}\right) \left(\frac{L_1}{M} (j\omega L_2 - \bar{Z}_L) - j\omega M\right)$$

$$\bar{V}_s - j\omega L_1 \bar{I}_s + j\omega (M+L_1) \bar{I}_L = 0$$

$$\bar{V}_s - j\omega L_1 \bar{I}_s + j\omega (M+L_1) \bar{I}_L = 0$$

$$\bar{V}_s \left[1 - \frac{j\omega (M+L_1)^2}{j\omega L_1 \bar{Z}_L - L_1 \bar{Z}_L - j\omega M^2}\right] = j\omega L_1 \bar{I}_s$$

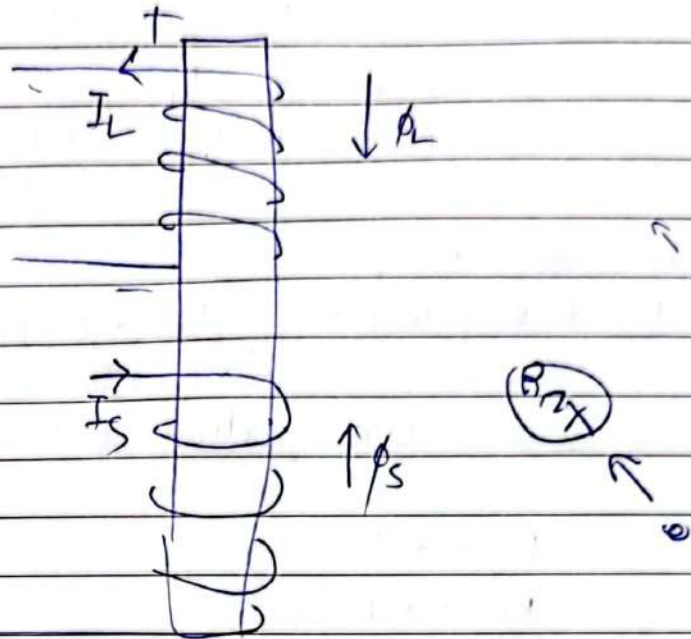
$$\bar{V}_s \left[1 + \frac{j\omega (M+L_1)^2}{L_1 \bar{Z}_L}\right] = j\omega L_1 \bar{I}_s$$

$$\bar{I}_s = \frac{j\omega L_1}{j\omega L_1 + \frac{(M+L_1)^2}{L_1 \bar{Z}_L}}$$

$$= \frac{1}{j\omega L_1} + \frac{1}{1 + \frac{121}{100 \bar{Z}_L}}$$

$$\frac{M}{L_1} = \frac{\sqrt{L_1 L_2}}{L_1} = \sqrt{\frac{L_2}{L_1}} = \frac{N_2}{N_1} = \frac{1}{10}$$

$$Z = \frac{100 \angle 30^\circ}{121}$$

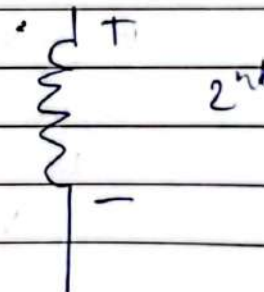


- M

$$Z_L \ll \omega L_1$$

$$I_L = 110 \angle -30^\circ \text{ A}$$

I_S



4-3-2025

Equivalent circuit of Transformer

From fig. 4 to fig. 5

Mesh equation in II loop is (multiplied by i_2)
 $(R_2 + R_L) i_2^2 + L_2 i_2 \frac{di_2}{dt} - M i_2 \frac{di_1}{dt} = 0$ (for fig. 4)

Replace i_2 by $i_2 c$ to get

$$\left(\frac{R_2}{c^2} + \frac{R_L}{c^2} \right) (i_2 c)^2 + \left(\frac{L_2}{c^2} \right) (i_2 c) \frac{d(i_2 c)}{dt} - \frac{M}{c} (i_2 c) \frac{di_1}{dt} = 0$$

Cancelling $i_2 c$,

$$\left(\frac{R_2}{c^2} + \frac{R_L}{c^2} \right) (i_2 c) + \left(\frac{L_2}{c^2} \right) \frac{d(i_2 c)}{dt} - \frac{M}{c} \frac{di_1}{dt} = 0$$

This is the mesh equation of II loop for fig. 5

Hence $R_2 \rightarrow R_2 / c^2$

$R_L \rightarrow R_L / c^2$

$L_2 \rightarrow L_2 / c^2$

$M \rightarrow M / c$

when $i_2 \rightarrow i_2 c$

and $v_2 \rightarrow v_2 / c$

($V_{i1} = V_2 i_2$)

From fig. 6 to fig. 7

Let $c = \sqrt{L_2 / L_1}$

$M = k \sqrt{L_1 L_2}$

coefficient of coupling

$$\therefore L_1 - \frac{M}{c} = L_1 - \frac{k \sqrt{L_1 L_2}}{\sqrt{L_2 / L_1}} = L_1 (1 - K)$$

$$\frac{L_2}{c^2} - \frac{M}{c} = L_1 - \frac{M}{c} = L_1 (1 - K) = \frac{L_2}{c^2} (1 - K)$$

$$\frac{M}{c} = K L_1$$

From fig-7 to fig-8

We assumed that leakage is not significant.

$$\Rightarrow \frac{L_2}{L_1} = \frac{N_2^2}{N_1^2}$$

$$\Rightarrow c = N_2/N_1$$

From fig-8 to fig-9 (10)

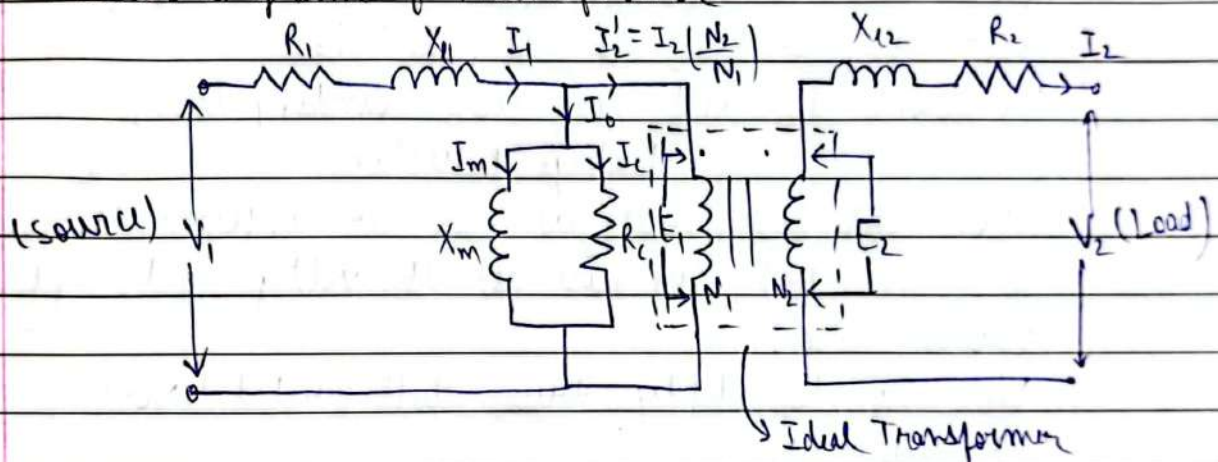
L_1 is very high (because of high μ_r) $\Rightarrow$

X_{m1} is very high $\Rightarrow$ we can shift X_{m1} to other side of X_{m1} to a reasonable approximation.

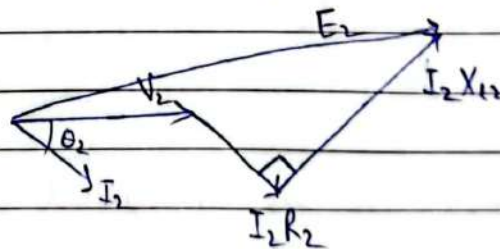
Also, R_c represents core loss. ($R_c \propto V_1$)

R_c is nonlinear. It includes into consideration hysteresis.

Phasor Diagram of Transformer

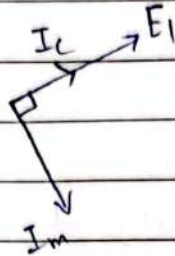


We take V_2 as reference phasor for secondary circuit



P.T.O

We take E_1 as reference for primary circuit
 $(E_1 = \frac{N_1}{N_2} E_2)$



Core loss, $P_c = I_c E_1 = I_c^2 R_c$

$$X_m = E_1 / I_m$$

We can derive the equivalent ckt parameters experimentally.

1) Open Ckt Test -

$$R_L \rightarrow \infty, I_2 = 0, I_2' = 0, I_1 = I_0$$

We want to find Core loss, Winding loss,
 Leakage Reactance

Open Ckt Test will give us core information
 $(X_m \text{ and } R_c)$, because all the voltage gets applied
 across core.

We have to apply ~~full~~ full (rated) voltage
 at source (V_1).

$$\cos \theta_0 = \frac{P_c}{V_1 I_0}$$

$$R_c = \frac{P_c}{I_c^2}$$

where P_c is measured using wattmeter

$$I_c = I_0 \cos \theta_0$$

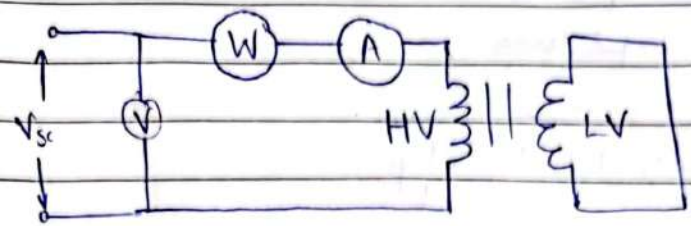
$$I_m = I_0 \sin \theta_0$$

where I_0 is measured experimentally

$$X_m = \frac{V_1}{I_m}$$

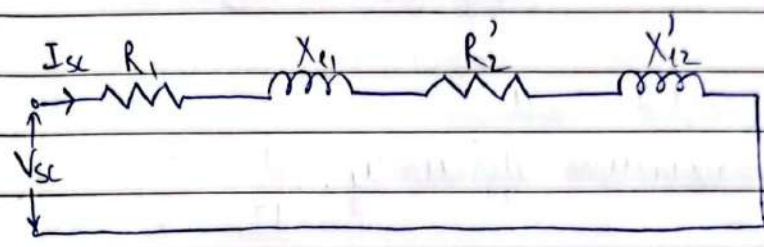
3-25

2) Short Circuit Test



W is wattmeter

HV = high voltage, LV = low voltage



$$R_{eq1} = R_1 + R_2' ; X_{eq1} = X_1 + X_2'$$

$$Z_{eq1} = \frac{V_{sc}}{I_{sc}}$$

$$P = R_{eq1} I_{sc}^2 \Rightarrow R_{eq1} = \frac{P}{I_{sc}^2}$$

$$X_{eq1} = \sqrt{Z_{eq1}^2 - R_{eq1}^2}$$

In order to drive rated current in secondary,
 $V_{sc} = k V_{rated1}$, where k = coefficient of coupling

Voltage Regulation

$$\text{Regulation (\%)} = \frac{(V_{L(OC)} - V_L) \times 100}{V_{L(OC)}} \quad \begin{matrix} \text{terminal voltage (for rated} \\ \text{load condition)} \end{matrix}$$

(voltage drop)

$$\text{Regulation} \approx \frac{I_2 R_{eq2} \cos \phi_2 + I_2 X_{eq2} \sin \phi_2}{V_{L(OC)}} \times 100$$

→ Deviation out of scope.

Efficiency of Transformer

$$\eta(\%) = \frac{\text{Output Power}}{\text{Input Power}} \times 100\%$$

$$\eta = \frac{V_2 I_2 \cos \theta_2}{V_2 I_2 \cos \theta_2 + P_c + I_2^2 R_{eq2}}$$

Core Loss
Coil loss

Core Loss $\rightarrow$ Hysteresis
 $\rightarrow$ Eddy current loss

I_2 is called load.

For maximum efficiency, $\frac{d\eta}{dI_2} = 0$

Solving, we get $P_c = I_2^2 R_{eq2}$

This is the maximum efficiency condition

Example

A transformer rated 50Hz, 50.6kVA, $V_1 = 220V$,
 $a = N_2/N_1 = 0.1$, $I_1(\text{rated}) = 230A$

It is a step-down transformer

$$L_{e1} = L'_{e2} = 127 \mu H$$

$$R_1 = R'_2 = 0.02 \mu H$$

$$L_m = 80mH \Rightarrow X_m = 25 \Omega$$

1) No load current = $\frac{V_1}{R_1 + j\omega L_{e1}} \cdot \frac{V_1}{R_1 + j(X_{e1} + X_m)}$

$$\text{Let } V_1 = 220 \angle 0^\circ V$$

$$\therefore \text{No load current} = 8.8 \angle -90^\circ A$$

2) open circuit secondary voltage as seen on primary side.

$$V_2'(\text{oc}) = I_1 \times jX_m \approx 220\text{V}$$

$$V_2(\text{oc}) = 220 \times 0.1 = 22\text{V}$$

3) R-L load, $R_L' = 2\Omega$, $X_L' = L_{\text{load}} = 6.37\text{mH}$

V_2' and $I_1 = ?$

$$I_1 = (52.85 - j53.91)\text{A} = 75.5 \angle -45.56^\circ$$

$$V_2 =$$

Skim Effect

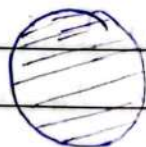
DC current is uniformly distributed across entire cross section of wire.

However, AC current is concentrated near the surface of the conductor. More the frequency, more the current goes towards surface.

Then S_{eff} decreases $\Rightarrow R = \frac{\rho l}{S_{\text{eff}}}$ increases

At 50 Hz, $R_{\text{ac}} = 1.1$ to $1.5 R_{\text{dc}}$

$$R = \frac{\rho l}{S_{\text{eff}}} \neq \frac{\rho l}{S_{\text{true}}}$$

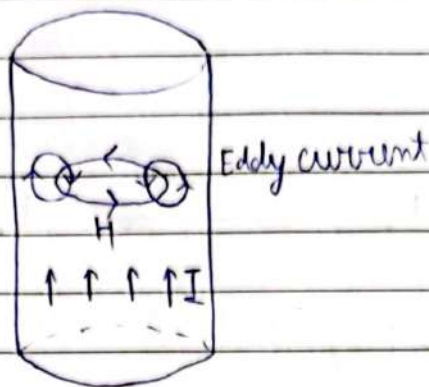


DC current



AC current

Cause of Skin Effect



Eddy current loops tend to increase current near surface and decrease near center.

Alternative explanation -

Consider radial layers of conductor cross section as different conductors in parallel. Inner conductor will have greater inductive impedance. Hence it will pass less current.

Physical understanding of X_m

More μ_r of core $\Rightarrow$ less magnetizing current I_m
 $\Rightarrow$ More X_m

Physical understanding of R_c

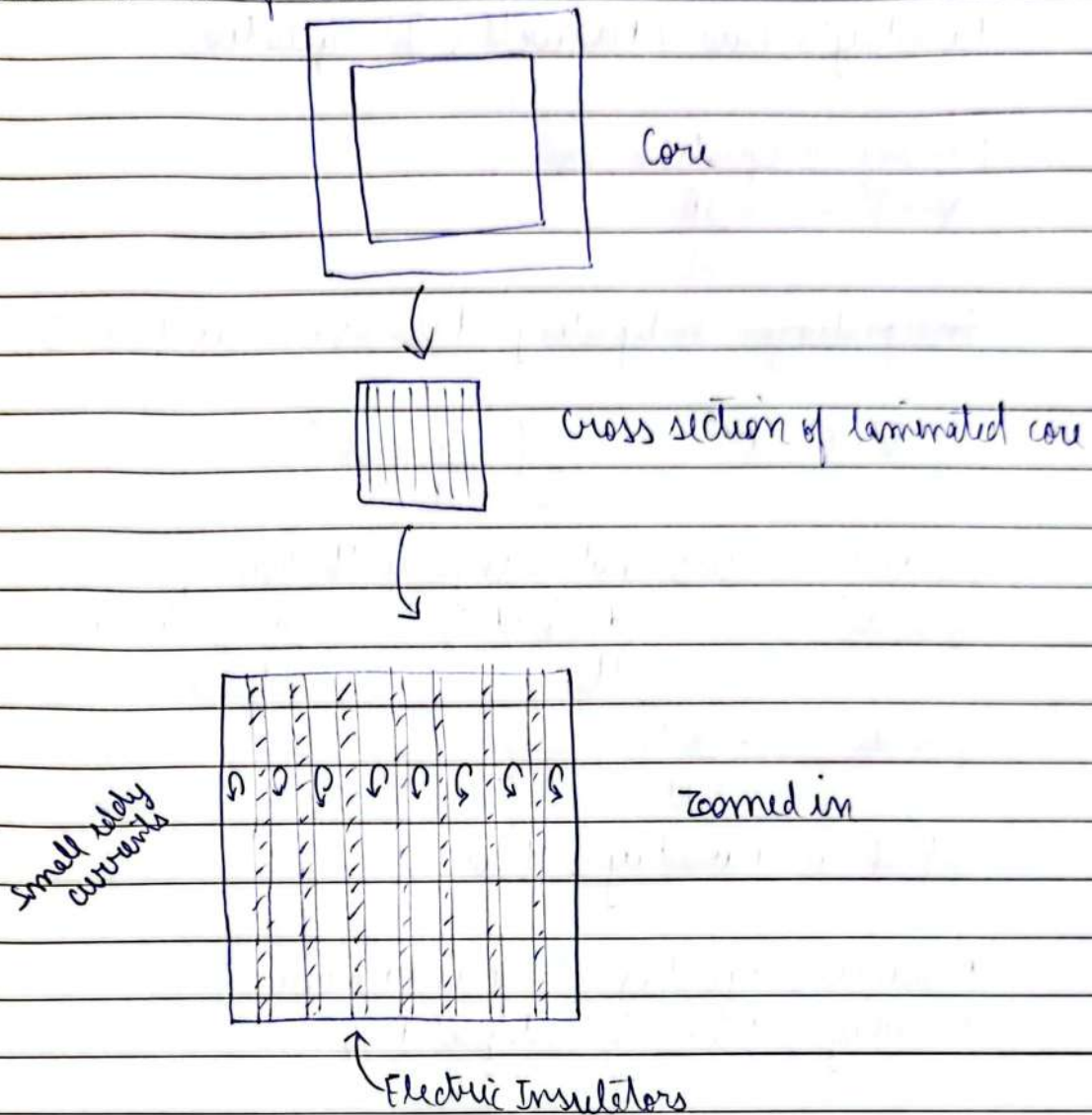
Core is made of steel / iron (good electric conductor).

Alternating flux causes eddy currents in core.

These eddy currents cause resistance to magnetic flux.

Therefore we need to use laminated core.

Lamination of core



Less
~~More~~ The core loss $\Rightarrow$ less magnetizing current $\Rightarrow$ more R_c

In case of both X_m and R_c , they are shunt resistance operating at (almost) constant voltage.

$\therefore$ Power dissipated $= P = \frac{V^2}{R}$

$P \propto \frac{1}{R}$ or $\frac{1}{Z}$

Faraday's Law / Maxwell's III equation

Maxwell's equation is

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

~~Integrating~~ Integrating this over surface S ,

$$\iint_S (\vec{\nabla} \times \vec{E}) \cdot d\vec{S} = \iint_S \left(-\frac{\partial \vec{B}}{\partial t}\right) \cdot d\vec{S}$$

Assuming S does not change with time,

$$\oint \vec{E} \cdot d\vec{l} = - \iint_S \frac{d(\vec{B} \cdot d\vec{S})}{dt} = - \frac{d}{dt} \iint_S \vec{B} \cdot d\vec{S}$$

$$\oint \vec{E} \cdot d\vec{l} = - \frac{d\phi_m}{dt}$$

which is Faraday's Law.

$\therefore$ Maxwell's equation $\Rightarrow$ Point Form

Faraday's law $\Rightarrow$ Integral Form

$$\mathcal{E} = - \frac{d\phi_m}{dt}$$

Quiz, 2, 10 marks, 30 minutes
Everything covered from Midterms to class before
quiz

DC Motor

The inductance of stator windings (field) is substantially higher than motor (armature) windings

This is because current is flipped in armature windings.

Transient Analysis of DC Motor

$$\cancel{T_e = K \phi I_a}$$

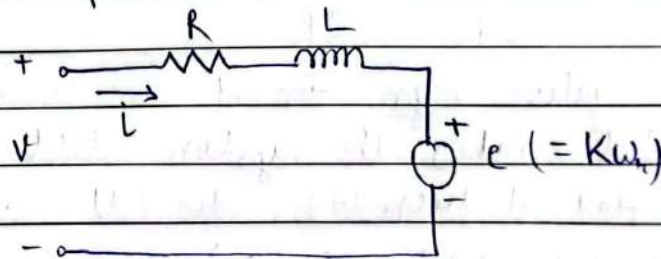
$$T_e = k \phi I_a = K I_a$$

$$E = k \phi \omega_m = K \omega_m$$

K is torque constant in units of Nm A^{-1}

Equivalently, K is the Back emf constant in units of $\text{V/(rad s}^{-1}\text{)}$

Equivalent ckt of DC Motor



R is resistance of armature windings

L is self-inductance of armature windings

e is back emf.

$$V = Ri + L \frac{di}{dt} + K\omega_m \quad (\text{Electrical equation})$$

$$T_e = \underbrace{\mu \omega_m}_{\text{Friction}} + \underbrace{J \frac{d\omega_m}{dt}}_{\text{Load Torque}} + T_L \quad (\text{Mechanical equation})$$

Both the equations are coupled with each other
 (2nd order differential system)

→ Let us neglect L and μ

$$V = R i + K \omega_r$$

$$T_e = J \frac{d\omega_r}{dt} + T_l$$

$$\therefore K i = J \frac{d\omega_r}{dt} + T_l$$

This reduced the original 2nd order system to a 1st order system.

Assume $T_l = 0$.

Consider what happens when a voltage V is applied as a step.

$$J \frac{d\omega_r}{dt} = K i$$

$$V = R i + K \omega_r$$

$$\Rightarrow \frac{d\omega_r}{dt} = -\frac{K^2}{JR} \omega_r + \frac{KV}{JR} \quad \text{--- (1)}$$

The negative sign in of ω_r term on RHS is what makes the system stable.

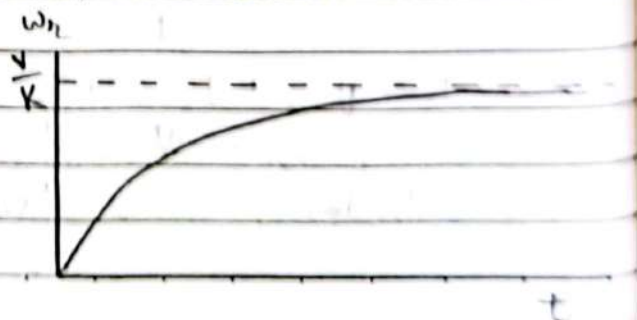
Had it been (+), $d\omega_r/dt$ would be positive for positive $\omega_r \Rightarrow \omega_r$ would increase boundlessly $\Rightarrow$ not physically feasible.

Let us define $\frac{JR}{K^2} = T_m$

T_m is the electromechanical time constant

Solving (1), we get,

$$\omega_r(t) = \frac{V}{K} (1 - e^{-t/T_m})$$



Note that at steady state, $\omega_n \propto \frac{1}{K}$

$$K \propto \phi$$

$$\omega_n \propto \frac{1}{\phi}$$

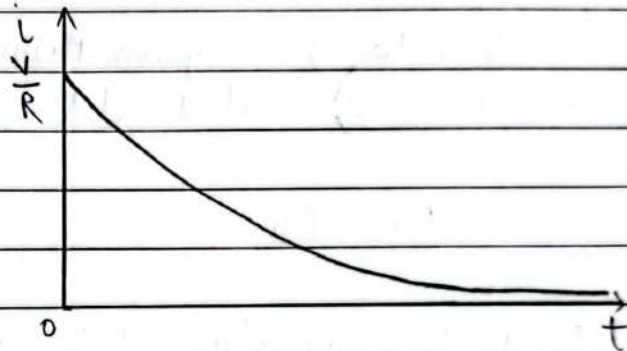
Armature current decays with time.

As $t \rightarrow \infty$, $i \rightarrow 0$.

$$Ri = V - K\omega_n$$

$$i = \frac{V}{R} - \frac{K}{R}\omega_n = \frac{V}{R} - \frac{K}{R} \cdot \frac{V}{K} (1 - e^{-t/\tau_m})$$

$$i = \frac{V}{R} e^{-t/\tau_m}$$



Some points to note -

- Typically in real DC motors, R is small, so initial current ($= V/R$) is very large. i.e., there is a spike in current when it is turned on.
 $\Rightarrow$ Can cause damage $\Rightarrow$ Need for a proper safe mechanism to start a motor gradually.

- $\tau_m \propto R$. Hence if R is small, τ_m will be small, i.e., transient will be short (desirable).

But R cannot be made very small because of the former note.

Superconductive materials can be used for field winding, but not for armature. R_a is critical.

Analysis of Transient without neglecting L & μ (Under-damped case)

$$\frac{di}{dt} = -\frac{R}{L}i - \frac{k}{L}\omega_n + \frac{V}{L}$$

$$\frac{d\omega_n}{dt} = \frac{K}{J}i - \frac{\mu}{J}\omega_n - \frac{T_L}{J}$$

Writing the system in matrix form.

$$\frac{d}{dt} \begin{bmatrix} i \\ \omega_n \end{bmatrix} = \begin{bmatrix} -R/L & -k/L \\ K/J & -\mu/J \end{bmatrix} \begin{bmatrix} i \\ \omega_n \end{bmatrix} + \begin{bmatrix} V/L \\ -T_L/J \end{bmatrix}$$

$\uparrow$ State Vector (x) $\uparrow$ System Matrix (A) $\uparrow$ Input matrix (u)
 i and ω_n are called state variables

$$\therefore \dot{x} = Ax + Bu$$

Find the eigenvalues of A. (Assume $\mu = 0$)

$$\lambda = \frac{-R \pm \sqrt{R^2 - 4K^2}}{2L} \quad (\text{Assume } T_L = 0)$$

$$\omega_n = \omega_h + \omega_p$$

$\uparrow$ Complete Solution $\uparrow$ Homogeneous Solution $\uparrow$ Particular Solution

$$\omega_h = Ae^{\alpha t} \cos \beta t + Be^{\alpha t} \sin \beta t$$

$$\omega_p = \frac{V}{K}$$

$$\text{where } \lambda = \alpha \pm j\beta$$

$$\alpha = -R/2L$$

Here $\alpha < 0$ is critical for this to be a physically feasible system.

$$\therefore w_n = A e^{\alpha t} \cos \beta t + B e^{\alpha t} \sin \beta t + \frac{V}{K}$$

A and B depend on initial conditions.

In our case, $w_n(t=0) = 0$

$$i(t=0) = 0 \Rightarrow \left. \frac{dw_n}{dt} \right|_{t=0} = 0$$

$i(t=0)=0$ was not true in the previous case.

We can make this assumption now because now there is L to prevent sudden in i value.

Putting these initial conditions, we get

$$0 = A + \frac{V}{K} \Rightarrow A = -\frac{V}{K}$$

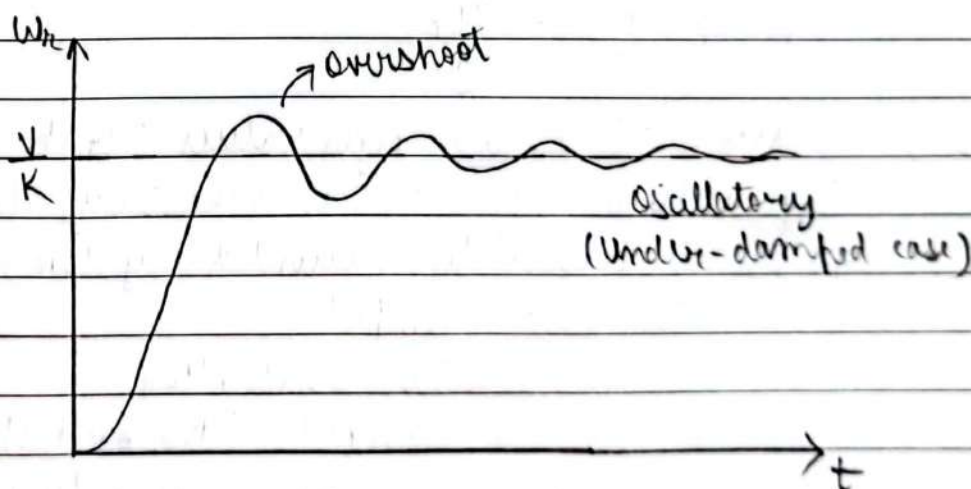
$$\left. \frac{dw_n}{dt} \right|_{t=0} = 0 = A\alpha + B\beta \Rightarrow B = -\frac{\alpha A}{\beta}$$

$$\left. \frac{dw_n}{dt} \right|_{t=0} = 0$$

$$A\alpha + B\beta = 0$$

$$B = -\frac{\alpha A}{\beta}$$

$$\therefore w_n = \frac{V}{K} - \frac{V}{K} e^{\alpha t} \left(\cos \beta t - \frac{\alpha}{\beta} \sin \beta t \right)$$



Note that slope of the curve is 0 at $t=0$

$$\text{i.e., } \left. \frac{dw_n}{dt} \right|_{t=0} = 0$$

In DC Motor, armature (motor) has uniformly distributed, full-pitched (180°) windings.

Induction motors has non-uniform (sinusoidal) distribution of windings.

This is the natural behaviour of the system (DC Machine). Its knowledge is needed in order to design a controller for the system, that forces the system to behave some other more desired way.

e.g., smaller overshoot, small time-constant etc.

$$i(t) = \frac{T}{K} \frac{d\omega}{dt}$$

$$i(t) = \frac{JV}{K^2 \beta} (\alpha^2 + \beta^2) e^{\alpha t} \sin \beta t$$

$$\text{As } t \rightarrow \infty, i(t) \rightarrow 0$$

* The natural response of motor is determined by sign of $\frac{R^2}{L^2} - \frac{4K^2}{JL}$

Case 1) (Positive) $\Rightarrow$ 2 real eigenvalues (both still $-ve$)
 Eigenvalues being $-ve$ is necessary for physical feasibility (bounded system)

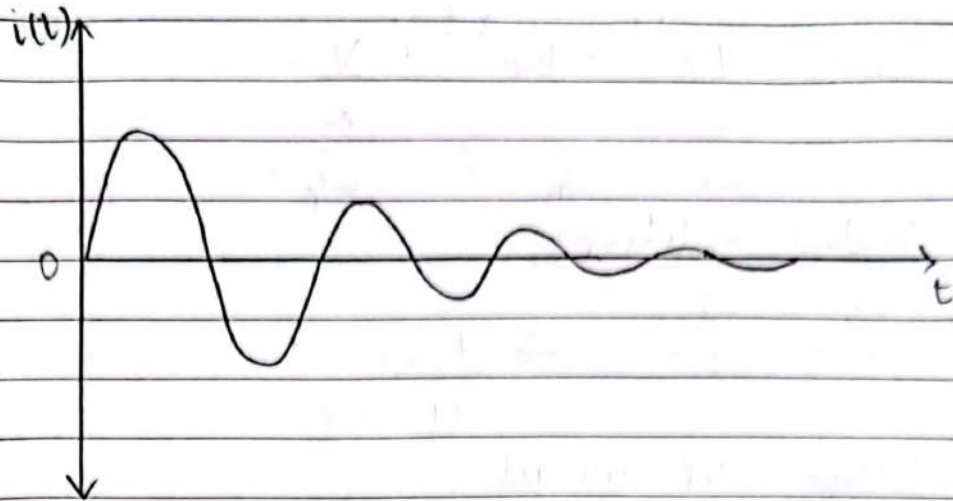
$\Rightarrow$ Over-damped

$\Rightarrow$ No solution oscillation

Case 2) $= 0 \Rightarrow$ Critically-damped $\Rightarrow$ No oscillation

Real and equal eigenvalues

Case 3) (Negative) $\Rightarrow$ Complex eigenvalues $\Rightarrow$ Under-damped
 $\Rightarrow$ Oscillatory



Steady-State Behaviour of DC Machines

3 cases -

- (i) Supply currents to field and armature windings separately (Separately-excited motor)
- (ii) Connect both windings in parallel (shunt-excited motor)
- (iii) Connect them in series (series excited motor)

Transient Analysis without neglecting L (contd.) (over-damped case)

Take $V = 100 \text{ V}$ step at $t = 0$

$$R = 0.017 \Omega$$

$$L = 1.1 \text{ mH}$$

$$J = 2.7 \text{ kg m}^2$$

$$K = 0.4 \text{ Nm A}^{-1}$$

$$\text{Eigenvalues, } \lambda = \frac{-R \pm \sqrt{R^2 - 4KL}}{2L}$$

$$\text{Inputting values, } \lambda_1 = -5.31 \text{ s}^{-1}$$

$$\lambda_2 = -10.14 \text{ s}^{-1}$$

For lower value of K , why does system become non-oscillatory? Explain using physical intuition.

$$\omega_n(t) = \underbrace{Ae^{\lambda_1 t} + Be^{\lambda_2 t}}_{\omega_h} + \underbrace{\frac{V}{K}}_{\omega_p}$$

Initial conditions,

$$\omega_n(t=0) = 0$$

$$i(t=0) = 0 \Rightarrow \left. \frac{d\omega_n}{dt} \right|_{t=0} = 0 \text{ since } T_L = 0$$

Solving IVP, we get

$$A = -524.11 \text{ s}^{-1}$$

$$B = 274.11 \text{ s}^{-1}$$

$$V/K = 250 \text{ s}^{-1}$$

$$\therefore \omega_n(t) = (-524.11 e^{-5.312t} + 274.11 e^{-10.14t} + 250) \text{ rad s}^{-1}$$

$\therefore$ In the over-damped case, system is non-oscillatory.

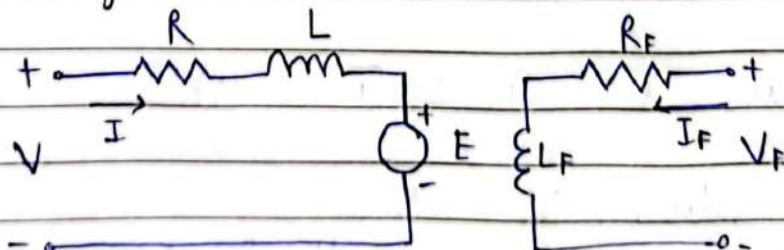
$$i(t) = \frac{T}{K} \frac{d\omega_n}{dt} = (18765 e^{-5.31t} - 18765 e^{-10.14t}) \text{ A}$$

The motor coil carries AC. Hence there are losses in the motor magnetic core.

Types of DC Motors

Steady-State Analysis (Speed-Torque Characteristic)

- Separately Excited DC Machine.



$$E = K\omega_n$$

$$\frac{d\omega_r}{dt} = \frac{K i}{J} - \frac{T_L}{J}$$

$$\frac{di}{dt} = \frac{-R i}{L} - \frac{K \omega_r}{L} + \frac{V}{L}$$

At steady state, $\frac{d\omega_r}{dt} = 0$ and $\frac{di}{dt} = 0$

$$V = R i + K \omega_r$$

$$i = \frac{T_e}{K}$$

$\therefore$ At steady state $T_e = T_L$ (because $d\omega_r/dt = 0$)

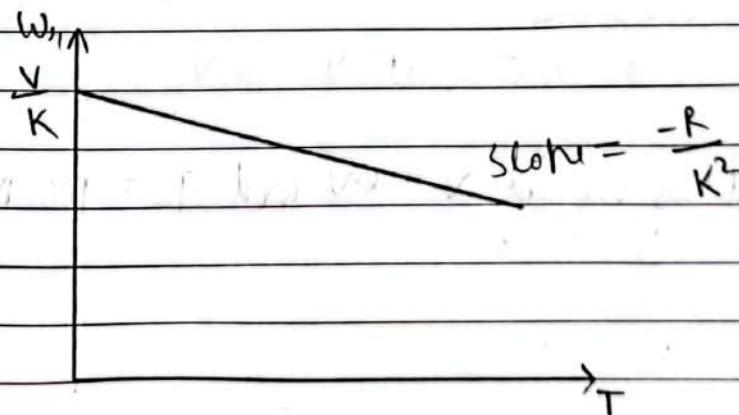
$$\omega_r = \frac{V}{K} - \frac{R}{K^2} T_e$$

$$T_e = \frac{K V}{R} - \frac{K^2 \omega_r}{R}$$

$\therefore$ The speed-torque characteristic is

$$\omega_r = \frac{V}{K} - \frac{R}{K^2} T_e$$

for a separately excited machine.



The intersection point of this curve with $T_L(\omega)$ is the operation point.

No-load speed, $\omega_0 = \frac{V}{K}$

Maximum Torque, $T_{e,max} = \frac{V K}{R}$ when $\omega_r = 0$

$T_{e,max}$ is also called 'Breakdown torque'

If we want to increase ω_r for the same T_e , we can increase V . $\Rightarrow$ Preferred Method

V (armature) is independent of field windings

$\therefore V$ is ~~not~~ independent of K .

(in case of separately excited machine)

What if we increase K ? ω_r may increase or decrease depending on value of T_e , because y-intercept goes up but the curve also becomes steeper. $\Rightarrow$ Not preferred method to increase ω_r

Numerical Examples

Let Φ_f be constant

$R = 0.05 \Omega$

$\omega_{r0} = 80 \text{ rad/s}$, at $V = 100 \text{ V}$

a) What is ω_r for $V = 80 \text{ V}$ and $T_e = 100 \text{ Nm}$

$\omega_{r0} = \frac{V}{K} = 80$

$\therefore K = 1.25$

$\therefore \omega_r = \frac{80}{1.25} - \frac{0.05 \times 100}{(1.25)^2}$

$= 64 - \frac{5 \times 16}{25} = 64 - 3.2 = 60.8 \text{ rad/s}$

b) ~~What is~~ For $T_L = 100 \text{ Nm}$, at what Terminal voltage V will the motor speed be 80 r/s .

$$80 = \frac{4}{5} V - \frac{16 \times 5}{25} = \frac{4}{5} V - 3.2$$

$$V = \frac{5 \times 83.2}{4} = 5 \times 20.8 = \underline{\underline{104 \text{ V}}}$$

c) Let ~~the~~ $T_L = 1.25 W_n$
What is W_n for $V = 100 \text{ V}$?

$$W_n = 80 - \frac{(3.2)}{100} \times 1.25 W_n$$

$$W_n = 76.92 \text{ r/s}$$

Capability Curves of the Machine
2 constraints

1) Current should not exceed beyond a certain maximum value.

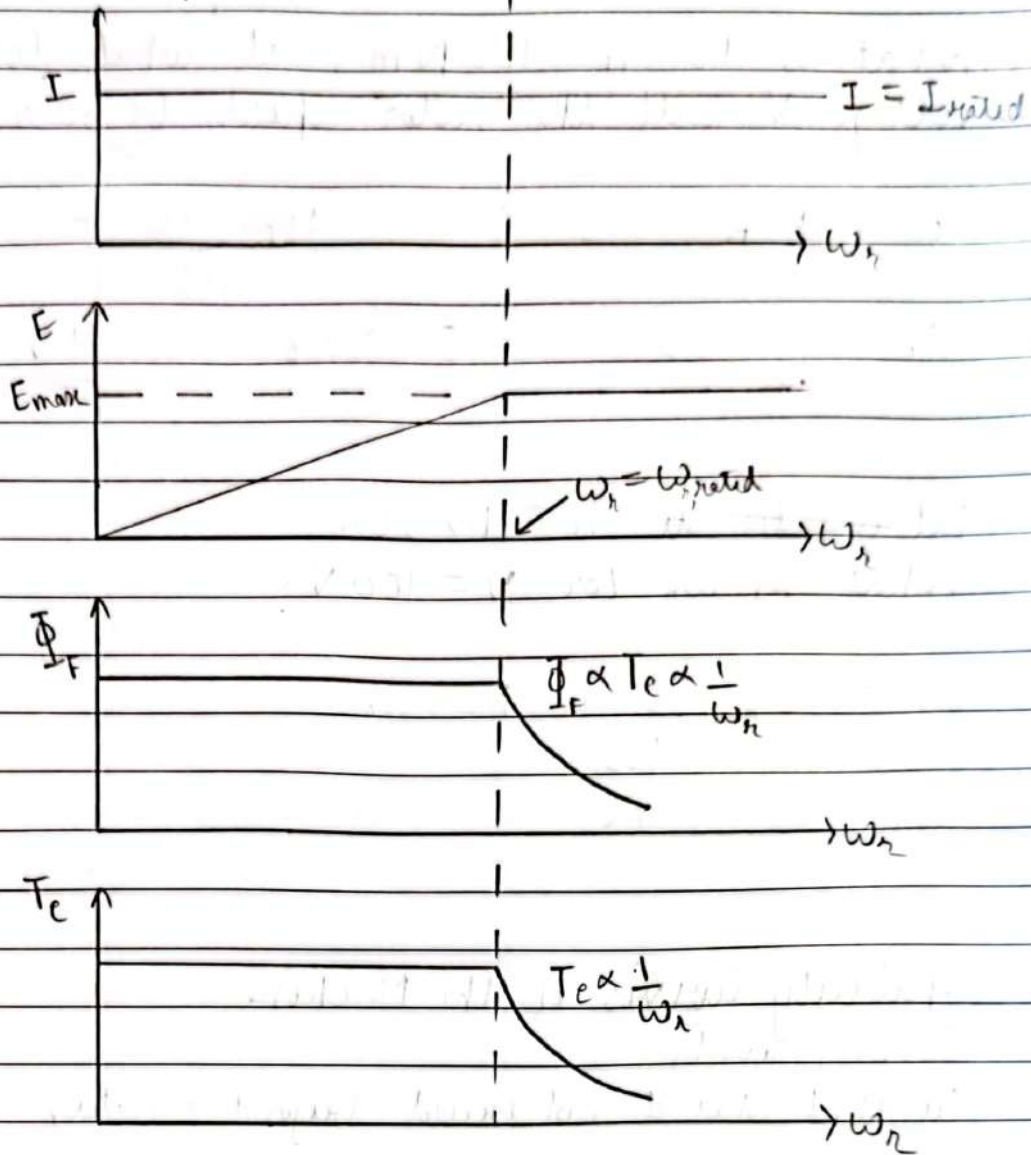
Assume that we hold current at its rated value.
(Rated value = Maximum allowed current)

2) Also, voltage V applied ~~cannot~~ must not exceed a certain maximum value.

P10

Constant Torque Region

Constant Power Region



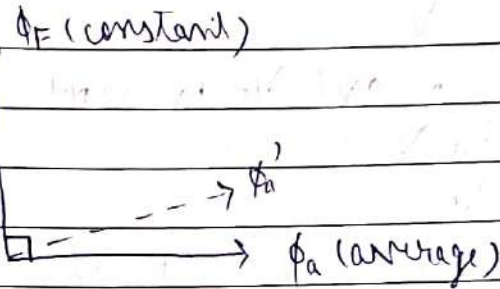
Beyond ω_n , since E becomes constant, and I is constant,

Input Power = $E I = \text{constant}$

$\therefore T_e \omega = \text{constant}$

Field Orientation in DC Machine

The average orientation of armature flux is 90° away from the field windings' flux (which is constant).



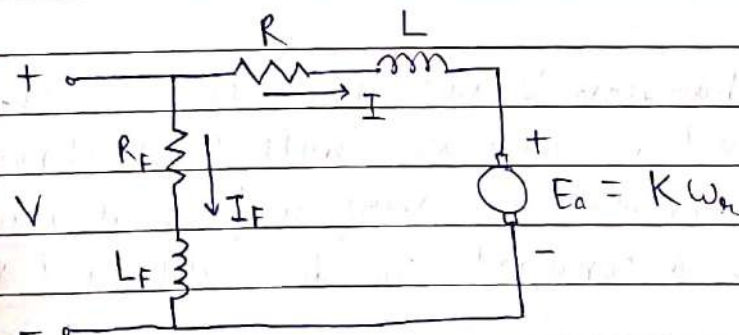
If, say, the average orientation was ϕ_a' , the armature flux would add to field winding flux.

Hence $(\phi_F)_{eff}$ ~~too~~ would have dependence on I_a i.e. the armature current. In this ~~at~~ case, the dynamics would become very complex to analyze, since $K = k_e(\phi_F)_{eff}$ is no longer constant.

Hence ϕ_a is kept perpendicular to ϕ_F in order to decouple field flux from armature current.

To ensure this, the brushes are placed appropriately on the rotor.

- Shunt Connected DC Machine



Note that $K \propto \phi_F$

If we assume linear magnetic characteristics,

$$K \propto \phi_F \propto I_F \propto V$$

$$K \propto V$$

$$K = \frac{V}{V_0}$$

$$K_0 \quad V_0$$

where K_0 and V_0 is some test point

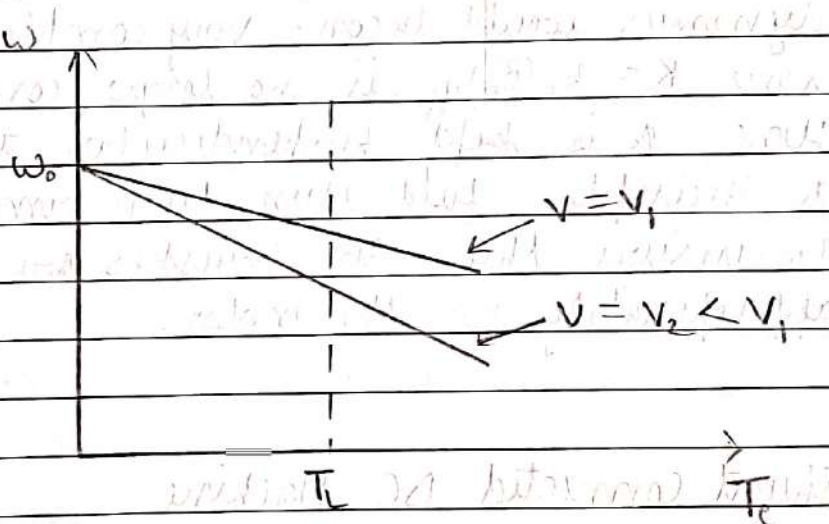
$$\therefore K = \frac{K_0 V}{V_0}$$

$$\text{Also, } \omega_n = \frac{V}{K} - \frac{R T_e}{K^2}$$

$$= \frac{V V_0}{K_0 V} - \frac{R T_e V_0^2}{K_0^2 V^2}$$

$$\omega_n = \omega_0 - T_e \left(\frac{\omega_0^2 R}{V^2} \right)$$

ω_0 is the no-load speed



Since changing V will only change slope of curve, it is not an effective method of changing ω_n .

Shunt motors are very useful if we want more or less a constant ω_n for different values of T_e and V .

How to control ω_n then?
change R_F .

If R_F increases, for same V , I_F decreases,
 ϕ_F decreases, K decreases $\Rightarrow \omega_n$ increases
but (-slope) also increases.

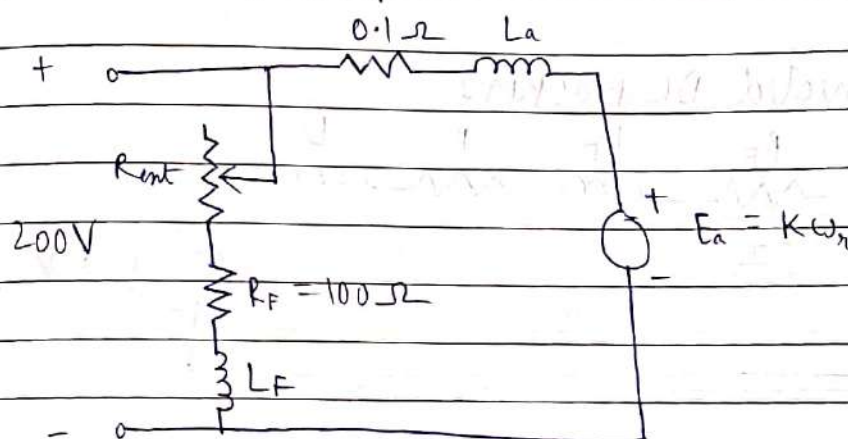
How to change R_F ?

Add a resistor in series with field windings.

$\therefore$ We can only increase R_F .

$\therefore$ We can only increase the no load speed ω_n .

Numerical Examples



R_{ext} can be varied from 0 to 100Ω

When $R_{ext} = 0$, $\omega_n = 100 \text{ rad/s}$

i) At what value of R_{ext} will the no-load speed
 $\omega_n = 125 \text{ rad/s}$?

$$\omega_n = \frac{V}{K}$$

$$100 = \frac{200}{K} \Rightarrow K = 2 \text{ Nm A}^{-1}$$

$$125 = \frac{200}{K'} \Rightarrow K' = \frac{8}{5} = 1.6 \text{ Nm A}^{-1}$$

$$K \propto \frac{1}{R_f + R_{ext}}$$

$$\therefore \frac{K}{K'} = \frac{100 + R_{ext}}{100} = \frac{2}{1.6} = \frac{20}{16} = \frac{5}{4}$$

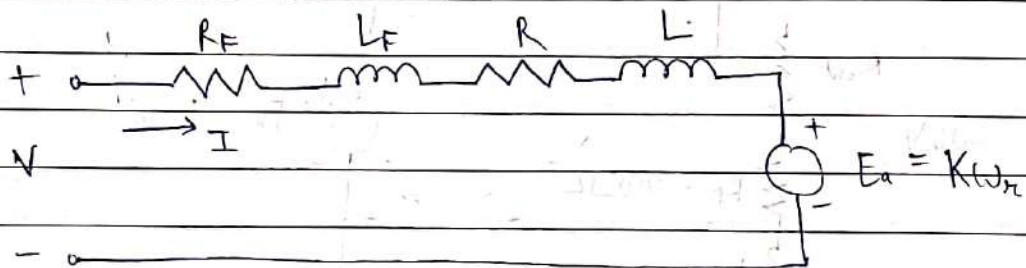
$$1 + \frac{R_{ext}}{100} = 1.25$$

$$R_{ext} = 25 \Omega$$

ii) If $R_{ext} = 25 \Omega$, $T_L = 3.2 \text{ Nm}$, find ω_r

~~ω_r~~ = Homework

- Series Connected DC Machine



Remember: The magnetic axes of L_f and L_a are perpendicular.

Let $K = K_0$ at $I = I_0$ at some test point.

Note that $K \propto \phi \propto I$ (assuming linear magnetics)

$K \propto I$ (assuming linear magnetics)

$$K = I \times \left(\frac{K_0}{I_0} \right)$$

$$T_e = KI = \frac{K_0 I^2}{I_0}$$

$$E = K\omega_r = \frac{K_0 I \omega_r}{I_0}$$

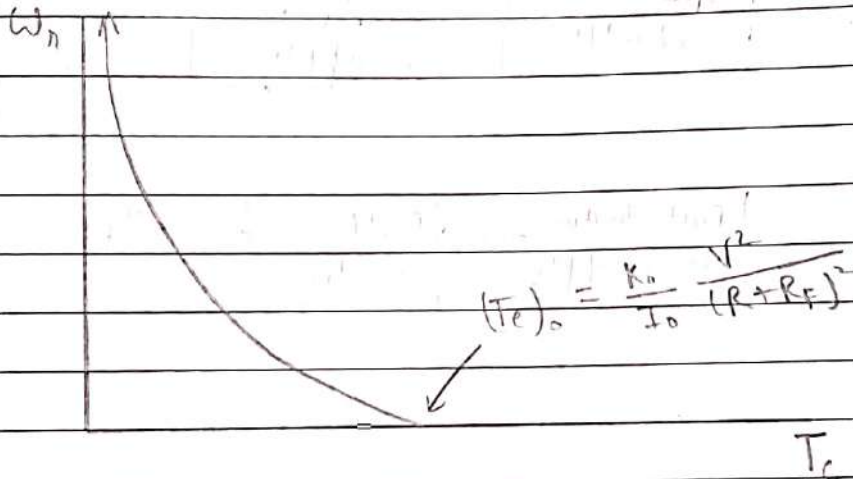
$$V = (R + R_F) I + \frac{K_o}{I_o} I \omega_n$$

$$I = \frac{V}{R + R_F + \left(\frac{K_o}{I_o}\right) \omega_n}$$

$$T_e = KI = \frac{K_o}{I_o} I^2$$

$$\therefore T_e = \frac{K_o}{I_o} \times \frac{V^2}{\left(R + R_F + \frac{K_o}{I_o} \omega_n\right)^2}$$

This is the torque-speed characteristic for a series DC Machine



Starting torque, $(T_e)_o = \frac{K_o V^2}{I_o (R + R_F)^2}$ at $\omega_n = 0$

As $T_e \rightarrow 0$, $\omega_n \rightarrow \infty$

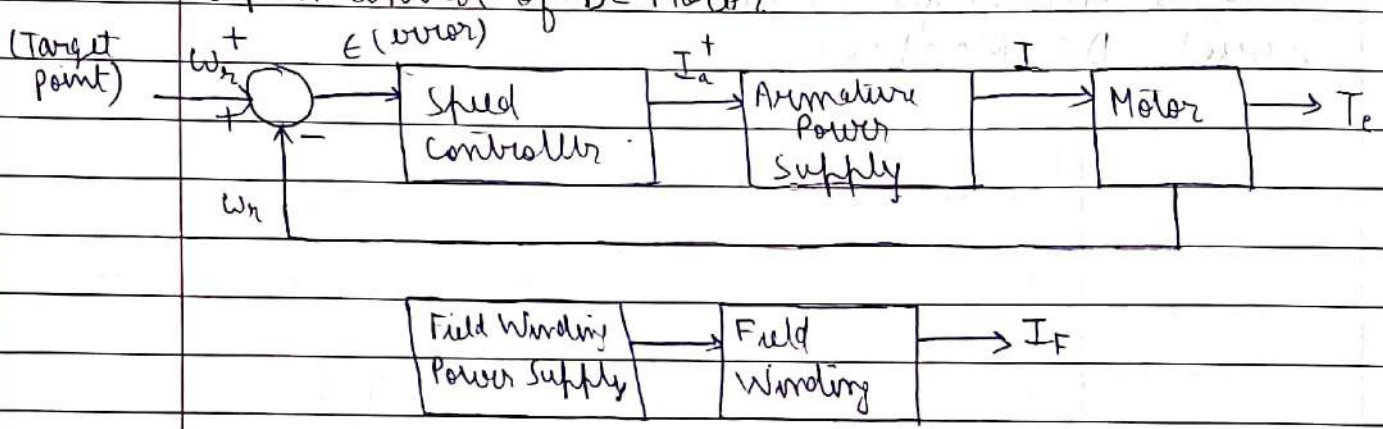
~~Used~~ At no load ($T_e = 0$), speed is extremely high. When load is present, $(T_e)_o \propto V^2$, i.e., can provide a very large torque.

Applications - Very commonly used in kitchen-appliances such as mixers and in hand-held drills etc. Can also be used almost equally well with AC supply, since both ϕ_F and ϕ_a change polarity.

* Series machine also works almost equally well with AC. But Best on DC.

Series machine is most commonly used in the DC motor type out of the 3 types in the real world.

Speed Control of DC Motor



Also by β -angle displaced 2 coils.

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Induction Motor

Rotor is not connected to a power supply.
Current in rotor is produced because of induction due to flux produced by stator (alternating flux).

The ~~rotor~~ ^{stator} has sinusoidally distributed windings.
i.e., the turns density is a sinusoidal function of ϕ .

Each arm of coil has its corresponding return 180° away.

The rotor windings are shorted at both ends.

Turns density $\eta(\phi)$ is number of turns per unit angle.
 $\eta(\phi) = \eta_m \sin \phi$ (ideal case)

$$\int_0^\pi \eta(\phi) d\phi = N$$

$$\therefore \eta_m = \frac{N}{2}$$

$$\therefore \eta(\phi) = \frac{N}{2} \sin \phi$$

The return path of every turn is 180° away from it.

It can be shown that in the air gap.

$$B_a(\phi) = \frac{\mu \cdot N i \cos \phi}{2g} = B_m \cos \phi$$

Flux density distribution is sinusoidal in space for a constant current.

Three-phase sinusoidally distributed windings

There are 3 windings, a, b and c phase. b and c phase windings are identical to a-phase, except they are angularly displaced $\frac{2\pi}{3}$ from it in phase.

The incoming current of a-phase shares a conductor with outgoing current of b-phase and so on.

$$i_a = I_m \cos(\omega t)$$

$$i_b = I_m \cos(\omega t - \frac{2\pi}{3})$$

$$i_c = I_m \cos(\omega t + \frac{2\pi}{3})$$

$$K = \frac{\mu_0 N}{2g}$$

$$B_a(\varphi, t) = K I_m \cos(\omega t) \cos(\varphi)$$

$$B_b(\varphi, t) = K I_m \cos(\omega t - \frac{2\pi}{3}) \cos(\varphi - \frac{2\pi}{3})$$

$$B_c(\varphi, t) = K I_m \cos(\omega t + \frac{2\pi}{3}) \cos(\varphi + \frac{2\pi}{3})$$

$$B(\phi, t) = B_a + B_b + B_c$$

$$= \frac{3}{2} K I_m \cos(\omega t - \phi)$$

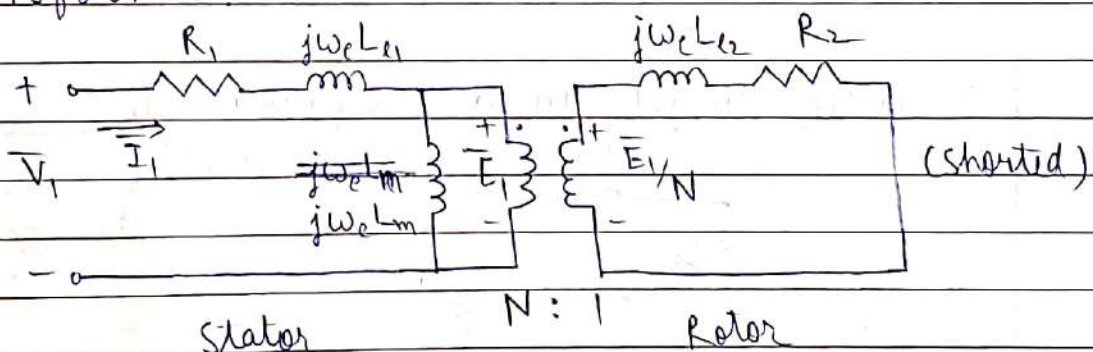
- ∴ The position of net maximum B rotates with angular velocity ω .
- ∴ This is called rotating magnetic field.

The motor rotates with an angular velocity different from that of magnetic field. This causes relative motion of B w.r.t to motor coil $\Rightarrow$ induces emf.

Since $\omega_r \neq \omega$, this motor is called asynchronous motor. It is also called squirrel-cage motor because of its structure.

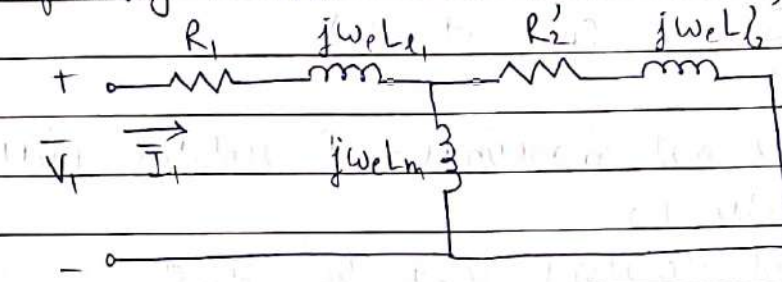
Equivalent circuit of 3- ϕ Induction Machine

- If we consider that machine is not rotating, the per-phase equivalent is basically a Transformer.



N is the effective turns ratio (measured experimentally). For stationary rotor, the frequency of current in rotor is same as that in stator.

Reflecting R_2 and L_2 to stator side,



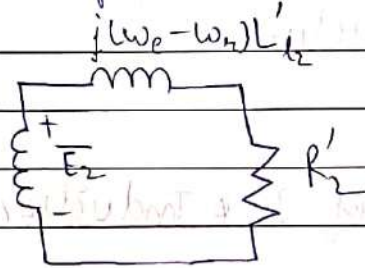
$$R_2' = N^2 R_2$$

$$L_2' = N^2 L_2$$

(Stator's frame of reference)

Now, if we allow motor to rotate, then the B rotates only $\omega_e - \omega_r$ relative to stator motor.

Equivalent ckt from motor's perspective,

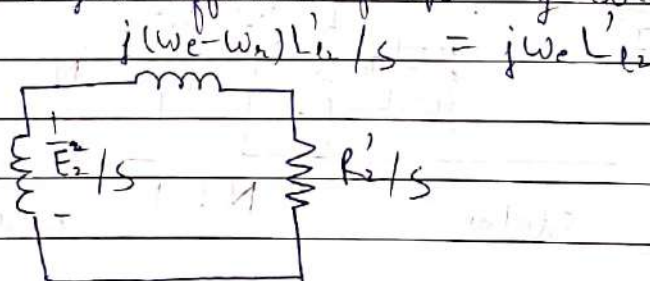


Rotor side circuit
(E_2 and I_2 have frequency $\omega_e - \omega_r$)

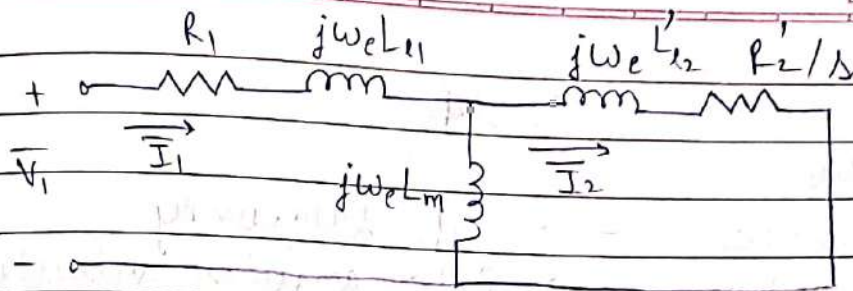
Let us define, slip, $s = \frac{\omega_e - \omega_r}{\omega_e}$

Slip frequency, $s\omega_e = \omega_e - \omega_r$

In order to get effective frequency ω_e ,



This is the motor circuit reflected onto stator side.



* R_2/s has speed dependence
This is the final equivalent ckt (per-phase) of induction-machine.

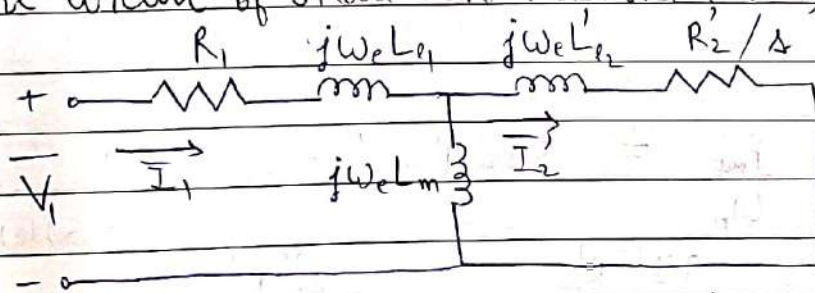
The power input from V_1 is consumed by R_2/s . Part of that is ^{actual} resistive loss. Remaining part is converted to mechanical energy, which is ~~represented by~~ represented by R_2/s in equivalent ckt. This is why $R_2 = R_2/s$ has speed dependence.

$\therefore R_2$ is essential for functioning of induction machine. It cannot be made zero.

Hence, there cannot be a loss-less induction machine, even in principle.

This is why rotor coils are made from Al and not Cu, because Al has higher resistance.

Equivalent circuit of Induction Machine (contd)



(per-phase equivalent circuit)
as viewed from stator side

$$s = \frac{\omega_e - \omega_r}{\omega_e} \rightarrow \text{Slip}$$

$$s\omega_e = \omega_e - \omega_r \rightarrow \text{Slip frequency}$$

$$\omega_r = (1-s)\omega_e \rightarrow \text{Rotor angular velocity}$$

Only R_1 and R_2'/s consume power.

s depends on ω_r .

$\therefore$ Power consumed depends on ω_r .

$\therefore$ Part of power input is dissipated in stator & rotor resistances. Rest is converted into mechanical energy in the form of ω_r .

The power that goes across the air gap (between stator & rotor) is P_{gap} .

It is essentially the power dissipated in rotor.

$$P_{gap} = 3 I_2'^2 R_2' \quad (3 \text{ because } 3-\phi)$$

$$s P_{gap} = 3 I_2'^2 R_2' \text{ is the power dissipated in rotor resistance.}$$

$$\therefore P_{r, \text{loss}} = s P_{gap} = 3 I_2'^2 R_2'$$

$$\therefore \text{Mechanical Power output} = P_{gap} - P_{r, \text{loss}} = (1-s) P_{gap}$$

$$P_{out} = 3 I_2'^2 R_2' \left(\frac{1-s}{s} \right)$$

$$T_e = \frac{P_{out}}{\omega_r} = \frac{3 I_2'^2 R_2' (1-s)}{s} \times \frac{1}{(1-s)\omega_e}$$

$$\therefore T_e = \frac{3 I_2'^2 R_2'}{s\omega_e} = \frac{3 I_2'^2 R_2'}{\omega_e - \omega_r}$$

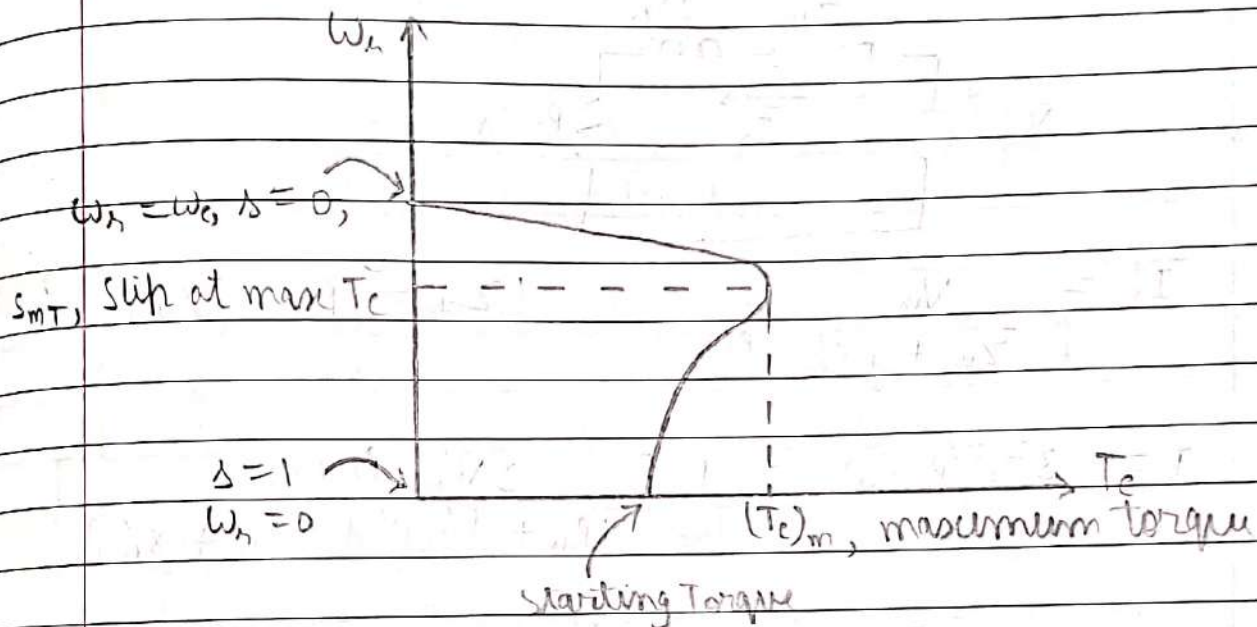
This is the Torque-Speed Characteristic.

$\therefore \cancel{T_c \propto R_2^2} \quad T_c \propto P_{r, loss}$

$\therefore$ If there is no motor resistance loss, there cannot be any torque produced

Torque - Speed Characteristic

If we keep V_1 and ω_c fixed, and vary ω_r , we get the curve,

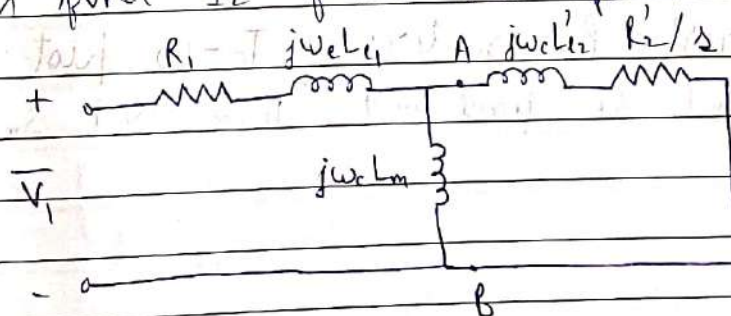


Now, we shall see its derivation.

$$T_c = \frac{3 I_2'^2 R_2'}{s \omega_c}$$

Here when ω_c changes, I_2' also changes.

We can find I_2' from the equivalent circuit.

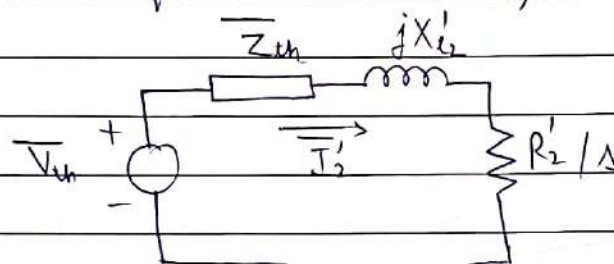


Find thevenin equivalent of stator circuit of across A-B.

$$\overline{V_{th}} = \overline{V_1} \left[\frac{jX_m}{R_1 + j(X_{e1} + X_m)} \right]$$

$$\overline{Z_{th}} = \frac{jX_m (R_1 + jX_{e1})}{R_1 + j(X_{e1} + X_m)}$$

Thevenin equivalent circuit,



$$\overline{I'_2} = \frac{\overline{V_{th}}}{\overline{Z_{th}} + jX'_{e2} + R'_2/s} \quad (\overline{Z_{th}} = R_{th} + jX_{th})$$

$$T_e = \frac{3 I_2'^2 (R'_2/s)}{\omega_e} = \frac{1}{\omega_e} \left[\frac{3 V_{th}^2 (R'_2/s)}{(R_{th} + (R'_2/s))^2 + (X_{th} + X'_{e2})^2} \right]$$

$$T_e = \frac{1}{\omega_e} \left[\frac{3 V_{th}^2 (R'_2/s)}{(R_{th} + (R'_2/s))^2 + (X_{th} + X'_{e2})^2} \right]$$

This is the required torque-speed characteristic

Characteristic Parameters of $T_e - \omega$ plot

We want to find out $(T_e)_m$ and s_{MT} and T_{start}

It can be shown that P_{gap} is maximum when

$$R_2' = |R_{th} + j(X_{th} + X_{e2}')|$$

(Maximum Power Transfer Theorem)

$$R_{2T}' = \sqrt{R_{th}^2 + (X_{th} + X_{e2}')^2}$$

$$\Delta_{mT} = \frac{R_2'}{\sqrt{R_{th}^2 + (X_{th} + X_{e2}')^2}}$$

$$(T_e)_m = \frac{3}{2\omega_e} \left[\frac{V_{th}^2}{R_{th} + \sqrt{R_{th}^2 + (X_{th} + X_{e2}')^2}} \right]$$

At starting torque, $\omega_r = 0 \Rightarrow s = 1$

$$(T_e)_s = \frac{1}{\omega_e} \left[\frac{3 V_{th}^2 R_2'}{(R_{th} + R_2')^2 + (X_{th} + X_{e2}')^2} \right]$$

Let us make some simplifying assumptions.

R_1 is negligibly small
 X_m is large.

$$\bar{Z}_{th} = \frac{jX_m(R_1 + jX_{e1})}{R_1 + j(X_{e1} + X_m)} \approx jX_{e1}$$

$$\bar{Z}_{th} \approx jX_{e1}$$

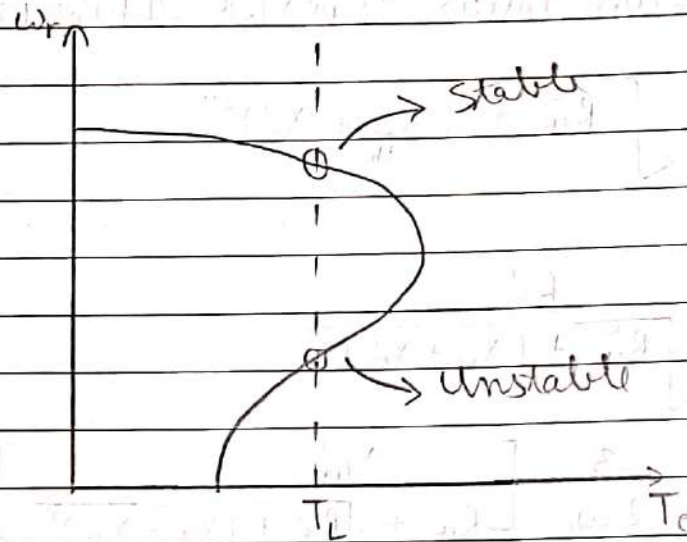
$$R_{th} \approx 0$$

$$X_{th} \approx X_{e1}$$

$$\bar{V}_{th} = V_1 \left(\frac{jX_m}{R_1 + j(X_{e1} + X_m)} \right) \approx V_1$$

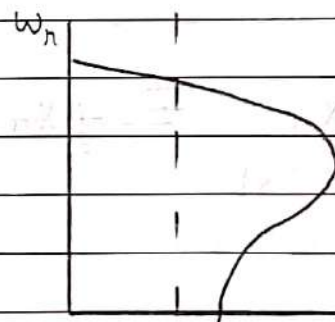
$$jX_m(R_1 + jX_{e1})(R_1 - j(X_{e1} + X_m)) = j(R_1^2 + X_{e1}(X_{e1} + X_m))$$

Below (using approximations)
 $T_c \propto S w_c$ for small $S w_c$



For a given T_c , there are 2 solutions for S (we get a quadratic in S).
The larger one is realized in the real world because it is a stable equilibrium.

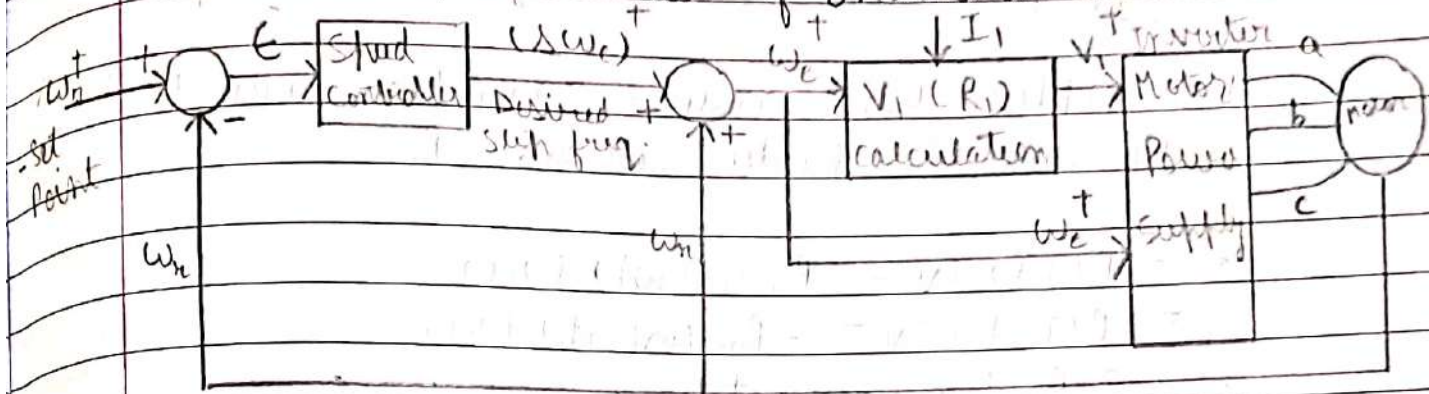
Moreover, the $w_r - T_c$ curve actually goes below the x -axis.



(w_r in opposite direction)

For speed control, we want to control ω_c , but not possible. $\therefore$ we control V_1 , ($V_1 \propto \omega_c$) $\Rightarrow$ we get slight change in T_m

Automatic Speed Control of Induction Machine



Scalar Control

Scalar control does not take special care to ensure that $\phi_r \perp \phi_a$ (Field winding flux $\perp$ armature flux)

Vector control ensures that, but is significantly more complex.

Scalar control can be implemented using analog circuits (using opamps). Vector control needs digital processors.

In the calculation block, we may include the $I_1 R_1$ voltage drop in mind (by having an estimate of R_1). Hence, we are able to control V_1 itself, instead of V .

$$T_e = \frac{3 V_1^2 (R_2'/s)}{\omega_c ((R_2'/s)^2 + (X_{e1} + X_{e2}')^2)}$$

$$\approx \frac{3 V_1^2 s}{\omega_c R_2'}$$

PTO

Analyzing Induction Motor using First Principles

For simplification, assume, armature current is too small to produce its own B .

$$e_2 = B(0, t) l v = B_m \cos(\omega t) l r \omega$$

$$e_1 = B(\pi, t) l v = -B_m \cos(\omega t) l r \omega$$

$$\therefore e = 2 B_m l r \omega \cos(\omega t)$$

Let the impedance of rotor turn be $\bar{Z} = Z \angle \beta$

$$\therefore i = \frac{2 B_m l r \omega \cos(\omega t - \beta)}{Z}$$

Force on each conductor,

$$f = B_m \cos(\omega t) \times \frac{2 B_m l r \omega \cos(\omega t - \beta)}{Z} \times l$$

$$f = \frac{B_m^2 l^2 r \omega}{Z} \{ \cos(2\omega t - \beta) + \cos \beta \}$$

$$\therefore \text{Average Force, } F = \frac{B_m^2 l^2 r \omega}{Z} \cos \beta$$

$$\text{Average Torque, } T_e = \frac{2 B_m^2 l^2 r^2 \omega}{Z} \cos \beta$$

$$\text{where } \beta = \tan^{-1} \left(\frac{\omega L}{R} \right)$$

$\therefore$ We need R to produce T_e . If $R=0$, $\beta = \pi/2$, $T_e = 0$

$\therefore$ i.e., if coil is purely inductive, current can not interact with ~~its own~~ the B that produced it. Presence of R introduces the necessary phase shift in i to do so.

Note that we can consider this $\omega = \omega_r - \omega_m$, because in this simplified case, motor is at rest.

$$T_e = \frac{2 B_m^2 L^2 r^2 \omega \cos \beta}{Z}$$

Here Z is a function of ω
 β is also a function of ω
 Hence, dependence of T_e on ω is not very simple

It actually turns out to be same as before